KINETIC ACCELERATION OF NON-EQUILIBRIUM STEADY STATES

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Abstract. We ask whether kinetic acceleration (lifting) survives away from equilibrium, and how the convergence rate deteriorates with the strength η of non-equilibrium driving. We study generators $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ on a Hilbert space, where $\gamma > 0$ is friction, $\mathcal{R} \leq 0$ is a self-adjoint dissipator, $\mathcal{V}_{\text{skew}}$ is skew-adjoint, and $\eta \geq 0$ measures departure from detailed balance.

Within the Dolbeault–Mouhot–Schmeiser hypocoercivity framework [13, 14], we construct a gap-adapted Lyapunov functional and obtain an explicit rate lower bound with a certified perturbation threshold $\eta^*(\gamma)$. For model families whose structural constants remain uniformly controlled as $\lambda(Q_O) \downarrow 0$, the gap-adapted choice of the DMS corrector recovers the kinetic scale $\mathcal{O}(\sqrt{\lambda(Q_O)})$ at optimal friction, improving on the diffusive $\mathcal{O}(\lambda(Q_O))$ scale. A complementary conditional overdamped ceiling shows that this acceleration is genuinely a finite-friction effect and that the $\sqrt{\lambda(Q_O)}$ exponent is optimal whenever the ceiling assumptions hold. We also give a constructive reverse-adiabatic-elimination recipe for building lifted generators. Applications to a convex-potential underdamped Langevin family and a boundary-driven quantum model are developed, providing rigorous infinite- and finite-dimensional verifications of the $\sqrt{\lambda(Q_O)}$ mechanism.

1. Introduction

1.1. **Motivation and main question.** Many systems of interest—forced diffusions [22, 32], boundary-driven quantum systems [26], and actively controlled open-system settings such as active reservoir engineering [23]—operate away from detailed balance. Their generators contain a non-reversible component of strength η and converge to a non-equilibrium steady state (NESS). For reversible systems ($\eta = 0$), lifting to a larger phase space can accelerate mixing [10, 16, 5, 29]: at optimal intermediate friction, the rate improves from the diffusive scale to its square root (quadratic speedup). More recently, non-reversible lifting mechanisms beyond the reversible-target setting have also been studied from a different viewpoint in Markov-process theory [15]. In this paper we ask:

> For a system that is η away from equilibrium, under what conditions does the kinetic acceleration (lifting) persist, and how does the convergence rate degrade as a function of η?

Concretely, we consider generators on a Hilbert space $\mathcal{H}$ of the form

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}, \tag{1}$$

where $\gamma > 0$ is a friction (scale-separation) parameter, $\mathcal{R} \leq 0$ is self-adjoint with nontrivial kernel $\mathcal{H}_S := \ker(\mathcal{R})$, $\mathcal{V}_{\text{skew}}$ is skew-adjoint, and $\eta\mathcal{V}_{\text{pert}}$ is a non-reversible perturbation whose strength $\eta \geq 0$ is *given by the physics*. In the singular limit $\gamma \to \infty$, adiabatic elimination [25, 37] reduces the dynamics to an effective slow generator

$$\mathcal{L}_O = -P_S\mathcal{V}\mathcal{R}^+\mathcal{V}P_S$$

acting on $\mathcal{H}_S$, where P_S is the orthogonal projection onto $\mathcal{H}_S$ and $\mathcal{R}^+$ is the Moore–Penrose pseudoinverse. This overdamped limit gives a decay rate of order $\lambda(\mathcal{L}_O)/\gamma$, i.e., a diffusive bottleneck.

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We then ask: (i) whether the quadratic acceleration at finite friction survives the non-equilibrium perturbation, and (ii) how large η can be before the accelerated regime breaks down.

1.2. **Main results.** The decomposition (1) isolates a self-adjoint *reversible backbone*

$$Q_O := -P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} P_S, \tag{2}$$

which captures the reversible skeleton of the slow dynamics. Let $\lambda(Q_O)$ denote its spectral gap on the mean-zero slow subspace.

Within the DMS hypocoercivity framework [13, 14], we construct a modified Lyapunov functional $\mathcal{E}_{\alpha,\varpi}(f) = \frac{1}{2}\|f\|^2 + \alpha\, \Re\langle \mathcal{A}_\varpi f, f\rangle$ with an explicit corrector operator $\mathcal{A}_\varpi$ adapted from [22, 6]. Our two main results are a constructive lower bound and a conditional overdamped ceiling.

In the formulas below, $\Lambda_{\mathrm{hypo}}(\gamma)$ is the unperturbed rate profile, $B_{\mathrm{pert}}(\gamma)$ the perturbative loss, and $\delta_*(\gamma)$ the Lyapunov-norm distortion.

Theorem A (Theorem 5.1). *Under Assumptions 4.1–4.2 (and Assumption 4.3 in the unbounded setting), for every $\gamma > 0$ and every perturbation strength within the certified range $\eta \le \eta^*(\gamma)$, the semigroup decays exponentially on the mean-zero subspace:*

$$\|e^{t\mathcal{L}} f\| \le C_*(\gamma)\, e^{-\lambda_\eta t}\|f\|, \qquad \forall\, f \in \mathcal{H}_0 \text{ (mean-zero subspace)},\ t \ge 0,$$

The rate is fully explicit and satisfies

$$\lambda_\eta \ge \frac{\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma)}{1 + \delta_*(\gamma)}.$$

Moreover, the Lyapunov distortion is uniformly controlled: $\delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\| \le 2^{-1/2}$, hence $C_*(\gamma) \le 1 + \sqrt{2}$, and therefore

$$\lambda_\eta \ge (2 - \sqrt{2})(\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma)).$$

The gap-adapted choice $\varpi_* := \lambda(Q_O)$ is the one that restores $\sqrt{\lambda(Q_O)}$ scaling; the certified threshold

$$\eta^*(\gamma) := \min\left\{\eta_{\mathrm{sym}}(\gamma),\ \frac{\Lambda_{\mathrm{hypo}}(\gamma)}{2B_{\mathrm{pert}}(\gamma)}\right\}, \qquad \eta_{\mathrm{sym}}(\gamma) := \begin{cases} \infty, & a_2 = 0, \\ 2\gamma/a_2, & a_2 > 0, \end{cases}$$

is completely computable once the structural constants of the model are known.

Theorem B (Theorem 5.7). *Under Assumption 5.6 (the structural assumptions plus contraction, eigenfunction attainment, and corrector regularity), any admissible L^2 exponential rate ν with prefactor C obeys the overdamped ceiling*

$$\nu \le (1 + \log C)\left[\frac{(1+\kappa_{\mathrm{ell}})\lambda(Q_O)}{\gamma} + \eta(b_1{+}b_2) + \eta(a_1{+}a_2)\frac{\sqrt{\lambda(Q_O)}}{\gamma}\right].$$

In particular, for $\gamma \to \infty$ and fixed η, the rate is at most $\mathcal{O}(\gamma^{-1}) + \mathcal{O}(\eta)$: the diffusive bottleneck persists, so acceleration is necessarily a finite-friction phenomenon.

Theorem A is fully constructive; Theorem B is a conditional complement requiring norm-level contraction of the lifted semigroup.

How to read the main statements. There are three logically distinct layers in the results:

(i) Theorem A gives a fully explicit *model-wise* decay bound under Assumptions 4.1–4.2.
(ii) Corollary 5.2 extracts the $\sqrt{\lambda(Q_O)}$ scaling for *model families* whose structural constants remain uniformly controlled as $\lambda(Q_O) \downarrow 0$.
(iii) Theorem B is a *conditional* overdamped ceiling. In the examples below, both the convex-potential Langevin family and the boundary-driven toy model verify the structural hypotheses behind Theorem A and Corollary 5.2, with the Langevin example providing the family-uniform verification of the kinetic mechanism for general convex potentials. For both natural NESS

examples, the additional contraction hypothesis required by Theorem B remains a separate issue.

A concrete illustration of the gap-adapted shift that makes this work within the DMS framework is given for a lifted random walk on the discrete cycle in Section 2, before the abstract development. Within the certified regime $\eta \le \eta^*(\gamma)$, the lower bound deteriorates linearly in η; outside that regime, Theorem B shows that simply moving to the overdamped regime cannot recover kinetic scaling.

Remark 1.1 (Near-optimality of the accelerated rate). Within the scope of Theorem B, any model family satisfying the additional uniformity hypotheses of Corollary 5.2 can achieve at most the $\sqrt{\lambda(Q_O)}$ scaling in the exponent. Indeed, Theorem B shows that any admissible L^2 rate satisfies $\nu = \mathcal{O}(\lambda(Q_O)/\gamma) + \mathcal{O}(\eta)$; at intermediate friction $\gamma \asymp \sqrt{\lambda(Q_O)}$, this gives $\nu = \mathcal{O}(\sqrt{\lambda(Q_O)})$. Whenever Corollary 5.2 applies, Theorem A attains $\Theta(\sqrt{\lambda(Q_O)})$ in the same regime, so the exponent is sharp. The same square-root ceiling has been established independently by Eberle–Lörler [16, 17] and Brigati–Lörler–Wang [5], confirming that it is intrinsic to kinetic lifting. We emphasize that optimality is in the *exponent* $\lambda(Q_O)^{1/2}$; the multiplicative constants are not claimed to be sharp.

1.3. **Kinetic acceleration and non-equilibrium stability.** Our contribution has three parts. First, for families satisfying the uniformity hypotheses of Corollary 5.2, lifting gives a *quadratic speedup*: the rate improves from $\lambda(Q_O)$ to $\sqrt{\lambda(Q_O)}$ at intermediate friction $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$. Second, we quantify how non-equilibrium driving modifies this acceleration within a DMS-type slow–fast framework built around the self-adjoint backbone Q_O; earlier analyses closest in spirit [10, 16, 5, 29] focus on reversible targets. Third, we propose a concrete *algorithmic lift* (Section 6.1, Algorithm 1): given a target $\mathcal{L}_O$ for which the reverse adiabatic-elimination consistency relation can be solved, we build a lifted generator, identify conditions for zero projection bias (Proposition 6.1), and obtain $\sqrt{\lambda(Q_O)}$ acceleration once the structural hypotheses are verified.

Theorem 5.1 shows that the certified lower bound degrades *linearly* in η, with slope $B_{\mathrm{pert}}(\gamma)$. A key cancellation makes $B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) = \mathcal{O}(1)$, so $\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}$: faster reversible mixing allows stronger non-equilibrium forcing while keeping kinetic acceleration.

The lifted NESS h_η and the overdamped NESS g_η may differ at finite friction: $\|P_S h_\eta - g_\eta\| = \mathcal{O}(\eta/\gamma)$ in general. For lifts built via reverse adiabatic elimination (Section 6.1), $P_F \mathcal{V} P_F = 0$ can be enforced by design (Proposition 6.1), eliminating this bias entirely; a two-phase extension handles the case $P_F \mathcal{V} P_F \neq 0$ (Remark 6.3).

1.4. **Related work.** The L^2 hypocoercive method based on modified Lyapunov functionals was developed by Dolbeault–Mouhot–Schmeiser [13, 14], building on Villani's H^1 approach [38]; its adaptation to Langevin dynamics using resolvent-type correctors is due to Iacobucci–Olla–Stoltz [22], Roussel–Stoltz [35], and Cao–Lu–Wang [6]. In a quantum direction, Fang–Lu–Tong [18] derive hypocoercive mixing-time estimates separating Hamiltonian and dissipative effects; our boundary-driven example differs in that $\mathcal{L}_O$ is already non-self-adjoint.

The lifting perspective—embedding slow dynamics into an enlarged space—originates with Chen–Lovász–Pak [10]; fundamental limits were identified by Apers–Ticozzi–Sarlette [3]. Extensions to continuous time are due to Eberle–Lörler [16, 17] and Brigati–Lörler–Wang [5]; the quantum analogue was developed by Li–Lu [29]; complementary flow-Poincaré methods appear in [15]. Our framework differs in being an operator-theoretic slow–fast perturbation theory centered on the self-adjoint backbone Q_O, tailored to NESS.

In the non-equilibrium setting, Dietert [12] develops L^2-hypocoercivity for kinetic equations with non-equilibrium forcing, and Monmarché [31] establishes geometric ergodicity for kinetic Langevin with NESS via a Lyapunov technique. We differ in providing an abstract framework with explicit η-dependence and a certified stability threshold. For boundary-driven quantum systems, see Carlen–Huse–Lebowitz [8] and the Zeno-limit analysis in [7].

Among hypocoercivity approaches, the variational method of [2] yields sharp regularity for kinetic Fokker–Planck; explicit matrix analyses apply in structured settings [1, 27]; and operator-theoretic precedents include [20, 4]. We adopt the DMS corrector framework for its natural extension to NESS. Our key novelty is the gap-adapted shift $\varpi_* = \lambda(Q_O)$, which upgrades the classical fixed-shift bound from $\mathcal{O}(\lambda^{5/2})$ to $\mathcal{O}(\sqrt{\lambda})$.

1.5. **Outline.** Section 2 gives a self-contained motivating example on the discrete cycle, previewing the gap-adapted shift mechanism. Section 3 sets up the abstract framework and adiabatic elimination. Section 4 formulates the structural assumptions and introduces the DMS corrector. Section 5 proves the main decay-rate bounds. Section 6 presents the convex-potential Langevin family, the reverse adiabatic elimination recipe, and a boundary-driven toy model. Section 7 discusses extensions and open problems.

2. Motivating example: a discrete lifted walk and the gap-adapted DMS corrector

This section gives a finite-dimensional warm-up that isolates the role of the resolvent shift in the DMS corrector. The model is a lifted random walk on the discrete cycle [10, 11]: the full Lyapunov argument reduces to an explicit 2×2 matrix estimate. With the fixed shift $\varpi = 1$, DMS gives a rate of order $\lambda^{3/2}$; the gap-adapted choice $\varpi = \lambda$ upgrades it to $\sqrt{\lambda}$. The genuinely nonequilibrium features—the perturbation $\eta\mathcal{V}_{\mathrm{pert}}$, stability threshold, and effective generator—enter in the abstract sections that follow.

2.1. **Setup.** Fix an odd integer $N \geq 3$. The state space is

$$E := \{(x, v) : x \in \{0, 1, \ldots, N-1\},\ v \in \{+1, -1\}\}.$$

The process moves one step in the current direction at rate 1 and flips the sign of the velocity at rate $\gamma > 0$. We work in the finite-dimensional Hilbert space

$$\mathcal{H} := L^2_0(E, \mu),$$

where μ is the uniform probability measure on E and the subscript 0 denotes the mean-zero subspace.

The slow and fast subspaces are

$$\mathcal{H}_S := \{f(x, v) : f(x, +1) = f(x, -1)\}, \tag{3}$$

$$\mathcal{H}_F := \{f(x, v) : f(x, +1) + f(x, -1) = 0\}, \tag{4}$$

with orthogonal projections P_S and P_F. Every $f \in \mathcal{H}$ decomposes uniquely as

$$f = f_S + f_F, \qquad f_S := P_S f, \quad f_F := P_F f,$$

and one has $\|f\|^2 = \|f_S\|^2 + \|f_F\|^2$.

The generator is written in the standard slow–fast form

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}, \tag{5}$$

where

$$\mathcal{R}f(x, v) := \tfrac{1}{2}\big(f(x, +1) + f(x, -1)\big) - f(x, v), \tag{6}$$

$$\mathcal{V}f(x, v) := \frac{v}{2}\big(f(x + 1, v) - f(x - 1, v)\big). \tag{7}$$

Here $x \pm 1$ is taken modulo N. Note that $\mathcal{V}$ is exactly the skew-adjoint part of the standard lifted walk jump generator.

The operator $\mathcal{R}$ is self-adjoint and non-positive, with

$$\mathcal{R}f_S = 0, \qquad \mathcal{R}f_F = -f_F.$$

Hence the fast spectral gap is exactly

$$-\langle \mathcal{R}f, f\rangle = \|f_F\|^2, \tag{8}$$

so $\lambda_R = 1$ and $M_R = 1$ in the notation of the abstract theory.

The operator $\mathcal{V}$ is skew-adjoint and centered:

$$P_S \mathcal{V} P_S = 0.$$

Indeed, if $f_S(x,v) = h(x)$ then

$$\mathcal{V} f_S(x,v) = \frac{v}{2}\big(h(x+1) - h(x-1)\big),$$

which is explicitly of the form $vg(x)$ and therefore belongs to $\mathcal{H}_F$.

Define the slow-to-fast flux operator

$$T := P_F \mathcal{V} P_S : \mathcal{H}_S \to \mathcal{H}_F.$$

Then $T^* = -P_S \mathcal{V} P_F$ and the slow diffusion is

$$Q := T^* T \quad \text{on } \mathcal{H}_S.$$

For $h = h(x) \in \mathcal{H}_S$, a direct computation gives

$$(Qh)(x) = -\frac{1}{4}\big(h(x+2) - 2h(x) + h(x-2)\big). \tag{9}$$

Thus Q is a discrete Laplacian $-\frac{1}{4}\Delta_2$ on the cycle. Because N is odd, the step-2 walk is irreducible. Its eigenvalues are

$$\sin^2(2\pi k/N), \qquad k = 1, \dots, N-1,$$

and because multiplication by 2 permutes the nonzero residues modulo N, the minimum positive eigenvalue is attained when $2k \equiv \pm 1 \pmod N$. Thus the slow spectral gap is

$$\lambda := \lambda(Q) = \sin^2(\pi/N) \sim \frac{\pi^2}{N^2}. \tag{10}$$

Remark 2.1 ($-Q$ as a special case of Q_O). In the abstract framework, the reversible backbone is

$$Q_O = -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S.$$

For the present discrete walk, $\mathcal{R} = -P_F$ on $\mathcal{H}_F$, hence $-\mathcal{R}^+ = \text{id}$ on $\mathcal{H}_F$. Therefore

$$Q_O = P_S \mathcal{V} P_F \mathcal{V} P_S = -T^* T = -Q.$$

So this warm-up (where $Q > 0$ acts as $-\Delta$) corresponds to the exact equilibrium instance of the abstract negative semi-definite slow dissipator Q_O used later in the paper.

2.2. **The classical corrector and its bottleneck.** The DMS corrector is

$$\mathcal{A}_\varpi := (\varpi + Q)^{-1} T^*, \qquad \varpi > 0.$$

If one follows the classical fixed-shift choice and sets $\varpi = 1$, then the slow coercive term is

$$\langle \mathcal{A}_1 \mathcal{V} f_S, f_S\rangle = \Big\langle \frac{Q}{1+Q} f_S, f_S \Big\rangle \geq \frac{\lambda}{1+\lambda}\|f_S\|^2. \tag{11}$$

Consequently, the coercive contribution in the Lyapunov derivative, $-\varepsilon\langle \mathcal{A}_1 \mathcal{V} f_S, f_S\rangle$, is of size $\varepsilon\lambda$ rather than ε. When $\lambda \ll 1$, this is exactly the bottleneck: the corrector restores only a very small amount of coercivity on the slow sector.

2.3. **The gap-adapted corrector.** The natural fix is to choose the shift equal to the slow spectral gap,

$$\varpi = \lambda.$$

Then the spectral multiplier is evaluated at its midpoint:

$$\langle \mathcal{A}_\lambda \mathcal{V} f_S, f_S\rangle = \Big\langle \frac{Q}{\lambda+Q} f_S, f_S\Big\rangle \geq \frac{\lambda}{\lambda+\lambda}\|f_S\|^2 = \frac{1}{2}\|f_S\|^2. \tag{12}$$

So the coercive contribution $-\varepsilon\langle \mathcal{A}_\lambda \mathcal{V} f_S, f_S\rangle$ becomes an order-one multiple of ε. This is the whole improvement.

The structural properties needed for the Lyapunov argument are especially simple in this model.

Lemma 2.2 (Corrector estimates for the discrete walk)**.** *For every $\varpi > 0$, the corrector $\mathcal{A}_\varpi = (\varpi + Q)^{-1}T^*$ satisfies:*

(i) $\|\mathcal{A}_\varpi \phi\| \leq \frac{1}{2\sqrt{\varpi}}\|\phi\|$ *;*
(ii) $\|\mathcal{V}\mathcal{A}_\varpi \phi\| \leq \|P_F\phi\| \leq \|\phi\|$ *;*
(iii) $\mathcal{A}_\varpi \mathcal{V} P_F = 0$ *;*
(iv) for every $\varepsilon > 0$,

$$\frac{1}{2}\Big(1 - \frac{\varepsilon}{2\sqrt{\varpi}}\Big)\|\phi\|^2 \leq \mathsf{E}_\varpi(\phi) := \frac{1}{2}\|\phi\|^2 - \varepsilon\,\Re\langle \mathcal{A}_\varpi\phi, \phi\rangle \leq \frac{1}{2}\Big(1 + \frac{\varepsilon}{2\sqrt{\varpi}}\Big)\|\phi\|^2.$$

In particular, if $\varepsilon \leq \sqrt{\varpi}/2$ then

$$\frac{1}{4}\|\phi\|^2 \leq \mathsf{E}_\varpi(\phi) \leq \frac{3}{4}\|\phi\|^2.$$

Proof. For (i), $\|\mathcal{A}_\varpi\|^2 = \sup_{q\geq 0} q/(\varpi+q)^2 = 1/(4\varpi)$. For (ii), write $T = UQ^{1/2}$ with U a partial isometry, so $\mathcal{V}\mathcal{A}_\varpi = U\frac{Q}{\varpi+Q}U^*$; since $q/(\varpi+q) \leq 1$, this gives $\|\mathcal{V}\mathcal{A}_\varpi\phi\| \leq \|P_F\phi\|$. For (iii), if $f_F = v\,g(x) \in \mathcal{H}_F$ then $\mathcal{V}f_F \in \mathcal{H}_S$, so $T^*(\mathcal{V}P_F\phi) = 0$. Part (iv) follows from (i) and Cauchy–Schwarz:

$$\big|\Re\langle \mathcal{A}_\varpi\phi, \phi\rangle\big| \leq \|\mathcal{A}_\varpi\|\,\|\phi\|^2 \leq \frac{1}{2\sqrt{\varpi}}\|\phi\|^2.$$

Substituting this into the definition of E_ϖ gives the displayed two-sided bound, and the special $\frac{1}{4}$–$\frac{3}{4}$ constants follow when $\varepsilon \leq \sqrt{\varpi}/2$. □

2.4. **Convergence rate.** Define the Lyapunov functional

$$\mathsf{E}_\varpi(f) := \frac{1}{2}\|f\|^2 - \varepsilon\,\Re\langle \mathcal{A}_\varpi f, f\rangle.$$

Differentiating along the flow $\partial_t f = \mathcal{L}f$ and using Lemma 2.2, one obtains

$$\frac{d}{dt}\mathsf{E}_\varpi(f) \leq -\mathbf{w}^T\mathbf{M}\,\mathbf{w}, \qquad \mathbf{w} = \begin{pmatrix}\|f_F\| \\ \|f_S\|\end{pmatrix}, \tag{13}$$

where the symmetric 2×2 dissipation matrix is

$$\mathbf{M} = \begin{pmatrix} \gamma - \varepsilon & -\dfrac{\varepsilon\gamma}{4\sqrt{\varpi}} \\ -\dfrac{\varepsilon\gamma}{4\sqrt{\varpi}} & \varepsilon\dfrac{\lambda}{\varpi+\lambda}\end{pmatrix}. \tag{14}$$

The derivation is straightforward: fast dissipation contributes $-\gamma\|f_F\|^2$, the slow coercive term is $-\varepsilon\langle Q(\varpi+Q)^{-1}f_S, f_S\rangle$, the transport remainder satisfies $|\Re\langle \mathcal{V}\mathcal{A}_\varpi f, f\rangle| \leq \|f_F\|^2$ (since $\mathcal{V}\mathcal{A}_\varpi f \in \mathcal{H}_F$), the term $\mathcal{A}_\varpi\mathcal{V}P_F$ vanishes by Lemma 2.2(iii), and $|\langle \mathcal{R}f_F, \mathcal{A}_\varpi^* f_S\rangle| \leq (2\sqrt{\varpi})^{-1}\|f_F\|\,\|f_S\|$.

We now choose the gap-adapted shift $\varpi = \lambda$. Then the slow block becomes $\varepsilon/2$ and the matrix simplifies to

$$\mathbf{M} = \begin{pmatrix} \gamma - \varepsilon & -\dfrac{\varepsilon\gamma}{4\sqrt{\lambda}} \\ -\dfrac{\varepsilon\gamma}{4\sqrt{\lambda}} & \dfrac{\varepsilon}{2} \end{pmatrix}. \tag{15}$$

Optimising ε to maximise the determinant gives the following explicit lower bound.

Proposition 2.3. *For the gap-adapted choice $\varpi = \lambda$, the DMS decay rate satisfies, for every $\gamma > 0$,*

$$\Lambda_{\mathrm{DMS},\lambda}(\gamma) \geq \frac{\gamma\lambda}{\gamma^2 + 8\lambda}. \tag{16}$$

The optimum over γ is attained at $\gamma_{\mathrm{opt}} = 2\sqrt{2\lambda}$, yielding

$$\Lambda_{\mathrm{DMS},\lambda}(\gamma_{\mathrm{opt}}) \geq \frac{\sqrt{\lambda}}{4\sqrt{2}}. \tag{17}$$

Since $\lambda \sim \pi^2/N^2$, this gives the kinetic scale $\Lambda_{\mathrm{DMS},\lambda}(\gamma_{\mathrm{opt}}) = \mathcal{O}(1/N)$.

Proof. With $\varpi = \lambda$, the matrix is

$$\mathbf{M} = \begin{pmatrix} \gamma - \varepsilon & -\dfrac{\varepsilon\gamma}{4\sqrt{\lambda}} \\ -\dfrac{\varepsilon\gamma}{4\sqrt{\lambda}} & \dfrac{\varepsilon}{2} \end{pmatrix}.$$

Its determinant is

$$\det \mathbf{M} = \frac{\varepsilon}{2}\Big[\gamma - \varepsilon\Big(1 + \frac{\gamma^2}{8\lambda}\Big)\Big].$$

Hence the maximizing weight is

$$\varepsilon_* = \frac{\gamma}{2\big(1 + \gamma^2/(8\lambda)\big)}.$$

In particular,

$$\frac{\varepsilon_*}{\sqrt{\lambda}} = \frac{4x}{x^2 + 8}, \qquad x := \frac{\gamma}{\sqrt{\lambda}},$$

and the right-hand side is maximized at $x = 2\sqrt{2}$, where it equals $1/\sqrt{2} < 1$. Therefore

$$2\varepsilon_* \|\mathcal{A}_\lambda\| \leq \frac{\varepsilon_*}{\sqrt{\lambda}} \leq 2^{-1/2} < 1,$$

so Lemma 2.2(iv) yields the uniform norm equivalence

$$\frac{1}{2}(1 - 2^{-1/2})\|f\|^2 \leq \mathsf{E}_\lambda(f) \leq \frac{1}{2}(1 + 2^{-1/2})\|f\|^2.$$

The special $\frac{1}{4}$–$\frac{3}{4}$ bounds from Lemma 2.2(iv) need not hold at the optimizing weight. Since $\mathrm{Tr}(\mathbf{M}) = \gamma - \varepsilon_*/2 \leq \gamma$, the determinant–trace estimate gives

$$\lambda_{\min}(\mathbf{M}) \geq \frac{\det(\mathbf{M})}{\mathrm{Tr}(\mathbf{M})} \geq \frac{\gamma\lambda}{\gamma^2 + 8\lambda}.$$

Setting $x = \gamma/\sqrt{\lambda}$, we obtain

$$\Lambda_{\mathrm{DMS},\lambda}(\gamma) \geq \sqrt{\lambda}\,\frac{x}{x^2 + 8}.$$

The function $x/(x^2 + 8)$ is maximized at $x = 2\sqrt{2}$, yielding $\gamma_{\mathrm{opt}} = 2\sqrt{2\lambda}$ and (17). □

For comparison, the classical fixed-shift choice $\varpi = 1$ gives

$$M_{22} = \varepsilon \frac{\lambda}{1+\lambda},$$

and the same optimisation yields a DMS lower bound of order $\lambda^{3/2}$. The loss comes entirely from the small spectral multiplier $\lambda/(1+\lambda)$.

2.5. **From the concrete to the abstract.** The four properties in Lemma 2.2 are the discrete analogues of the abstract ingredients used later. Property (i) is the general solver bound $\|\mathcal{A}_\varpi\| \lesssim \varpi^{-1/2}$. Property (ii) is the forward regularity estimate. Property (iii) is stronger than the abstract dual-regularity assumption: in the present model the operator $\mathcal{A}_\varpi \mathcal{V} P_F$ vanishes exactly, so the corresponding geometric constant is $K_{\text{geom}} = 0$. Property (iv) gives the usual norm equivalence for the modified Lyapunov functional.

Thus the discrete walk shows, with essentially no technical overhead, why the gap-adapted choice $\varpi_* = \lambda(Q_O)$ is the right one. It also explains why the warm-up fixed-shift exponent is $\lambda^{3/2}$ rather than the $\lambda^{5/2}$ that appears in the Langevin calculation: here one cross term disappears exactly, so the fixed-shift DMS estimate is less singular than in the PDE setting.

3. Preliminaries

We set up the abstract Hilbert-space framework and the adiabatic-elimination reduction used throughout.

3.1. **Hilbert-space setup.** Let $\mathcal{H}$ be a complex Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\|\cdot\|$. The generator $\mathcal{L}$ (possibly unbounded) generates a strongly continuous semigroup $(\mathcal{P}_t)_{t\geq 0}$ on $\mathcal{H}$, with adjoint $\mathcal{L}^\dagger$.

Concrete realizations. In the noncommutative (quantum) setting, $\mathcal{H} = L^2(\sigma)$ for a faithful state σ on a finite-dimensional space $\mathfrak{h}$, equipped with the KMS inner product $\langle X, Y\rangle_\sigma := \operatorname{tr}(\sigma^{1/2} X^\dagger \sigma^{1/2} Y)$. The generator is then a GKSL Lindbladian $\mathcal{L}(X) = i[H, X] + \sum_\alpha \mathcal{D}^\dagger[L_\alpha]X$. In the classical (PDE) setting, $\mathcal{H} = L^2(\mu)$ for a reference measure μ.

3.2. **Spectral gaps.** Let Π denote the projection onto the stationary direction and $\mathcal{H}_0 := \ker(\Pi)$ the mean-zero subspace. Write $\mathcal{S} := \frac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger) \leq 0$ for the symmetric part of the generator.

Definition 3.1 (Gap parameters)**.** On $\mathcal{H}_0$ we define the *coercive gap* $\lambda_S := \inf_{f\in\mathcal{H}_0\setminus\{0\}}(-\langle f, \mathcal{S}f\rangle/\|f\|^2)$ and the *spectral gap* $\lambda(\mathcal{L}) := \inf\{\Re\mu : \mu \in \operatorname{Spec}(-\mathcal{L}|_{\mathcal{H}_0})\}$.

When $\mathcal{L}$ is self-adjoint (detailed balance), $\lambda(\mathcal{L}) = \lambda_S$. In the non-reversible regime, it may happen that $\lambda_S = 0$ while $\lambda(\mathcal{L}) > 0$: the skew part $\frac{1}{2}(\mathcal{L} - \mathcal{L}^\dagger)$ couples degenerate directions to dissipative ones, producing exponential relaxation by hypocoercivity [38, 13]. The standard hypocoercive estimate asserts $\|\mathcal{P}_t f\| \leq Ce^{-\lambda t}\|f\|$ for $f \in \mathcal{H}_0$; when $C > 1$ is necessary, the relaxation is *strictly hypocoercive*.

Lemma 3.2 (Coercivity characterization)**.** *Assume* $\ker(\mathcal{L}) = \mathbb{C}\mathbf{1}$*. Then:*

(i) If $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$ *and* $-\mathcal{S}|_{\mathcal{H}_0}$ *has a positive spectral gap (automatic in finite dimensions), then the semigroup is coercive (*$C = 1$*). Conversely, coercivity implies* $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$*.*

(ii) If $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$*, any exponential decay estimate requires* $C > 1$*.*

3.3. **Slow–fast decomposition and adiabatic elimination.** Consider the generator $\mathcal{L}_\gamma = \gamma\mathcal{R} + \mathcal{V}$ with self-adjoint dissipator $\mathcal{R} \leq 0$ and interaction $\mathcal{V}$. Define the *slow* and *fast* subspaces

$$\mathcal{H}_S := \ker(\mathcal{R}), \qquad \mathcal{H}_F := \mathcal{H}_S^\perp,$$

with orthogonal projections P_S, $P_F := \mathrm{id} - P_S$. We assume $\mathcal{R}$ has spectral gap $\lambda_R > 0$ on $\mathcal{H}_F$. Any $f \in \mathcal{H}$ decomposes as $f = f_S + f_F$ with $f_S := P_S f$ (slow/macroscopic) and $f_F := P_F f$

(fast/microscopic). In the quantitative theory below, the stationary direction is generated by a vector $\mathbf{1} \in \mathcal{H}_S$, so Π is the orthogonal projection onto $\mathbb{C}\mathbf{1}$. Because $\mathcal{H}_F \perp \mathcal{H}_S$, this implies

$$\mathcal{H}_F \subset \mathcal{H}_0, \qquad \mathcal{H}_0 = (\mathcal{H}_S \cap \mathcal{H}_0) \oplus \mathcal{H}_F.$$

In particular, if $f \in \mathcal{H}_0$, then both $f_S := P_S f$ and $f_F := P_F f$ remain in $\mathcal{H}_0$.

Definition 3.3 (Centered coupling)**.** The interaction satisfies $P_S \mathcal{V} P_S = 0$.

This centering condition ensures that the effective slow dynamics is second-order in γ^{-1}—generated by slow–fast–slow excursions—rather than driven by an $\mathcal{O}(1)$ tangential drift [32, 7].

Definition 3.4 (Effective generator)**.** Let $\mathcal{R}^+$ denote the Moore–Penrose pseudoinverse of $\mathcal{R}$, bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \le \lambda_R^{-1}$. Under the centered coupling condition, we define

$$\mathcal{L}_O := -P_S \mathcal{V} \mathcal{R}^+ \mathcal{V} P_S. \tag{18}$$

In the context of lifting, $\mathcal{L}_O$ is the effective "target" dynamics one seeks to accelerate. On the macroscopic timescale $\tau = t/\gamma$, the full evolution converges to $e^{\tau \mathcal{L}_O}$ as $\gamma \to \infty$ (quantitative Zeno-limit error bounds are given in [7]). From the rate perspective, Theorem 5.7 shows that the decay is bounded by a diffusive γ^{-1} ceiling in this overdamped regime, motivating the finite-friction analysis of Section 5.

When the interaction is decomposed as $\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}$ (skew-adjoint part plus perturbation), we impose the stronger componentwise conditions $P_S \mathcal{V}_{\text{skew}} P_S = 0$ and $P_S \mathcal{V}_{\text{pert}} P_S = 0$. For a single fixed value of η, this is stronger than merely assuming $P_S \mathcal{V} P_S = 0$ for the full interaction; we adopt it because the quantitative estimates below are intended to hold uniformly over an admissible range of η, and because it isolates the reversible backbone Q_O from the η-dependent perturbation.

4. The non-equilibrium lifting framework

We now state the structural assumptions used for quantitative hypocoercivity at finite friction. The goal is to isolate the ingredients that transfer dissipation from fast modes to slow modes through the DMS corrector [13, 14, 22]: fast spectral gap, centering, flux regularity, and perturbation bounds. Standing convention on the stationary direction. Throughout Sections 4–6, we fix a unit vector $\mathbf{1} \in \mathcal{H}_S$ such that $\mathcal{L}\mathbf{1} = 0$, and denote by Π the orthogonal projection onto $\mathbb{C}\mathbf{1}$. We assume $\Pi e^{t\mathcal{L}} = e^{t\mathcal{L}} \Pi$ for all $t \ge 0$, so the mean-zero subspace $\mathcal{H}_0 := \ker(\Pi)$ is invariant under the semigroup. Because $\mathbf{1} \in \mathcal{H}_S$, both P_S and P_F preserve $\mathcal{H}_0$, and every $f \in \mathcal{H}_0$ decomposes as $f = f_S + f_F$ with $f_S \in \mathcal{H}_S \cap \mathcal{H}_0$ and $f_F \in \mathcal{H}_F$.

4.1. Structural assumptions.

Let $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$ be an orthogonal slow–fast decomposition. We study generators depending on friction $\gamma > 0$ and perturbation strength $\eta \ge 0$:

$$\mathcal{L} = \gamma \mathcal{R} + \mathcal{V}. \tag{19}$$

The transport term is decomposed as $\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}$, and P_S, P_F denote the orthogonal projections onto $\mathcal{H}_S, \mathcal{H}_F$.

Assumption 4.1 (Structural Conditions on the Non-Equilibrium Generator)**.** The system is defined on $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$. The generator is $\mathcal{L} = \gamma \mathcal{R} + \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}$. We assume the following structural conditions:

(1) **Microscopic Spectral Gap.** The fast dissipator $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$, with kernel $\ker(\mathcal{R}) = \mathcal{H}_S$. Moreover, it is strictly dissipative on $\mathcal{H}_F$: there exists $\lambda_R > 0$ such that

$$-\langle \mathcal{R} f, f \rangle \ge \lambda_R \|f\|^2, \qquad \forall f \in \mathcal{H}_F. \tag{20}$$

In particular, $(-\mathcal{R})^{1/2}$ is well-defined by functional calculus. This allows defining the Moore–Penrose pseudoinverse $\mathcal{R}^+$, which is a bounded self-adjoint operator on $\mathcal{H}_F$ with $-\mathcal{R}^+ \ge 0$ and $\|\mathcal{R}^+\| \le \lambda_R^{-1}$.

(2) **Centered Coupling.** We assume that the interaction does not generate an $\mathcal{O}(1)$ drift on the slow manifold in the macroscopic time scaling $\tau = t/\gamma$. Equivalently, the full interaction $\mathcal{V} := \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ satisfies

$$P_S \mathcal{V} P_S = 0. \tag{21}$$

Because the estimates below are designed to be uniform over an admissible interval of η, we impose the stronger componentwise conditions

$$P_S \mathcal{V}_{\text{skew}} P_S = 0, \qquad P_S \mathcal{V}_{\text{pert}} P_S = 0. \tag{22}$$

This is a modelling restriction rather than a formal consequence of $P_S \mathcal{V} P_S = 0$ for a single fixed η; it matches the centered coupling condition (Definition 3.3) used to define the effective generator $\mathcal{L}_O$ and is satisfied in our examples.

(3) **Fast-to-slow Regularity (Flux conditioning and flux regularity).** For $\psi \in \text{dom}(Q_O) \subset \mathcal{H}_S$ (where Q_O is the reversible backbone defined in (31) below), define its *microscopic flux* induced by transport as

$$g_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi \in \mathcal{H}_F. \tag{23}$$

In the bounded-operator (finite-dimensional) setting, Q_O is a bounded self-adjoint operator on $\mathcal{H}_S$ and thus $\text{dom}(Q_O) = \mathcal{H}_S$. In unbounded settings, the identities below are understood on the common analytic core from Assumption 4.3 and then extended by density to the relevant energy domains. We assume a **flux conditioning bound**: there exists a constant $M_R \in (0, \infty)$ such that

$$\|g_\psi\|^2 \le M_R \langle g_\psi, -\mathcal{R}^+ g_\psi \rangle, \qquad \forall \psi \in \text{dom}(Q_O) \subset \mathcal{H}_S. \tag{24}$$

Since (24) is monotone in M_R, if it holds for some constant then it also holds for any larger one; throughout, M_R may be taken to be any convenient upper bound (at the cost of more conservative constants in subsequent estimates). In the bounded-operator (finite-dimensional) setting, since $-\mathcal{R} \ge 0$ is bounded on $\mathcal{H}_F$, one may take $M_R := \|\mathcal{R}\|$ (equivalently $\|-\mathcal{R}\|$ on $\mathcal{H}_F$). In particular, for all $h \in \mathcal{H}_F$,

$$\langle (-\mathcal{R})h, h \rangle \le M_R \|h\|^2. \tag{25}$$

We also assume a **flux regularity bound**: applying the fast–fast component of the transport to the corrected flux is controlled by the induced slow operator. Concretely, there exists $\kappa_{\text{ell}} > 0$ such that for all $\psi \in \text{dom}(Q_O)$,

$$\|P_F \mathcal{V}_{\text{skew}} P_F \left(-\mathcal{R}^+\right) g_\psi\| \le \ \kappa_{\text{ell}} \|P_S \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) g_\psi\| \ = \ \kappa_{\text{ell}} \|(-Q_O)\psi\|. \tag{26}$$

(4) **Structural Decomposition of the Perturbation.** The perturbation operator $\mathcal{V}_{\text{pert}}$ admits a decomposition into a skew-symmetric part and a symmetric part, each controlled in the $(-\mathcal{R})^{1/2}$-energy norm $\|\cdot\|_{(1)}$:

$$\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}, \tag{27}$$

satisfying the following conditions:

(a) **Skew Perturbation Control:** The operator $\mathcal{K}_{\text{pert}}$ is skew-symmetric, i.e., for all $f \in \text{dom}(\mathcal{V}_{\text{pert}})$:

$$\Re\langle \mathcal{K}_{\text{pert}} f, f \rangle = 0. \tag{28}$$

Moreover, $\mathcal{K}_{\text{pert}}$ is relatively bounded with respect to $(-\mathcal{R})^{1/2}$: there exist constants $a_1, b_1 \ge 0$ such that for all $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{K}_{\text{pert}} f\| \le a_1 \|(-\mathcal{R})^{1/2} f\| + b_1 \|f\|. \tag{29}$$

(b) **Symmetric Perturbation Control:** The operator $\mathcal{S}_{\text{pert}}$ is symmetric and defined at least on the energy domain $\text{dom}((-\mathcal{R})^{1/2})$. There exist constants $a_2, b_2 \geq 0$ such that for all $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{S}_{\text{pert}} f\| \leq a_2 \|(-\mathcal{R})^{1/2} f\| + b_2 \|f\|. \tag{30}$$

Equivalently, $\mathcal{S}_{\text{pert}}$ is bounded as a map from $(\text{dom}((-\mathcal{R})^{1/2}), \|\cdot\|_{(1)})$ to $\mathcal{H}$. (In PDE models, $\mathcal{S}_{\text{pert}}$ is often an unbounded multiplier on plain L^2; (30) is the standard replacement allowing it to be controlled by the fast dissipation.)

The pair (λ_R, M_R) measures fast dissipation and flux conditioning, κ_{ell} controls the geometric transport remainder, and (a_i, b_i) quantify the perturbation in the energy norm; Section 5 combines them into the rate $\Lambda_{\text{hypo}}(\gamma)$ and the perturbative loss $B_{\text{pert}}(\gamma)$.

Under Assumption 4.1, we recall the effective generator from Definition 3.4, $\mathcal{L}_O = -P_S \mathcal{V} \mathcal{R}^+ \mathcal{V} P_S$. To construct the Lyapunov functional, we isolate the reversible contribution to the effective diffusion via the **reversible backbone** Q_O:

$$Q_O := -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S. \tag{31}$$

From the skew-adjointness of $\mathcal{V}_{\text{skew}}$ and the negativity of $\mathcal{R}^+$, it follows that for any $f \in \mathcal{H}_S$,

$$\langle f, Q_O f\rangle = -\langle f, P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S f\rangle = -\langle \mathcal{V}_{\text{skew}}^\dagger f, \mathcal{R}^+ \mathcal{V}_{\text{skew}} f\rangle = \langle \mathcal{V}_{\text{skew}} f, \mathcal{R}^+ \mathcal{V}_{\text{skew}} f\rangle \leq 0. \tag{32}$$

Thus, Q_O is a self-adjoint, dissipative operator on $\mathcal{H}_S$.

In infinite-dimensional kinetic models, the composition defining Q_O requires a common core on which $\mathcal{V}_{\text{skew}}$ and $\mathcal{R}^+$ are compatible; Assumption 4.3 isolates this bookkeeping.

Assumption 4.2 (Kernel-Compatible Spectral Gap of Effective Dynamics)**.** We assume that the stationary direction is the whole kernel of the reversible backbone on the slow space:

$$\ker(Q_O) = \mathbb{C}\mathbf{1} \subset \mathcal{H}_S.$$

Equivalently, on the mean-zero slow subspace $\mathcal{H}_{S,0} := \mathcal{H}_S \cap \mathcal{H}_0$, the operator Q_O admits a **spectral gap** $\lambda(Q_O) > 0$, i.e.

$$-\langle f, Q_O f\rangle \geq \lambda(Q_O)\|f\|^2, \qquad \forall f \in \mathcal{H}_{S,0}. \tag{33}$$

Consequently, for any shift parameter $\varpi > 0$, the operator $(\varpi - Q_O)$ is strictly positive definite on $\mathcal{H}_S$ and invertible, with operator norm $\|(\varpi - Q_O)^{-1}\| \leq \varpi^{-1}$.

In unbounded settings, we additionally assume (as part of the model-dependent analytic hypotheses) that $(\varpi - Q_O)^{-1}$ maps $\mathcal{H}_S$ into $\text{dom}(Q_O)$.

Assumption 4.3 (Analytic Core in Unbounded Settings)**.** This assumption is vacuous in finite dimensions. In unbounded settings, we fix a dense subspace

$$\mathcal{C} \subset \text{dom}(\mathcal{L}) \cap \text{dom}((-\mathcal{R})^{1/2})$$

such that:

(1) $\mathcal{C}$ is invariant under P_S, P_F, $\mathcal{R}^+$, $\mathcal{V}_{\text{skew}}$, and $\mathcal{V}_{\text{pert}}$, and $\mathcal{C} \cap \mathcal{H}_S$ is a core for Q_O;
(2) for every $\varpi > 0$, the resolvent $(\varpi - Q_O)^{-1}$ maps $\mathcal{H}_S$ into $\text{dom}(Q_O)$ and preserves $\mathcal{C} \cap \mathcal{H}_S$;
(3) all operator compositions appearing in the definitions of Q_O, $\mathcal{A}_\varpi$, and in the Lyapunov and singular-value computations of Appendices B–D are well defined on $\mathcal{C}$, and the corresponding bilinear forms extend continuously from $\mathcal{C}$ to the energy spaces on which the theorems are stated.

Remark 4.4 (Why kernel compatibility is needed). The quantitative estimate in Theorem 5.1 is an L^2 decay statement on the whole mean-zero space $\mathcal{H}_0$. For that reason, the reversible backbone must have no neutral directions there: if $\ker(Q_O) \cap \mathcal{H}_0 \neq \{0\}$, then the DMS coercivity term controls only the quotient by those extra zero modes, and one cannot conclude exponential decay on all of

$\mathcal{H}_0$ without adding further structure. Section 6.3 provides a finite-dimensional illustration of this obstruction.

Remark 4.5 (Perturbative comparison: $\mathcal{L}_O$ vs. Q_O). By the expansion (70), $\mathcal{L}_O - Q_O = \mathcal{O}(\eta)$. Standard perturbation theory [24, Chapter V] gives $(1-\theta)\lambda(Q_O) \le \lambda(\mathcal{L}_O) \le (1+\theta)\lambda(Q_O)$ whenever $\|\mathcal{L}_O - Q_O\| \le \theta\,\lambda(Q_O)$ with $0 < \theta < 1$. In particular, in this perturbative regime the scaling $\lambda_\eta \asymp \sqrt{\lambda(Q_O)}$ is equivalent to $\lambda_\eta \asymp \sqrt{\lambda(\mathcal{L}_O)}$. Possible Liouvillian exceptional points belong to the non-self-adjoint generator $\mathcal{L}_O$, not to the self-adjoint backbone Q_O; compare [34] for a recent two-qubit study of such non-self-adjoint phenomena.

4.2. **The DMS corrector.** To close the hypocoercive estimate, we introduce a bounded corrector that solves a shifted slow-space Poisson problem. Following the Dolbeault–Mouhot–Schmeiser (DMS) approach [13, 14], extended to L^2 settings by Iacobucci–Olla–Stoltz [22] and Cao–Lu–Wang [6], this operator captures the response of the slow component to fast transport excursions and provides the cross term that restores coercivity at the Lyapunov level. This is the abstract counterpart of the concrete corrector from Section 2: in the discrete warm-up the gap-adapted choice is $\varpi = \lambda(Q)$, and in the abstract theory it becomes $\varpi_* = \lambda(Q_O)$.

Definition 4.6 (DMS Corrector)**.** Let $\varpi > 0$ be a shift parameter, to be optimized in Section 5. We define the **corrector** $\mathcal{A}_\varpi : \mathcal{H} \to \mathcal{H}_S$ by

$$\mathcal{A}_\varpi := (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+) P_F. \tag{34}$$

The key obstruction to a direct energy estimate is that dissipation is degenerate on $\mathcal{H}_S$: for any $\phi \in \mathcal{H}_S$ with $\phi \neq 0$, one has $\Re\langle \mathcal{L}\phi, \phi\rangle = 0$, so the plain energy $\frac{1}{2}\|f\|^2$ cannot produce a spectral gap. The remedy is to augment it with a cross term involving the corrector $\mathcal{A}_\varpi$; differentiating $\Re\langle \mathcal{A}_\varpi f, f\rangle$ along the flow generates a slow coercive contribution tied to $-Q_O$, together with controlled remainders from the fast and perturbative components.

The following lemma establishes the critical boundedness properties of $\mathcal{A}_\varpi$, including the control over non-equilibrium perturbations. The reader may compare with the concrete discrete analogues in Lemma 2.2.

Lemma 4.7 (Structural Estimates for the DMS Corrector)**.** *Under Assumption 4.1, together with Assumption 4.3 in the unbounded setting, the operator $\mathcal{A}_\varpi$ satisfies the following structural estimates, with constants depending solely on the structural parameters $(M_R, \lambda_R, \kappa_{\text{ell}}, a_1, b_1, a_2, b_2)$ and* not *on ϖ:*

(1) ***Corrector Boundedness:***

$$\|\mathcal{A}_\varpi\| \le k_{\mathcal{A}} \varpi^{-1/2}, \quad \textit{with } k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}. \tag{35}$$

(2) ***Regularity Estimates (Forward and Dual):*** *The corrector acts as a bounded regularizer for the skew-symmetric transport in both directions:*

$$\|\mathcal{V}_{\text{skew}} \mathcal{A}_\varpi\| \le k_{\text{reg}}, \quad \textit{with } k_{\text{reg}} := \sqrt{M_R/\lambda_R}, \tag{36a}$$

$$\|\mathcal{A}_\varpi \mathcal{V}_{\text{skew}} P_F\| \le k_{\text{dual}}, \quad \textit{with } k_{\text{dual}} := \kappa_{\text{ell}}. \tag{36b}$$

The dual estimate (36b) *is a direct consequence of the Macroscopic Flux Regularity (Assumption 4.1.3).*

(3) ***Perturbation Control:*** *The interaction with the non-equilibrium perturbation is controlled in both directions:*

$$\|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\varpi\| \le k_P(\varpi), \tag{37a}$$

$$\|\mathcal{A}_\varpi \mathcal{V}_{\text{pert}}\| \le \tilde{k}_P(\varpi). \tag{37b}$$

with control constants

$$k_P(\varpi) := (b_1 + b_2)k_{\mathcal{A}}\,\varpi^{-1/2}, \tag{38}$$

$$\tilde{k}_P(\varpi) := \Big((b_1 + b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\varpi^{-1/2}. \tag{39}$$

Moreover, the centered perturbation coupling in Assumption 4.1.2 implies

$$P_S\mathcal{V}_{\text{pert}}P_S = 0 \qquad \Longrightarrow \qquad P_S\mathcal{V}_{\text{pert}}^{\dagger}P_S = 0, \tag{40}$$

and therefore

$$\mathcal{V}_{\text{pert}}\,\mathcal{H}_S \subset \mathcal{H}_F, \qquad \mathcal{V}_{\text{pert}}^{\dagger}\,\mathcal{H}_S \subset \mathcal{H}_F. \tag{41}$$

Proof. See Appendix B. □

4.3. **The modified Lyapunov functional.** We define a modified squared norm that explicitly incorporates the correlation between slow and fast degrees of freedom mediated by the corrector $\mathcal{A}_\varpi$.

Definition 4.8 (Modified Lyapunov functional [13, 14])**.** For any $f \in \mathcal{H}$ and a parameter $\alpha \in (0, 1)$, we define the functional:

$$\mathcal{E}_{\alpha,\varpi}(f) := \frac{1}{2}\|f\|^2 + \alpha\Re\langle\mathcal{A}_\varpi f, f\rangle. \tag{42}$$

This functional defines a norm equivalent to the underlying Hilbert norm, provided the weight α is sufficiently small relative to the bounded solver $\mathcal{A}_\varpi$.

Proposition 4.9 (Norm equivalence)**.** *If the parameter α satisfies $\alpha\|\mathcal{A}_\varpi\| \le 1/2$, then for all $f \in \mathcal{H}$:*

$$\frac{1}{2}(1 - 2\alpha\|\mathcal{A}_\varpi\|)\|f\|^2 \le \mathcal{E}_{\alpha,\varpi}(f) \le \frac{1}{2}(1 + 2\alpha\|\mathcal{A}_\varpi\|)\|f\|^2. \tag{43}$$

Proof. The proof follows directly from the Cauchy–Schwarz inequality $|\Re\langle\mathcal{A}_\varpi f, f\rangle| \le \|\mathcal{A}_\varpi\|\|f\|^2$ and the triangle inequality. □

5. Convergence bounds for the lifted semigroup

We now derive quantitative decay bounds for the semigroup on the mean-zero subspace. The strategy has two parts: a constructive lower bound from a 2×2 dissipation matrix, and a complementary overdamped upper ceiling from singular values. Together they identify the finite-friction acceleration window.

5.1. **Lower bound via modified hypocoercivity.** We derive an explicit lower bound by optimizing the Lyapunov parameters (α, ϖ); the full matrix estimates are in Appendix C. We package the structural input from Assumption 4.1 into derived quantities: a coupling scale $Z(\gamma)$, an optimal Lyapunov weight $\alpha_*(\gamma)$, an unperturbed decay profile $\Lambda_{\text{hypo}}(\gamma)$, and a perturbative loss rate $B_{\text{pert}}(\gamma)$.

Notation for the rate functions. Define the structural scalars

$$\Theta := \frac{\sqrt{M_R}}{2}, \qquad K_{\text{geom}} := \kappa_{\text{ell}}, \qquad b_{2,\text{eff}} := b_2 + \frac{a_2}{2}(1 + \lambda_R), \tag{44}$$

where κ_{ell} is the flux regularity constant from (26); here Θ measures slow–fast flux amplitude and K_{geom} collects the cross-term transport geometry.

For each friction $\gamma > 0$, define the *coupling scale*

$$Z(\gamma) := \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}, \tag{45}$$

which measures effective slow–fast interaction strength as a function of γ, and set:

$$\alpha_*(\gamma) := \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\mathrm{reg}}}, \tag{46}$$

$$\Lambda_{\mathrm{hypo}}(\gamma) := \frac{\frac{1}{4}\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\mathrm{reg}}}, \tag{47}$$

$$B_{\mathrm{pert}}(\gamma) := b_{2,\mathrm{eff}} + \alpha_*(\gamma)\, C_{\mathrm{sing}}\, \lambda(Q_O)^{-1/2}, \tag{48}$$

where $C_{\mathrm{sing}} := 2(b_1 + b_2)\, k_{\mathcal{A}} + (a_1 + a_2)$ is the perturbation interaction constant (see Appendix C for the derivation), with $k_{\mathcal{A}}$ the resolvent norm bound from Lemma 4.7. Here $\alpha_*(\gamma)$ is the optimal Lyapunov mixing weight, $\Lambda_{\mathrm{hypo}}(\gamma)$ is the lower-rate profile, and $B_{\mathrm{pert}}(\gamma)$ quantifies perturbation destabilization. Although B_{pert} contains an apparent $\lambda(Q_O)^{-1/2}$ factor, it is cancelled at optimal friction because $\alpha_*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}$ (Corollary 5.2).

We also isolate the explicit admissibility restriction coming from the symmetric perturbation in the basic energy estimate:

$$\eta_{\mathrm{sym}}(\gamma) := \begin{cases} +\infty, & a_2 = 0, \\ 2\gamma/a_2, & a_2 > 0. \end{cases} \tag{49}$$

Equivalently, $\eta \le \eta_{\mathrm{sym}}(\gamma)$ is exactly the condition ensuring that the coefficient of $\|(-\mathcal{R})^{1/2} f\|^2$ remains nonnegative after the symmetric perturbation is absorbed into the dissipative part; see Appendix C.

With the gap-adapted choice $\varpi_* := \lambda(Q_O)$, define the associated Lyapunov-distortion parameter and norm-equivalence prefactor by

$$\delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|, \tag{50}$$

$$C_*(\gamma) := \sqrt{\frac{1 + \delta_*(\gamma)}{1 - \delta_*(\gamma)}}. \tag{51}$$

Main result. The structural properties established in Appendix C yield an explicit norm equivalence with distortion parameter $\delta_*(\gamma) < 1$ for all $\gamma > 0$. In fact the appendix proves the uniform bound $\delta_*(\gamma) \le 2^{-1/2}$; equivalently,

$$\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\| = \frac{\delta_*(\gamma)}{2} \le 2^{-3/2} < \frac{1}{2},$$

so Proposition 4.9 applies to the optimized Lyapunov functional, and hence $C_*(\gamma) \le 1 + \sqrt{2}$.

Theorem 5.1 (Explicit quantitative hypocoercivity)**.** *Suppose Assumptions 4.1 and 4.2 hold, and in the unbounded setting also Assumption 4.3. Set the explicit certified threshold*

$$\eta^*(\gamma) := \min\left\{\eta_{\mathrm{sym}}(\gamma),\ \frac{\Lambda_{\mathrm{hypo}}(\gamma)}{2B_{\mathrm{pert}}(\gamma)}\right\}.$$

For any $\gamma > 0$, if $\eta \le \eta^*(\gamma)$, then $e^{t\mathcal{L}}$ is exponentially stable on the invariant mean-zero subspace $\mathcal{H}_0 = \ker(\Pi)$:

$$\|e^{t\mathcal{L}} f\| \le C_*(\gamma)\, e^{-\lambda_\eta t}\|f\| \le (1 + \sqrt{2})\, e^{-\lambda_\eta t}\|f\|, \quad \forall t \ge 0,\ \forall f \in \mathcal{H}_0. \tag{52}$$

The global decay rate λ_η satisfies the strict lower bound:

$$\lambda_\eta \ge \frac{\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma)}{1 + \delta_*(\gamma)} \ge (2 - \sqrt{2})\left(\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma)\right). \tag{53}$$

Sketch of Proof. The full computation is in Appendix C. With the resolvent shift fixed at $\varpi_* = \lambda(Q_O)$, the Lyapunov derivative satisfies $\frac{d}{dt}\mathcal{E}_{\alpha,\varpi_*} \le -\mathbf{v}^T\mathbf{M}(\alpha)\mathbf{v}$ for the vector $\mathbf{v} = (\|P_F f\|, \|P_S f\|)^T$

and a 2×2 positive-definite dissipation matrix $\mathbf{M}(\alpha)$. Maximising over α yields the optimal weight $\alpha_*(\gamma)$ from (46) and the rate $\Lambda_{\rm hypo}(\gamma)$. The distortion parameter $\delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|$ satisfies

$$\delta_*(\gamma) \le \frac{\sqrt{\lambda_R}}{\sqrt{M_R}\big(K_{\rm geom} + \sqrt{K_{\rm geom}^2 + 2k_{\rm reg}}\big)} \le 2^{-1/2} \qquad \forall\, \gamma > 0, \tag{54}$$

by the solver bound $\|\mathcal{A}_{\varpi_*}\| \le (2\sqrt{\lambda_R})^{-1}\lambda(Q_O)^{-1/2}$, the formula for α_* in (46), and the elementary extremum $\sup_{x>0} x\big[(x+K_{\rm geom})^2 + 2k_{\rm reg}\big]^{-1} = 1/\big(2(K_{\rm geom} + \sqrt{K_{\rm geom}^2 + 2k_{\rm reg}})\big)$. This gives the explicit norm equivalence encoded by $C_*(\gamma)$, together with the uniform simplification $C_*(\gamma) \le 1+\sqrt{2}$ and the lower bound in (53). □

Corollary 5.2 (Kinetic acceleration and singularity cancellation)**.** *Assume the setting of Theorem 5.1. Assume furthermore that the structural constants Θ, $K_{\rm geom}$, $k_{\rm reg}$, λ_R, $b_{2,\rm eff}$, and $C_{\rm sing}$ are independent of $\lambda(Q_O)$ (in particular, they remain $\mathcal{O}(1)$ as $\lambda(Q_O)\downarrow 0$), and that $b_{2,\rm eff} > 0$.*

Then the unperturbed rate function $\Lambda_{\rm hypo}(\gamma)$ defined in (47) attains its global maximum at

$$\gamma_{\rm opt} = \frac{\sqrt{\lambda(Q_O)}}{\Theta}\sqrt{K_{\rm geom}^2 + 2k_{\rm reg}} \asymp \sqrt{\lambda(Q_O)}. \tag{55}$$

At this friction scale, the maximal lower bound exhibits kinetic scaling:

$$\max_{\gamma>0} \Lambda_{\rm hypo}(\gamma) \asymp \sqrt{\lambda(Q_O)}. \tag{56}$$

Moreover, the mixing weight in (48) satisfies

$$\alpha_*(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}, \qquad \alpha_*(\gamma_{\rm opt})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1), \tag{57}$$

so the solver singularity $\lambda(Q_O)^{-1/2}$ in $B_{\rm pert}(\gamma)$ is cancelled at the optimal friction. Consequently, $B_{\rm pert}(\gamma_{\rm opt}) = \mathcal{O}(1)$ and the certified threshold satisfies

$$\eta^*(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}. \tag{58}$$

Proof. Write

$$\Lambda_{\rm hypo}(\gamma) = \frac{\lambda_R}{4}\,\frac{\gamma}{\big(\gamma\Theta\lambda(Q_O)^{-1/2} + K_{\rm geom}\big)^2 + 2k_{\rm reg}}. \tag{59}$$

Set $x := \gamma\Theta\lambda(Q_O)^{-1/2}$ and $C_0 := 2k_{\rm reg}$. Then $\Lambda_{\rm hypo}(\gamma) = \frac{\lambda_R}{4\Theta}\sqrt{\lambda(Q_O)}\,\frac{x}{(x+K_{\rm geom})^2+C_0}$. A direct derivative computation shows that the function $x \mapsto \frac{x}{(x+K_{\rm geom})^2+C_0}$ is maximized at $x = \sqrt{K_{\rm geom}^2 + C_0}$, giving the stated $\gamma_{\rm opt}$ and the scaling $\max_\gamma \Lambda_{\rm hypo}(\gamma) \asymp \sqrt{\lambda(Q_O)}$.

For the perturbation term, by (48), $B_{\rm pert}(\gamma) = b_{2,\rm eff} + \alpha_*(\gamma)\,C_{\rm sing}\,\lambda(Q_O)^{-1/2}$. At $\gamma = \gamma_{\rm opt}$ we have $x = \mathcal{O}(1)$ and $Z(\gamma_{\rm opt}) = x + K_{\rm geom} = \mathcal{O}(1)$, hence $\alpha_*(\gamma_{\rm opt}) = \frac{\gamma_{\rm opt}\lambda_R}{Z(\gamma_{\rm opt})^2+2k_{\rm reg}} = \mathcal{O}(\sqrt{\lambda(Q_O)})$, which implies $\alpha_*(\gamma_{\rm opt})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1)$ and therefore $B_{\rm pert}(\gamma_{\rm opt}) = \mathcal{O}(1)$. Finally, the matrix-controlled threshold satisfies $\Lambda_{\rm hypo}(\gamma_{\rm opt})/(2B_{\rm pert}(\gamma_{\rm opt})) \asymp \sqrt{\lambda(Q_O)}$. If $a_2 = 0$, then $\eta_{\rm sym}(\gamma_{\rm opt}) = +\infty$ and therefore $\eta^*(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}$. If $a_2 > 0$, then $\eta_{\rm sym}(\gamma_{\rm opt}) = 2\gamma_{\rm opt}/a_2 \asymp \sqrt{\lambda(Q_O)}$. In either case, $\eta^*(\gamma_{\rm opt}) = \min\{\eta_{\rm sym}(\gamma_{\rm opt}), \Lambda_{\rm hypo}(\gamma_{\rm opt})/(2B_{\rm pert}(\gamma_{\rm opt}))\} \asymp \sqrt{\lambda(Q_O)}$. □

The certified threshold $\eta^*(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}$ admits a transparent interpretation: the dimensionless ratio $\eta/\sqrt{\lambda(Q_O)}$ compares forcing to the kinetic relaxation scale, and the condition $\eta \le \eta^*$ requires this ratio to be $\mathcal{O}(1)$. This perturbative character is shared by prior approaches: Iacobucci–Olla–Stoltz [22] and Roussel–Stoltz [35] require the forcing to be small relative to the spectral gap, and Eberle–Lörler [16, 17] impose analogous contractivity conditions. The advantage of our threshold is that it is *explicit* and *computable once those structural constants are available.* In the explicit examples below, the structural constants can be computed directly from the model. In the

finite-dimensional boundary-driven toy model, the numerical data track the 1/2 law very closely and show that acceleration persists well beyond the certified perturbative threshold.

Remark 5.3 (Exact recovery of the warm-up constants). For the discrete walk of Section 2 ($\eta = 0$), one has $\lambda_R = M_R = 1$, $\Theta = \frac{1}{2}$, $k_{\mathrm{reg}} = 1$, $K_{\mathrm{geom}} = 0$, and $\varpi_* = \lambda$. The abstract formula (47) then gives $\Lambda_{\mathrm{hypo}}(\gamma) = \gamma\lambda/(\gamma^2+8\lambda)$, exactly matching Proposition 2.3, with optimal friction $\gamma_{\mathrm{opt}} = 2\sqrt{2\lambda}$ and optimal rate $\sqrt{\lambda}/(4\sqrt{2})$.

Remark 5.4 (Why the gap-adapted shift matters). The $\sqrt{\lambda(Q_O)}$ scaling relies crucially on $\varpi_* = \lambda(Q_O)$. For a fixed shift $\varpi > 0$ independent of $\lambda := \lambda(Q_O)$, the spectral multiplier $\lambda/(\varpi + \lambda) \asymp \lambda$ as $\lambda \downarrow 0$. Writing the unperturbed dissipation matrix from Appendix C with this scaling, the slow coercive entry is $M^0_{22} = \alpha\lambda/(\varpi + \lambda)$, and the determinant–trace estimate gives

$$\Lambda_{\mathrm{hypo}} \asymp \frac{\gamma\lambda^2}{\gamma + \alpha\lambda} \asymp \lambda^{5/2} \qquad \text{when } \gamma \asymp \sqrt{\lambda}, \tag{60}$$

a loss of two powers of $\sqrt{\lambda}$ compared to the optimal rate. With the gap-adapted choice $\varpi_* = \lambda$, the multiplier becomes $\lambda/(\lambda + \lambda) = \frac{1}{2}$, so the slow block is of order α rather than $\alpha\lambda$; this restores $\Lambda_{\mathrm{hypo}} \asymp \sqrt{\lambda}$. The discrete warm-up of Section 2 illustrates the same mechanism with $K_{\mathrm{geom}} = 0$, where the fixed-shift loss is the milder $\lambda^{3/2}$.

Define the (rigorous) L^2 relaxation time as in [29] by

$$t_{\mathrm{rel}}(\mathcal{L}) := \inf\Big\{t \geq 0\,;\ \|e^{t\mathcal{L}} f\| \leq e^{-1}\,\|f\| \text{ for all } f \in \mathcal{H}_0\Big\}. \tag{61}$$

The following holds:

Corollary 5.5 (Acceleration in the L^2 relaxation time)**.** *Assume the setting of Theorem 5.1. Then for $\eta \leq \eta^*(\gamma)$, the exponential decay estimate from Theorem 5.1 implies*

$$t_{\mathrm{rel}}(\mathcal{L}) \leq \frac{1 + \log C_*(\gamma)}{\lambda_\eta} \leq \frac{1 + \log(1 + \sqrt{2})}{\lambda_\eta}. \tag{62}$$

In particular, at the optimal friction $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$ (Corollary 5.2) and for perturbations within the certified threshold, one has

$$t_{\mathrm{rel}}(\mathcal{L}) = \mathcal{O}\big(\lambda(Q_O)^{-1/2}\big). \tag{63}$$

Proof. By Theorem 5.1, for all $f \in \mathcal{H}_0$ and $t \geq 0$ one has $\|e^{t\mathcal{L}} f\| \leq C_*(\gamma)\, e^{-\lambda_\eta t}\, \|f\|$. Choose $t = (1 + \log C_*(\gamma))/\lambda_\eta$ so that $C_*(\gamma)\, e^{-\lambda_\eta t} = e^{-1}$. □

5.2. **Overdamped diffusive ceiling.** To establish that the high-friction γ^{-1} behavior is intrinsic and not an artifact of the lower-bound construction, we derive a universal upper bound on any admissible L^2 exponential rate in terms of the singular-value gap.

Define the *singular-value gap*

$$s(\mathcal{L}) := \inf_{f \in \mathcal{H}_0 \cap \mathrm{dom}(\mathcal{L}) \setminus \{0\}} \frac{\|\mathcal{L} f\|}{\|f\|}, \tag{64}$$

that is, the smallest singular value of $\mathcal{L}|_{\mathcal{H}_0}$ in finite dimensions, remaining meaningful as an infimum over the operator domain in unbounded settings. Quantities of this type are continuous-time analogues of *multiplicative reversibilizations* and the *pseudo spectral gap* used in non-reversible Markov chain theory; see, e.g., [33]. In the discrete-time finite-state setting, a closely related point of view is developed by Chatterjee, who defines a nonreversible "spectral gap" via singular values of the generator $I - P$ [9].

Assumption 5.6 (Overdamped ceiling hypotheses)**.** In addition to Assumptions 4.1, 4.2 (and 4.3 in the unbounded setting), we assume:

(i) $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$;
(ii) $\lambda(Q_O)$ is attained by a normalized eigenfunction $\phi \in \mathrm{dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0$ (automatic in finite dimensions and many compact-resolvent settings);
(iii) the first-order corrector $\psi := \mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi$ belongs to $\mathrm{dom}(\mathcal{L}) \cap \mathrm{dom}((-\mathcal{R})^{1/2})$ with $P_F\mathcal{V}_{\mathrm{skew}}\psi$ well defined;
(iv) $e^{t\mathcal{L}}$ is a contraction semigroup on $\mathcal{H}_0$.

Theorem 5.7 (Overdamped diffusive ceiling via singular value gap)**.** *Assume Assumption 5.6. Let $C \geq 1$ and $\nu > 0$ be such that*

$$\|e^{t\mathcal{L}} f\| \leq C\, e^{-\nu t}\, \|f\|, \qquad \forall t \geq 0,\ \forall f \in \mathcal{H}_0. \tag{65}$$

Then for every $\gamma > 0$,

$$\nu \leq (1 + \log C)\, s(\mathcal{L}) \leq (1 + \log C) \left[\frac{(1 + \kappa_{\mathrm{ell}})\, \lambda(Q_O)}{\gamma} + \eta\,(b_1 + b_2) + \eta\,(a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma} \right], \tag{66}$$

where κ_{ell} is the flux-regularity constant from Assumption 4.1.3 and a_1, a_2, b_1, b_2 are from Assumption 4.1.4. In particular, for $\eta \leq c/\gamma$ one obtains the diffusive overdamped scaling $\nu = \mathcal{O}(\gamma^{-1})$ as $\gamma \to \infty$.

Sketch of Proof. The proof is given in Appendix D. The key point is that one can test the singular value gap on a *first-order corrector* $f_\gamma := \phi - \gamma^{-1}\psi$ built from a slow eigenfunction $Q_O\phi = -\lambda(Q_O)\phi$ and $\psi := \mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi$. The additional domain hypothesis ensures that $f_\gamma \in \mathrm{dom}(\mathcal{L})$ and that all terms below are well defined. The flux regularity bound (26) controls the only nontrivial fast remainder term $P_F\mathcal{V}_{\mathrm{skew}}\psi$ without ever invoking $\|\mathcal{V}_{\mathrm{skew}}\|$. The perturbation contribution is controlled by the $(-\mathcal{R})^{1/2}$-energy bounds in Assumption 4.1.4 and the explicit identity $\|(-\mathcal{R})^{1/2}\psi\|^2 = \lambda(Q_O)$. The first inequality $\nu \leq (1+\log C)\, s(\mathcal{L})$ then follows from Lemma D.1, which uses the contraction bound $\|e^{t\mathcal{L}} f\| \leq \|f\|$ on $[0, (\log C)/\nu]$ and the exponential decay on $[(\log C)/\nu, \infty)$. □

Theorem 5.7 shows that in the overdamped regime the best achievable L^2 exponential decay rate (with any finite prefactor C) is bounded above by $(1 + \log C)$ times the macroscopic diffusion scale $\lambda(Q_O)/\gamma$, up to explicit perturbative contributions. Combined with Theorem 5.1 and, for the explicit $\sqrt{\lambda(Q_O)}$ scaling statement, Corollary 5.2, this shows that the accelerated regime must occur at *intermediate* friction $\gamma \asymp \sqrt{\lambda(Q_O)}$, rather than in the large-friction limit.

At equilibrium ($\eta = 0$), the contraction hypothesis (Assumption 5.6(iv)) holds unconditionally: $\mathrm{Re}\langle \mathcal{L}f, f\rangle = \gamma\langle \mathcal{R}f, f\rangle \leq 0$ by skew-adjointness of $\mathcal{V}_{\mathrm{skew}}$, so $e^{t\mathcal{L}}$ is a contraction semigroup on $\mathcal{H}_0$. Together with Theorem 5.1, this gives a matching characterization at equilibrium: the exponent 1/2 in the $\sqrt{\lambda(Q_O)}$ scaling is optimal. For $\eta \neq 0$, the generator is generically not dissipative in $L^2(\mu_0)$; however, recent works [12, 31] establish L^2 convergence in stationary-state-adapted norms where contraction holds by construction. Reformulating the ceiling in such a norm would make the hypothesis automatic, at the cost of modifying the structural constants; this gap is one of reference-measure choice rather than a fundamental obstruction (see Open Problem 3).

Remark 5.8 (Ceiling hypotheses for designed lifts). For lifts built via Section 6.1, the eigenfunction and domain requirements in Assumption 5.6 are straightforward: they hold in finite dimensions by compactness and in PDE settings by standard regularity. Hence Theorem 5.7 applies unconditionally at equilibrium ($\eta = 0$). For $\eta \neq 0$, only the contraction hypothesis remains conditional in the working Hilbert norms; see Open Problem 3.

Corollary 5.9 (Diffusive ceiling for the L^2 relaxation time)**.** *Assume the setting of Theorem 5.7. Define $t_{\mathrm{rel}}(\mathcal{L})$ as in Corollary 5.5. Then the singular-value gap satisfies the explicit ceiling*

$$s(\mathcal{L}) \leq \frac{(1 + \kappa_{\mathrm{ell}})\, \lambda(Q_O)}{\gamma} + \eta\,(b_1 + b_2) + \eta\,(a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{67}$$

Moreover, the relaxation time is bounded below in terms of the singular-value gap (see Lemma D.2):

$$t_{\mathrm{rel}}(\mathcal{L}) \geq \frac{1}{2\, s(\mathcal{L})}. \tag{68}$$

Consequently, if $\eta \leq c/\gamma$, then $s(\mathcal{L}) = \mathcal{O}(\gamma^{-1})$ as $\gamma \to \infty$ and hence

$$t_{\mathrm{rel}}(\mathcal{L}) = \Omega(\gamma) \qquad (\gamma \to \infty). \tag{69}$$

In particular, combining this ceiling with Corollary 5.5 shows that, at optimal friction, the relaxation-time upper bound is of kinetic order $t_{\mathrm{rel}}(\mathcal{L}) = \mathcal{O}(\lambda(Q_O)^{-1/2})$.

6. Examples

This section constructs lifted generators from target slow dynamics via *reverse adiabatic elimination*, then applies the construction to two examples: a convex-potential Langevin family (Section 6.2) verifying the full family-uniform scaling, and a finite-dimensional boundary-driven model (Section 6.3) illustrating all structural assumptions including kernel compatibility.

6.1. **Reverse adiabatic elimination: from effective dynamics to lifted dynamics.** Input: a non-equilibrium effective generator $\mathcal{L}_O$ on the slow space $\mathcal{H}_S$. Output: a lifted generator $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ on $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$ such that adiabatic elimination (Definition 3.4) gives back $\mathcal{L}_O$. We call this construction *reverse adiabatic elimination*: identify the reversible backbone Q_O, choose a fast space and skew coupling that reproduce it, then add the perturbation so that the full effective generator is exactly $\mathcal{L}_O$. In finite dimensions this is algebraic; in unbounded settings one must additionally verify domain properties.

Perturbation expansion of $\mathcal{L}_O$. By Definition 3.4, $\mathcal{L}_O = -P_S\mathcal{V}\mathcal{R}^+\mathcal{V}P_S$ with $\mathcal{V} = \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$. Expanding in powers of η gives

$$\mathcal{L}_O = \underbrace{-P_S\mathcal{V}_{\mathrm{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{skew}}\,P_S}_{Q_O} \underbrace{-\,\eta\,P_S\big(\mathcal{V}_{\mathrm{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{pert}}\;+\;\mathcal{V}_{\mathrm{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{skew}}\big)P_S}_{\text{cross terms } \mathcal{O}(\eta)} \underbrace{-\,\eta^2\,P_S\mathcal{V}_{\mathrm{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{pert}}\,P_S}_{\text{pure perturbation } \mathcal{O}(\eta^2)}. \tag{70}$$

The first term is the reversible backbone Q_O (the $\eta = 0$ limit). The second and third terms are the $\mathcal{O}(\eta)$ and $\mathcal{O}(\eta^2)$ corrections. In the boundary-driven example, all these terms are written explicitly in (97).

Step 1: Identify Q_O. Set $\eta = 0$ in the slow model. The resulting operator is Q_O. When the kernel-compatible assumption $\ker(Q_O) = \mathbb{C}\mathbf{1}$ holds, its spectral gap $\lambda(Q_O)$ is the key scale in Theorem 5.1 (both for the accelerated rate and for the admissible perturbation size).

Step 2: Choose fast space and coupling to match Q_O. The adiabatic-elimination identity implies

$$Q_O = T^*\,(\mathcal{R}_F)^{-1}\,T, \qquad T := P_F\mathcal{V}_{\mathrm{skew}}P_S, \tag{71}$$

where $\mathcal{R}_F := \mathcal{R}|_{\mathcal{H}_F} < 0$. So, after choosing $\mathcal{H}_F$ and $\mathcal{R}_F$, we pick $T : \mathcal{H}_S \to \mathcal{H}_F$ to satisfy $T^*\mathcal{R}_F^{-1}T = Q_O$. A universal choice is $\mathcal{H}_F = \mathcal{H}_S$, $\mathcal{R}_F = -\lambda_R\,\mathrm{id}$, $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$. Then we set $\mathcal{R} := 0 \oplus \mathcal{R}_F$ and $\mathcal{V}_{\mathrm{skew}} := \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$.

This universal choice is worth keeping in mind: it always matches the prescribed Q_O, it makes the slow–fast geometry especially transparent, and it is the version used in the algorithmic-lift comparisons later in the paper.

Step 3: Determine $\mathcal{V}_{\mathrm{pert}}$ from consistency. With $\mathcal{R}$ and $\mathcal{V}_{\mathrm{skew}}$ fixed, we choose $\mathcal{V}_{\mathrm{pert}}$ so that adiabatic elimination exactly reproduces $\mathcal{L}_O$ whenever the resulting consistency problem is solvable in the chosen class of operators. This is equivalent to the consistency relation

$$\mathcal{L}_O - Q_O \;=\; -\eta\,P_S\big(\mathcal{V}_{\mathrm{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{pert}} + \mathcal{V}_{\mathrm{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{skew}}\big)P_S \;-\; \eta^2\,P_S\mathcal{V}_{\mathrm{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{pert}}\,P_S, \tag{72}$$

with the centering constraint $P_S\mathcal{V}_{\mathrm{pert}}P_S = 0$. In the Langevin example, $\mathcal{V}_{\mathrm{pert}}P_S = 0$, so the formula simplifies strongly. In the boundary-driven example, all terms are present, and we write

them explicitly in (97). In finite dimensions this is an algebraic problem, typically with non-unique solutions; in infinite-dimensional settings, solvability is model-dependent and should be verified together with the required domain properties. The resulting lifted generator $\mathcal{L} = \gamma\mathcal{R}+\mathcal{V}_{\text{skew}}+\eta\mathcal{V}_{\text{pert}}$ automatically satisfies centering. To apply Theorem 5.1, one must in addition verify the remaining structural bounds and the kernel-compatible gap assumption $\ker(Q_O) = \mathbb{C}\mathbf{1}$. We now state the key structural advantage: for a broad class of designed lifts, the projection bias can be removed exactly.

Proposition 6.1 (Unbiasedness of designed lifts)**.** *Consider a lift $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ constructed via the reverse adiabatic elimination recipe above, with $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$ (Step 2) and $\mathcal{V}_{\text{pert}}$ satisfying the centering condition $P_S\mathcal{V}_{\text{pert}}P_S = 0$ and the consistency condition* (72)*. If additionally $P_F\mathcal{V}_{\text{pert}}P_F = 0$ (the perturbation has no fast-to-fast component), then:*

(i) $P_F\mathcal{V}P_F = 0$;
(ii) the Schur complement (74) *satisfies $\gamma\, S(\gamma) = \mathcal{L}_O$ for all $\gamma > 0$;*
(iii) whenever the lifted and effective stationary states are well defined, the lifted NESS satisfies $P_S h_\eta = g_\eta$ for all $\gamma > 0$, i.e., the projection bias $\Delta_{\text{bias}}(\gamma) \equiv 0$.

The condition $P_F\mathcal{V}_{\text{pert}}P_F = 0$ is a free design choice within this parametrization: it does not appear in the consistency condition (72) *(which constrains only $P_F\mathcal{V}_{\text{pert}}P_S$ and $P_S\mathcal{V}_{\text{pert}}P_F$), so once the consistency problem is solvable it can be enforced without affecting the effective dynamics.*

Proof. (i) Since $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$ is block-off-diagonal, $P_F\mathcal{V}_{\text{skew}}P_F = 0$. Combined with $P_F\mathcal{V}_{\text{pert}}P_F = 0$, we get $P_F\mathcal{V}P_F = P_F(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})P_F = 0$. (ii) With $P_F\mathcal{V}P_F = 0$, the Schur complement (74) reduces to $S(\gamma) = -P_S\mathcal{V}(\gamma\mathcal{R}_F)^{-1}\mathcal{V}\,P_S = \gamma^{-1}\mathcal{L}_O$. (iii) Since $\gamma S(\gamma) = \mathcal{L}_O$, the NESS equation $S(\gamma)u = 0$ becomes $\mathcal{L}_O u = 0$, so $u = g_\eta$. □

Proposition 6.1 gives a clean design rule: if we enforce $P_F\mathcal{V}_{\text{pert}}P_F = 0$, then the projected lifted steady state equals the overdamped steady state for all γ. This does *not* remove non-equilibrium effects, because $\mathcal{L}_O$ still depends on the off-diagonal perturbation blocks $P_S\mathcal{V}_{\text{pert}}P_F$ and $P_F\mathcal{V}_{\text{pert}}P_S$. So non-self-adjoint slow dynamics remain fully allowed.

Remark 6.2 (Kernel consistency and invariant states). We verify that the lifted and effective dynamics have the same equilibrium kernel at $\eta = 0$, and hence the same stationary state whenever that kernel is one-dimensional.

Equilibrium case ($\eta = 0$): exact kernel identity for all $\gamma > 0$. Let $f \in \ker(\mathcal{L}|_{\eta=0}) = \ker(\gamma\mathcal{R} + \mathcal{V}_{\text{skew}})$. Taking $\operatorname{Re}\langle f, \mathcal{L}f\rangle = \gamma\langle f, \mathcal{R}f\rangle = 0$ (using skew-adjointness of $\mathcal{V}_{\text{skew}}$) forces $f \in \ker(\mathcal{R}) = \mathcal{H}_S$. Then $\mathcal{V}_{\text{skew}}f = 0$, which entails $Tf = P_F\mathcal{V}_{\text{skew}}P_S f = 0$ and thus $Q_O f = -T^*\mathcal{R}_F^{-1}Tf = 0$. Conversely, if $f \in \ker(Q_O) \subset \mathcal{H}_S$, then

$$0 = -\langle f, Q_O f\rangle = \langle Tf, (-\mathcal{R}_F)^{-1}Tf\rangle,$$

and since $(-\mathcal{R}_F)^{-1} > 0$ on $\mathcal{H}_F$, this implies $Tf = 0$. Because $\mathcal{V}_{\text{skew}}$ is block-off-diagonal in the reverse adiabatic-elimination construction, $\mathcal{V}_{\text{skew}}f = Tf = 0$, hence $(\gamma\mathcal{R} + \mathcal{V}_{\text{skew}})f = 0$. Therefore

$$\ker(\mathcal{L}|_{\eta=0}) = \ker(Q_O), \qquad \forall\, \gamma > 0. \tag{73}$$

Under Assumption 4.2, this common kernel is exactly $\mathbb{C}\mathbf{1}$. This is the analogue of [29, Lemma 2.10, eq. (2.17)].

Non-equilibrium case ($\eta \neq 0$). When $\eta \neq 0$ and $P_F\mathcal{V}P_F \neq 0$, the NESS $h_\eta \in \ker(\mathcal{L})$ acquires a fast component $P_F h_\eta = \mathcal{O}(\eta/\gamma)$, and the slow marginal $u = P_S h_\eta$ satisfies the Schur complement equation

$$S(\gamma)\, u = 0, \qquad S(\gamma) := -P_S\mathcal{V}\big(\gamma\mathcal{R}_F + P_F\mathcal{V}P_F\big)^{-1}\mathcal{V}\,P_S. \tag{74}$$

Expanding in a Neumann series:

$$\gamma\, S(\gamma) = \mathcal{L}_O + \gamma^{-1}\, P_S\mathcal{V}\,\mathcal{R}^+\,(P_F\mathcal{V}P_F)\,\mathcal{R}^+\,\mathcal{V}\,P_S + \mathcal{O}(\gamma^{-2}). \tag{75}$$

Thus $P_S h_\eta \to g_\eta$ as $\gamma \to \infty$ (the Zeno/adiabatic limit). The Schur correction is of order $\mathcal{O}(\gamma^{-1})$. Under the standard perturbative setting (simple isolated zero mode and smooth dependence on η), this gives $\Delta_{\text{bias}}(\gamma) := \|P_S h_\eta - g_\eta\| = \mathcal{O}(\eta/\gamma)$. However, when $P_F \mathcal{V} P_F = 0$ (as guaranteed by Proposition 6.1 for designed lifts), the expansion collapses to $\gamma\, S(\gamma) = \mathcal{L}_O$ and $\Delta_{\text{bias}}(\gamma) \equiv 0$.

Since the lifted dynamics converges to h_η, the approximation error for the overdamped NESS g_η satisfies the triangle inequality

$$\|P_S e^{t\mathcal{L}} f - g_\eta\|_{\mathcal{H}_S} \;\le\; \underbrace{C_*(\gamma)\, e^{-\lambda_\eta t}\, \|f - h_\eta\|}_{\text{convergence error}} \;+\; \underbrace{\|P_S h_\eta - g_\eta\|_{\mathcal{H}_S}}_{\text{bias } \Delta_{\text{bias}}(\gamma)}. \tag{76}$$

For lifts satisfying the conditions of Proposition 6.1, $\Delta_{\text{bias}}(\gamma) = 0$ and the accelerated mixing time is

$$T_{\text{mix}}(\varepsilon) \;=\; \mathcal{O}\Big(\frac{\log(1/\varepsilon)}{\sqrt{\lambda(Q_O)}}\Big), \qquad \forall\, \varepsilon > 0. \tag{77}$$

Algorithm 1 summarises the procedure.

Algorithm 1 Accelerated NESS Approximation via Lifting

Require: Target effective generator $\mathcal{L}_O$ on $\mathcal{H}_S$, target accuracy $\varepsilon > 0$
Ensure: Approximation of g_η (overdamped NESS) to accuracy ε

Step 1: Construct lifted dynamics (Section 6.1)
Factor $Q_O = T^* \mathcal{R}_F^{-1} T$ and set $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$
Choose $\mathcal{V}_{\text{pert}}$ satisfying (72) with $P_F \mathcal{V}_{\text{pert}} P_F = 0$ ▷ Zero bias (Prop. 6.1)

Step 2: Compute optimal friction
$\gamma_{\text{opt}} \leftarrow \sqrt{\lambda(Q_O)} \cdot \Theta^{-1} \sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}}}$
$\lambda_\eta \leftarrow \big(\Lambda_{\text{hypo}}(\gamma_{\text{opt}}) - \eta\, B_{\text{pert}}(\gamma_{\text{opt}})\big)/(1 + \delta_*(\gamma_{\text{opt}}))$

Step 3: Run lifted dynamics (rate $\sim \sqrt{\lambda(Q_O)}$)
$t_{\text{run}} \leftarrow \log\big(C_*(\gamma_{\text{opt}}) \|f_0\|/\varepsilon\big)/\lambda_\eta$
return $P_S\big(e^{t_{\text{run}}\mathcal{L}} h_0\big)$ ▷ ε-approximation of g_η

Remark 6.3 (Two-phase extension for general models). For natural physical models where $P_F \mathcal{V} P_F \neq 0$ (and the conditions of Proposition 6.1 do not hold), a nonzero accuracy floor $\varepsilon_* := \Delta_{\text{bias}}(\gamma_{\text{opt}}) = \mathcal{O}(\eta/\sqrt{\lambda(Q_O)})$ may arise. Then the error bound (76) cannot go below ε_* using only lifted dynamics. A direct remedy is two-phase evolution: first run the lifted dynamics at γ_{opt} down to scale ε_*, then switch to $\mathcal{L}_O$ to reach the final target ε. The total time is at most the overdamped baseline and is strictly smaller whenever $\lambda_\eta > \lambda(\mathcal{L}_O)$. This extension is not needed for either of the examples in this paper (Remark 6.8).

Remark 6.4 (Connection to the literature). The lifting construction shares the spirit of previous acceleration methods in Markov-chain and MCMC settings [36, 27, 21, 29], cast here in the continuous-time operator framework of Assumption 4.1. For the universal algorithmic lift one has $M_R = \lambda_R$ and $\kappa_{\text{ell}} = 0$ (block-off-diagonal construction); for the convex-potential Langevin family of Section 6.2, $\lambda_R = M_R = 1$ and the Bochner inequality gives a family-uniform κ_{ell}. Both realize the same $\sqrt{\lambda(Q_O)}$ mechanism with different geometric constants.

6.2. **Example 1: Nonequilibrium Langevin family with convex potential.** We study the underdamped Langevin family on $\mathbb{R}^{2d}$ with a uniformly convex confining potential and a bounded non-gradient perturbation. Its primary role is to provide a classical infinite-dimensional example in which all structural assumptions hold *uniformly* as $m \downarrow 0$, yielding a fully rigorous verification of the $\sqrt{\lambda(Q_O)}$ kinetic acceleration in a genuinely nonequilibrium setting. Throughout this example we fix $d \geq 2$, $\beta > 0$, $m \in (0, 1]$, and take $U \in C^2(\mathbb{R}^d)$ to satisfy

$$\nabla^2 U(q) \geq m \quad \text{for all } q \in \mathbb{R}^d, \tag{78}$$

and $F \in C_b^\infty(\mathbb{R}^d; \mathbb{R}^d)$ to be bounded and not of gradient form; the non-gradient condition ensures that $\mathcal{L}_O$ is genuinely nonequilibrium for every $\eta \neq 0$.

Setup. Let

$$d\mu(q, p) = Z^{-1} \exp\Big(-\beta\big(\tfrac{1}{2}|p|^2 + U(q)\big)\Big)\, dq\, dp = d\mu_q(q)\, d\mu_p(p), \tag{79}$$

where $\mu_q(dq) \propto e^{-\beta U(q)}\, dq$, and work in $\mathcal{H} = L^2(\mu)$ with slow projection

$$P_S f = \Pi_p f := \int_{\mathbb{R}^d} f(\cdot, p)\, d\mu_p(p). \tag{80}$$

We use the Schwartz core

$$\mathcal{C}_{\text{Lan}} := \mathcal{S}(\mathbb{R}^d_q \times \mathbb{R}^d_p). \tag{81}$$

Generator decomposition. The Langevin generator is written as

$$\mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma \mathcal{R} + \eta \mathcal{V}_{\text{pert}}, \tag{82}$$

with

$$\mathcal{V}_{\text{skew}} := p \cdot \nabla_q - \nabla_q U \cdot \nabla_p, \qquad \mathcal{R} := -p \cdot \nabla_p + \beta^{-1} \Delta_p, \qquad \mathcal{V}_{\text{pert}} f := F(q) \cdot \nabla_p f. \tag{83}$$

Effective generator $\mathcal{L}_O$ (explicit formula). Since $\mathcal{R}(p_i) = -p_i$ and $\int p_i\, d\mu_p = 0$, Definition 3.4 gives, for $\phi = \phi(q)$,

$$\mathcal{L}_O \phi = \beta^{-1} \Delta_q \phi - (\nabla_q U - \eta F(q)) \cdot \nabla_q \phi, \tag{84}$$

and therefore

$$Q_O \phi = \beta^{-1} \Delta_q \phi - \nabla_q U \cdot \nabla_q \phi. \tag{85}$$

The slow reference measure is $\mu_q(dq) \propto e^{-\beta U(q)}\, dq$. The Bakry–Émery criterion, applied to the uniform convexity $\nabla^2 U \geq m > 0$, yields the Poincaré inequality

$$-\langle \phi, Q_O \phi \rangle_{L^2(\mu_q)} = \beta^{-1} \|\nabla_q \phi\|^2_{L^2(\mu_q)} \geq m\, \|\phi - \textstyle\int \phi\, d\mu_q\|^2_{L^2(\mu_q)}. \tag{86}$$

Hence

$$\ker(Q_O) = \mathbb{C}\mathbf{1}, \qquad \lambda(Q_O) = m. \tag{87}$$

Since F is not a gradient field, the drift $\nabla_q U - \eta F(q)$ is not of gradient form for $\eta \neq 0$, so the effective generator $\mathcal{L}_O$ is genuinely nonequilibrium.

Non-self-adjointness on the slow space. In $L^2(\mu_q)$, integration by parts gives

$$(F \cdot \nabla_q)^\dagger = -F \cdot \nabla_q - (\nabla \cdot F) + \beta\, \nabla_q U \cdot F. \tag{88}$$

Hence $\mathcal{L}_O$ is generically non-self-adjoint as soon as $\eta \neq 0$. For the concrete force used in the numerics below,

$$F(q) = (-\tanh q_2, \tanh q_1), \tag{89}$$

one has $\nabla \cdot F = 0$ but $\nabla_q U \cdot F \not\equiv 0$ for any non-radial U, so $\mathcal{L}_O$ is explicitly non-self-adjoint for every $\eta \neq 0$.

Verification of Assumptions 4.1 and 4.2. The following proposition collects the verification of all structural hypotheses, with structural constants that are uniform in m and depend only on $(\beta, d, \|F\|_{L^\infty})$.

Proposition 6.5 (Verification for the convex-potential family)**.** *For the family* (83) *with* $U \in C^2(\mathbb{R}^d)$ *satisfying* $\nabla^2 U \geq m > 0$ *and bounded non-gradient force* F*, Assumptions 4.1, 4.2, and 4.3 hold in* $L^2(\mu)$*. Moreover, one may choose structural constants satisfying*

$$\lambda_R = 1, \qquad M_R = 1, \qquad \kappa_{\mathrm{ell}} = \sqrt{2}, \tag{90}$$

and the perturbation constants (a_1, b_1, a_2, b_2) *depend only on* $(\beta, d, \|F\|_{L^\infty})$ *and are independent of* m*.*

(The boundedness $F \in C_b^\infty$ is a convenient sufficient condition; on compact position spaces such as the torus [22] it is automatic, and for growing forces the abstract framework still applies via the relative bounds in Assumption 4.1.4.)

Corollary 6.6 (Rigorous kinetic scaling for the family)**.** *For the convex-potential Langevin family above, the hypotheses of Corollary 5.2 are satisfied uniformly as* $m \downarrow 0$*. Consequently,*

$$\gamma_{\mathrm{opt}} \asymp \sqrt{m}, \qquad \max_{\gamma>0} \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{m}, \qquad \eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{m}. \tag{91}$$

Thus this family gives a fully rigorous classical nonequilibrium verifier of the $\sqrt{\lambda(Q_O)}$ *acceleration mechanism within the present theorem.*

The proof of Proposition 6.5 is given in Appendix E.

Numerical verification. We verify the predicted scaling on the concrete quadratic instance $U(q) = \frac{m}{2}|q|^2$ in $d = 2$, with $F(q) = (-\tanh q_2, \tanh q_1)$ and nonequilibrium coupling $\eta = c\sqrt{m}$, $c \in \{0, \frac{1}{2}, 1\}$. The generator is discretized in a tensor-product Hermite–Galerkin basis (truncation degree $n_q = n_p = 3$, $\dim \mathcal{H} = 256$; see [22] for a related spectral approach on the torus).

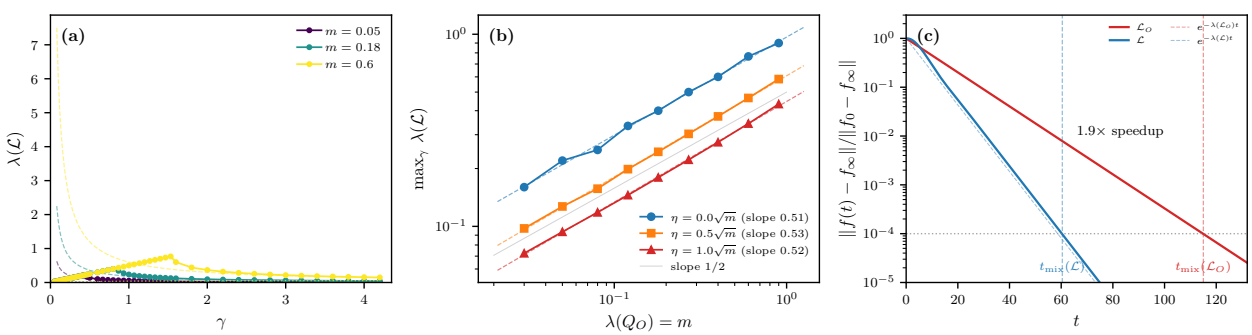

FIGURE 1. Numerical verification for Example 6.2 ($\beta = 1$, $m \in [0.03, 0.90]$, 45-point γ-grid). **(a)** Spectral gap versus γ at $\eta = 0$; dashed: overdamped envelope m/γ; dotted: gap-adapted DMS lower bound. **(b)** Peak gap versus m on log–log scale for $c \in \{0, \frac{1}{2}, 1\}$; fitted slopes ≈ 0.51–0.53. **(c)** Semigroup decay at $m = 0.08$, $c = \frac{1}{2}$, $\gamma^* = 0.5$; vertical markers: ε-mixing times ($\varepsilon = 10^{-4}$).

Figure 1 confirms the three-regime structure (underdamped, peaked, overdamped) and the predicted $\sqrt{\lambda(Q_O)}$ scaling: the fitted log–log slopes are 0.51, 0.53, and 0.52 for $c = 0, \frac{1}{2}, 1$. At $m = 0.08$, $c = \frac{1}{2}$, the lifted gap $\lambda(\mathcal{L}) \approx 0.157$ yields a 1.9× speedup over $\lambda(\mathcal{L}_O) \approx 0.080$, with mixing times $t_{\mathrm{mix}} \approx 60$ versus 115. Truncation accuracy is $O(10^{-4})$ relative to the next basis enlargement.

Algorithmic lift via generic square-root factorization. Here we use exactly the same three-step construction as in Example 6.3, now on the truncated slow Hermite space underlying Figure 1.

Step 1 (compute slow operators). On the 16-dimensional slow space we assemble the Galerkin matrices Q_O and $\mathcal{L}_O$ from the physical Langevin model.

Step 2 (build a generic lifted fast block). Set $\mathcal{H}_{F,\mathrm{alg}} = \mathcal{H}_S$, $\mathcal{R}_{F,\mathrm{alg}} = -\lambda_R \,\mathrm{id}$ with $\lambda_R = 1$, and $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$. This yields a 32×32 algorithmic lift, compared with the 256×256 physical lift.

Step 3 (match the non-equilibrium part). For Langevin, the perturbation satisfies $\mathcal{V}_{\mathrm{pert}} P_S = 0$, so we impose the same triangular structure in the designed lift by choosing $P_F \mathcal{V}_{\mathrm{pert}} P_S = 0$ and $P_F \mathcal{V}_{\mathrm{pert}} P_F = 0$. Then the consistency relation (72) reduces exactly to

$$\mathcal{L}_O - Q_O = \frac{\eta}{\lambda_R} \left(P_S \mathcal{V}_{\mathrm{pert}} P_F\right) T. \tag{92}$$

Since $\mathcal{L}_O - Q_O$ annihilates constants, we may set

$$P_S \mathcal{V}_{\mathrm{pert}} P_F = \frac{\lambda_R}{\eta} (\mathcal{L}_O - Q_O) T^+, \tag{93}$$

with T^+ the Moore–Penrose pseudoinverse, and the relation is solved exactly on the mean-zero slow space. On the matched grid used below the relative residual stays below 2×10^{-16}. Because $P_F \mathcal{V}_{\mathrm{pert}} P_F = 0$, Proposition 6.1 applies to the designed lift.

Figure 2 and Table 1 compare the physical (256×256) and algorithmic (32×32) lifts on the matched grid $m \in \{0.03, 0.05, 0.08, 0.12\}$, $c \in \{\frac{1}{4}, \frac{1}{2}\}$. The two constructions track each other closely: fitted log–log slopes are 0.50–0.51 for both. At $m = 0.08$, $c = \frac{1}{4}$, the mixing times ($\varepsilon = 10^{-4}$) are $t_{\mathrm{mix}} \approx 47.5$ (physical), 47.6 (algorithmic), and 115.2 (overdamped), confirming that the algorithmic lift reproduces the kinetic speedup while compressing the fast space from dimension 240 to 16.

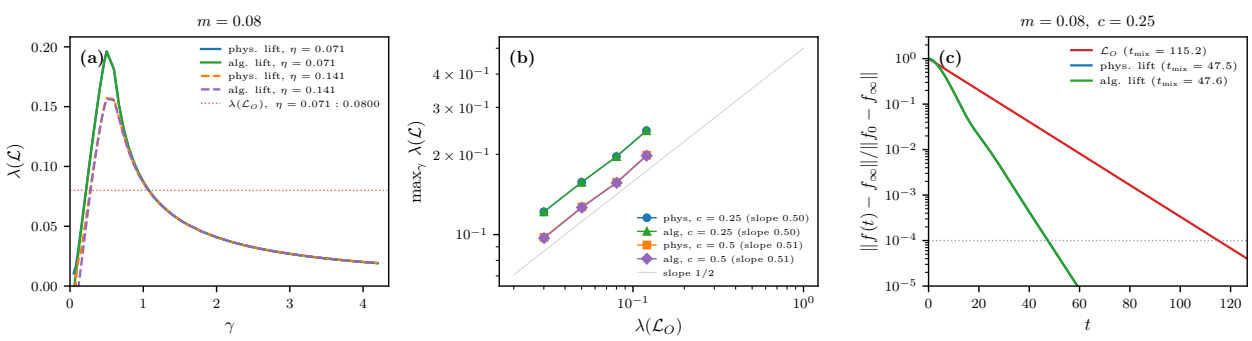

FIGURE 2. Comparison of the *physical* (256×256) and *algorithmic* (32×32, square-root factorization) lifted generators for Example 6.2. **(a)** Spectral gap versus γ at $m = 0.08$ for both lifts with $c = \frac{1}{4}$ and $c = \frac{1}{2}$ (that is, $\eta = c\sqrt{m}$); the overdamped reference is $\lambda(\mathcal{L}_O)$. **(b)** Log–log plot of peak spectral gap versus $\lambda(\mathcal{L}_O)$ on the matched grid $m \in \{0.03, 0.05, 0.08, 0.12\}$ for both lifts. **(c)** Convergence comparison at $m = 0.08$, $c = \frac{1}{4}$, $\varepsilon = 10^{-4}$: algorithmic lift, physical lift, and overdamped $\mathcal{L}_O$.

6.3. **Example 2: A Boundary-Driven Two-Qubit Model.** We now study a two-qubit boundary-driven model that satisfies all structural assumptions, including the kernel-compatible gap (Assumption 4.2). Its main role is to provide a finite-dimensional illustration in which the effective slow generator is genuinely nonequilibrium—not only the lifted generator but also the adiabatically eliminated slow dynamics is away from detailed balance for $\eta \neq 0$—and in which the reverse

TABLE 1. Peak spectral gaps $\max_\gamma \lambda(\mathcal{L})$ for the physical and algorithmic lifts on the matched grid. The final column reports the physical-lift speedup $\max_\gamma \lambda(\mathcal{L}_{\text{phys}})/\lambda(\mathcal{L}_O)$.

m	c	$\lambda(\mathcal{L}_O)$	phys. peak	alg. peak	phys. speedup
0.03	$\frac{1}{4}$	0.03001	0.12175	0.12163	4.06×
0.03	$\frac{1}{2}$	0.03010	0.09757	0.09701	3.24×
0.05	$\frac{1}{4}$	0.05001	0.15718	0.15703	3.14×
0.05	$\frac{1}{2}$	0.05016	0.12696	0.12629	2.53×
0.08	$\frac{1}{4}$	0.08002	0.19628	0.19613	2.45×
0.08	$\frac{1}{2}$	0.08024	0.15707	0.15640	1.96×
0.12	$\frac{1}{4}$	0.12002	0.24546	0.24519	2.05×
0.12	$\frac{1}{2}$	0.12032	0.19855	0.19765	1.65×

adiabatic-elimination algebra can be written out explicitly. Because $\dim \mathcal{H} = 16$, all operators are bounded and there are no domain subtleties.

Setup. Both fast and slow subsystems are qubits: $\mathfrak{h}_F \cong \mathfrak{h}_S \cong \mathbb{C}^2$. We work in $\mathcal{H} = L^2(\sigma)$ with $\sigma = \frac{1}{2}\mathbf{1}_F \otimes \frac{1}{2}\mathbf{1}_S$. The slow subspace is $\mathcal{H}_S = \mathbf{1}_F \otimes \mathcal{B}(\mathfrak{h}_S)$, with projection $P_S(X) = \mathbf{1}_F \otimes \mathrm{tr}_F(\frac{1}{2}\mathbf{1}_F \cdot X)$.

Generator decomposition. The decomposition is $\mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma\mathcal{R} + \eta\mathcal{V}_{\text{pert}}$. Here $\mathcal{R} = \mathcal{R}_F \otimes \mathrm{id}_S$ with $\mathcal{R}_F X = \Gamma \sum_{j\in\{x,y,z\}} \mathcal{D}^\dagger[\sigma_j]X$, so $\lambda_R = 4\Gamma$. The transport part uses a *two-channel* coupling,

$$\mathcal{V}_{\text{skew}} X = -i[g(\sigma_x \otimes \sigma_x + \sigma_z \otimes \sigma_z), X], \tag{94}$$

and the perturbation is $\mathcal{V}_{\text{pert}} = \sigma_z \otimes \mathcal{G}_S$, $\mathcal{G}_S(X) = \mathcal{D}^\dagger[\sigma_-]X$. Because $\mathrm{tr}(\frac{1}{2}\mathbf{1}_F \cdot \sigma_z) = 0$, both terms map $\mathcal{H}_S$ into $\mathcal{H}_F$, so centering is satisfied.

Effective generator $\mathcal{L}_O$ (explicit formula). Let $X = \mathbf{1}_F \otimes X_S \in \mathcal{H}_S$.
Step 1 (first excursion to fast space). The two-channel coupling gives

$$(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})X = \sum_{a\in\{x,z\}} \sigma_a \otimes \Big(- ig\,[\sigma_a, X_S]\Big) + \sigma_z \otimes \eta\,\mathcal{G}_S(X_S). \tag{95}$$

Step 2 (Poisson equation on the fast ancilla). Since $\mathcal{R}_F(\sigma_a) = -\kappa\,\sigma_a$ with $\kappa := 4\Gamma$ for each $a \in \{x, y, z\}$, we have $\mathcal{R}^+(\sigma_a \otimes Y) = -(1/\kappa)\sigma_a \otimes Y$. Therefore

$$\mathcal{R}^+(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})X = -\frac{1}{\kappa} \sum_{a\in\{x,z\}} \sigma_a \otimes \Big(- ig\,[\sigma_a, X_S]\Big) - \frac{1}{\kappa}\sigma_z \otimes \eta\,\mathcal{G}_S(X_S). \tag{96}$$

Step 3 (return map and projection). Using $\mathcal{V}_{\text{skew}}(\sigma_a \otimes Y) = -ig \sum_{b\in\{x,z\}} \sigma_a\sigma_b \otimes [\sigma_b, Y]$, the cross-channel products $\sigma_x\sigma_z = -i\sigma_y$ and $\sigma_z\sigma_x = i\sigma_y$ produce σ_y factors, which vanish under P_S (since $\mathrm{tr}(\sigma_y) = 0$). Only same-channel terms survive. Projecting gives

$$\begin{aligned}\mathcal{L}_O(\mathbf{1}_F \otimes X_S) = -\frac{1}{\kappa}\,\mathbf{1}_F \otimes \Big(g^2 &\sum_{a\in\{x,z\}} [\sigma_a, [\sigma_a, X_S]] \\ &+ i\eta g\,\big[\sigma_z, \mathcal{G}_S(X_S)\big] \\ &+ i\eta g\,\mathcal{G}_S([\sigma_z, X_S]) \\ &- \eta^2\,\mathcal{G}_S^2(X_S)\Big).\end{aligned} \tag{97}$$

The reference operator is the $\eta = 0$ part,

$$Q_O(\mathbf{1}_F \otimes X_S) = -\frac{g^2}{\kappa}\,\mathbf{1}_F \otimes \sum_{a\in\{x,z\}} [\sigma_a, [\sigma_a, X_S]]. \tag{98}$$

Non-self-adjointness on the slow space. Since the slow marginal of σ is proportional to the identity, the $L^2(\sigma)$ inner product on $\mathcal{H}_S$ reduces to the Hilbert–Schmidt inner product, and one checks that $(\mathcal{D}^\dagger[\sigma_-])^\dagger = \mathcal{D}[\sigma_-] \neq \mathcal{D}^\dagger[\sigma_-]$. Thus $\mathcal{G}_S$ is not self-adjoint on $\mathcal{H}_S$. Consequently, the $\mathcal{O}(\eta)$ cross terms in $\mathcal{L}_O - Q_O$ involve the mixed operators $[\sigma_z, \mathcal{G}_S(\cdot)] + \mathcal{G}_S([\sigma_z, \cdot])$, which are generically not self-adjoint (unless additional commutation relations hold), so $\mathcal{L}_O$ is generically non-self-adjoint for $\eta \neq 0$. In particular, the effective slow dynamics is already a NESS-type evolution before any finite-friction acceleration is invoked; we are not merely lifting an equilibrium reduced model.

This can be seen explicitly on the Pauli basis. Using

$$\mathcal{G}_S(\mathbf{1}) = 0, \qquad \mathcal{G}_S(\sigma_x) = -\tfrac{1}{2}\sigma_x, \qquad \mathcal{G}_S(\sigma_y) = -\tfrac{1}{2}\sigma_y, \qquad \mathcal{G}_S(\sigma_z) = -(\mathbf{1} + \sigma_z), \tag{99}$$

the effective generator (97) acts as

$$\begin{aligned}
\mathcal{L}_O(\mathbf{1}) &= 0,\\
\mathcal{L}_O(\sigma_x) &= -\frac{4g^2 - \eta^2/4}{\kappa}\,\sigma_x - \frac{2\eta g}{\kappa}\,\sigma_y,\\
\mathcal{L}_O(\sigma_y) &= \frac{2\eta g}{\kappa}\,\sigma_x - \frac{8g^2 - \eta^2/4}{\kappa}\,\sigma_y,\\
\mathcal{L}_O(\sigma_z) &= \frac{\eta^2}{\kappa}\,\mathbf{1} - \frac{4g^2 - \eta^2}{\kappa}\,\sigma_z.
\end{aligned} \tag{100}$$

The σ_x–σ_y coupling and the $\sigma_z \to \mathbf{1}$ drift vanish when $\eta = 0$; for $\eta \neq 0$ they provide an explicit algebraic witness that the reduced dynamics itself is already away from detailed balance.

Verification of Assumptions 4.1 and 4.2. The following proposition collects the structural verification; all operators are bounded on the finite-dimensional space $\mathcal{H}$, so the analysis reduces to linear algebra.

Proposition 6.7 (Verification for the boundary-driven model)**.** *For the boundary-driven two-qubit family above with the two-channel coupling $H_{\text{int}} = g(\sigma_x \otimes \sigma_x + \sigma_z \otimes \sigma_z)$, Assumptions 4.1 and 4.2 hold. Moreover,*

$$\lambda_R = 4\Gamma, \qquad M_R = \|\mathcal{R}\|, \qquad \ker(Q_O) = \mathbb{C}\mathbf{1}, \qquad \lambda(Q_O) = g^2/\Gamma, \tag{101}$$

and the perturbation constants (a_1, b_1, a_2, b_2) satisfy $a_1 = a_2 = 0$, $b_j = \|\mathcal{K}_{\text{pert}}\|$ or $\|\mathcal{S}_{\text{pert}}\|$.

The proof is given in Appendix F.

Thus the boundary-driven toy is a fully rigorous verifier of the structural part of the theory (Theorem 5.1 and, for the family scaling, Corollary 5.2). By contrast, the additional contraction hypothesis in Assumption 5.6(iv) is not automatic here for $\eta \neq 0$ in the Hilbert–Schmidt/KMS norm used above, so Theorem 5.7 remains conditional for this example in its present formulation.

TABLE 2. Kernel compatibility and spectral gap for Q_O with the two-channel coupling $H_{\text{int}} = g(\sigma_x \otimes \sigma_x + \sigma_z \otimes \sigma_z)$. For all $g > 0$: $\dim\ker(Q_O) = 1$ $(= \mathbb{C}\mathbf{1})$ and $\lambda(Q_O) = g^2/\Gamma$.

g	$\dim\ker(Q_O)$	$\lambda(Q_O)$
0.3	1	0.1125
0.5	1	0.3125
1.0	1	1.2500

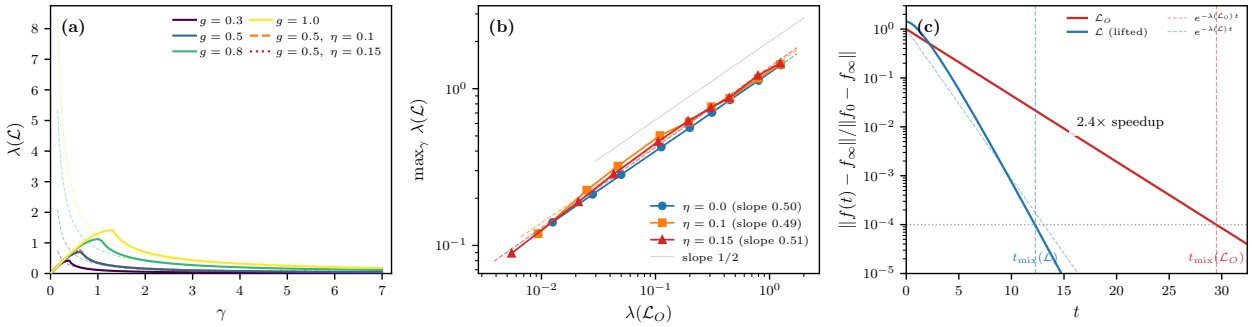

FIGURE 3. Numerical verification for Example 6.3. **(a)** Spectral gap $\lambda(\mathcal{L})$ vs. γ at $\eta = 0$ for several values of g; dashed curves show the overdamped envelope $\lambda(\mathcal{L}_O)/\gamma$. **(b)** Log–log plot of $\max_\gamma \lambda(\mathcal{L})$ versus $\lambda(\mathcal{L}_O)$. In this finite-dimensional setting the fitted slopes track the half-law closely, so the panel serves as a clean numerical benchmark for the mechanism. **(c)** Decay of the projected slow component for the lifted dynamics at γ^* and for $\mathcal{L}_O$ ($g = 0.5$, $\eta = 0$); dashed lines show the exponential envelopes. Vertical markers indicate the ε-level crossing times ($\varepsilon = 10^{-4}$). Parameters: $\Gamma = 0.8$, $\kappa = 3.2$, $g \in \{0.1, \ldots, 1.0\}$, 53-point γ-grid over $[0.02, 7.0]$.

Numerical verification. We assemble the exact 16×16 generator and scan spectral gaps over $g \in \{0.1, \ldots, 1.0\}$, $\Gamma = 0.8$, $\gamma \in [0.02, 7.0]$, and $\eta \in \{0, 0.10, 0.15\}$. Figure 3 confirms the three-regime structure and the $\sqrt{\lambda(Q_O)}$ scaling: fitted log–log slopes are 0.50, 0.49, 0.51 for $\eta = 0$, 0.10, 0.15. At $g = 0.5$, $\eta = 0$, the lifted gap yields a 2.4× speedup ($t_{\text{mix}} \approx 12.3$ versus 29.5, $\varepsilon = 10^{-4}$).

Algorithmic lift via generic square-root factorization. Here we use exactly the same three-step construction as above, now in the boundary-driven toy model.

Step 1 (compute slow operators). Compute $\mathcal{L}_O$ and Q_O from the physical model.

Step 2 (build a generic lifted fast block). Set $\mathcal{H}_{F,\text{alg}} = \mathcal{H}_S$, $\mathcal{R}_{F,\text{alg}} = -\lambda_R \,\text{id}$, and $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$. This produces an 8×8 lifted generator (the physical model is 16×16).

Step 3 (match the non-equilibrium part). Solve the consistency relation (72) numerically for the perturbation block. This linear algebra problem has multiple solutions; we choose the one with smallest matrix norm. The final residual is below 10^{-13}.

Figure 4 and Table 3 compare the physical (16×16) and algorithmic (8×8) lifts on the matched grid $g \in \{0.3, 0.5, 0.8, 1.0\}$, $\eta \in \{0.10, 0.15\}$. The two constructions track each other closely in both peak gaps and convergence curves; stationary states agree to machine precision.

TABLE 3. Peak spectral gaps $\max_\gamma \lambda(\mathcal{L})$ for the physical and algorithmic lifts (53-point γ-grid). The "speedup" column is $\max_\gamma \lambda(\mathcal{L}_{\text{phys}})/\lambda(\mathcal{L}_O)$.

g	η	$\lambda(\mathcal{L}_O)$	phys. peak	alg. peak	phys. speedup
0.3	0.10	0.1094	0.5024	0.5806	4.59×
0.3	0.15	0.1055	0.4600	0.5806	4.36×
0.5	0.10	0.3094	0.7673	0.8831	2.48×
0.5	0.15	0.3055	0.7552	0.8800	2.47×
0.8	0.10	0.7969	1.1663	1.4906	1.46×
0.8	0.15	0.7930	1.2163	1.4906	1.53×
1.0	0.10	1.2469	1.4291	1.9789	1.15×
1.0	0.15	1.2430	1.4534	1.9789	1.17×

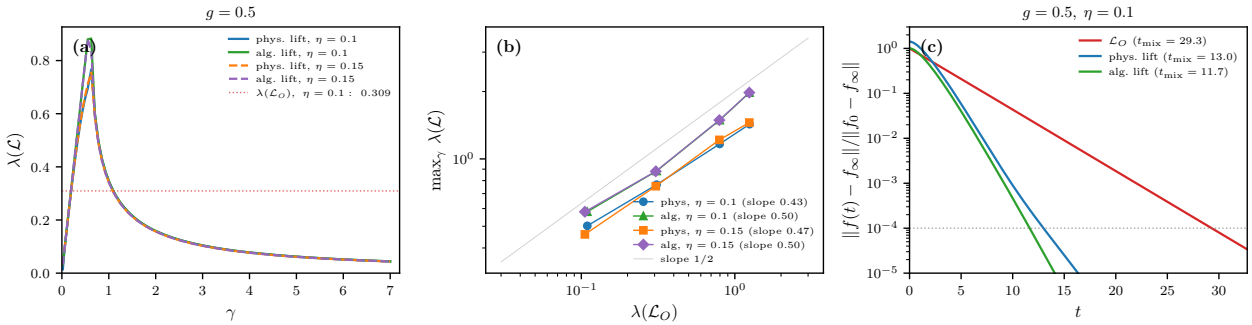

FIGURE 4. Comparison of the *physical* (16×16) and *algorithmic* (8×8, square-root factorization) lifted generators for Example 6.3. **(a)** Spectral gap $\lambda(\mathcal{L})$ vs. γ at $g = 0.5$ for both lifts ($\eta = 0.10$ and $\eta = 0.15$); the overdamped reference is $\lambda(\mathcal{L}_O)$. **(b)** Log–log plot of peak spectral gap vs. $\lambda(\mathcal{L}_O)$ for both lifts. **(c)** Convergence comparison ($g = 0.5$, $\eta = 0.10$, $\varepsilon = 10^{-4}$): algorithmic lift, physical lift, and overdamped $\mathcal{L}_O$.

Remark 6.8 (Bias behaviour in the two examples). For the *algorithmic* lifts in both examples, $P_F \mathcal{V}_{\text{pert}} P_F = 0$ by construction, so Proposition 6.1 predicts zero projection bias. Numerically, $\|P_S h_\eta^{\text{alg}} - g_\eta\|/\|g_\eta\| \lesssim 10^{-16}$ at all tested parameter values; the physical lifts give the same result on the adopted truncations. For models with $P_F \mathcal{V} P_F \neq 0$, use the two-phase strategy of Remark 6.3.

7. Conclusion and open problems

We have established explicit model-wise bounds for the interplay between friction, reversible backbone gap, and non-equilibrium forcing in lifted generators, recovering the kinetic $\sqrt{\lambda(Q_O)}$ scaling for uniformly controlled families. We close with several open directions.

(1) Sharp stability thresholds and non-convex potentials. The threshold η^* involves structural constants that may not be sharp; numerically, acceleration persists well beyond η^*. The present convex-potential Langevin family (Section 6.2) establishes family-uniform structural constants for all U with $\nabla^2 U \geq m > 0$, but extending the result to non-convex potentials (where the Bochner inequality fails) remains open. For models where forcing enhances the effective gap (e.g. optimally rotated diffusions [21]), one expects $\lambda(\mathcal{L}_O) > \lambda(Q_O)$, and a refined analysis could yield

$$\max_{\gamma>0} \lambda(\mathcal{L}_\gamma) \asymp \sqrt{\lambda(\mathcal{L}_O)} \qquad \text{(conjectured, when } \lambda(\mathcal{L}_O) > \lambda(Q_O)\text{)}. \tag{102}$$

(2) Optimal factorization. The reverse adiabatic elimination recipe lifts any target Q_O, but the structural constants depend on the auxiliary space $(\mathcal{H}_F, \mathcal{R}_F, T)$. Characterizing the factorization that maximises η^* in infinite-dimensional settings, and extending the framework to multilevel decompositions $\mathcal{H}_{S_1} \subset \mathcal{H}_{S_2} \subset \cdots$, remain open.

(3) Beyond the perturbative regime. Extending the analysis to large η may require space-time divergence methods [16, 17] or probabilistic contractivity arguments [31]. A related problem is to formulate the overdamped ceiling directly in a stationary-state-adapted norm, making the contraction hypothesis automatic.

(4) Quantum many-body NESS. The two-qubit toy model of Section 6.3 shows that the framework applies to boundary-driven Lindbladians [19, 30] when suitable noncommutative Poincaré inequalities hold; compare [28]. For many-body systems (e.g., boundary-driven XXZ chains), scaling the structural constants with system size and establishing meaningful $\lambda(Q_O)$ lower bounds remain open.

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Appendix A. Proofs for Preliminaries

This appendix collects the proofs from Section 3, carried out at Hilbert-space level with $\mathcal{H} = \mathbb{C}\mathbf{1} \oplus \mathcal{H}_0$.

A.1. **Proof of Lemma 3.2.** We first clarify the geometric relationship between the fixed point space $\mathcal{F}(\mathcal{L}) = \mathbb{C}\mathbf{1}$ and the kernel of the symmetric part $\mathcal{S}$.

Proposition A.1 (Kernel Inclusion)**.** *For any dissipative generator with symmetric part* $\mathcal{S} := \frac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger) \leq 0$*, one has*

$$\ker(\mathcal{L}) \subseteq \ker(\mathcal{S}). \tag{103}$$

Proof. Let $X \in \ker(\mathcal{L})$. Then $\mathcal{L}(X) = 0$. Consequently,

$$\langle X, \mathcal{S}(X)\rangle = \Re\langle X, \mathcal{L}(X)\rangle = 0. \tag{104}$$

Since $\mathcal{S}$ is a dissipative operator (negative semi-definite, $\mathcal{S} \leq 0$), the equality $\langle X, \mathcal{S}X\rangle = 0$ implies $\mathcal{S}X = 0$ (essentially due to the Cauchy–Schwarz inequality for the semi-inner product defined by $-\mathcal{S}$). Thus, $X \in \ker(\mathcal{S})$. □

Proof of Lemma 3.2. We prove the coercivity characterization and show that, if $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$, then no exponential decay estimate can hold with prefactor $C = 1$.

(⇐): Assume $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$ and $-\mathcal{S}|_{\mathcal{H}_0}$ has a positive spectral gap $\lambda_S > 0$.

$$\langle X, \mathcal{S}X\rangle \leq -\lambda_S \|X\|^2, \qquad \forall X \in \mathcal{H}_0. \tag{105}$$

For any $X_0 \in \mathrm{dom}(\mathcal{L}) \cap \mathcal{H}_0$, set $X_t = \mathcal{P}_t X_0$. Then $t \mapsto X_t$ is differentiable in $\mathcal{H}_0$, and the evolution of the norm satisfies:

$$\frac{1}{2}\frac{d}{dt}\|X_t\|^2 = \Re\langle X_t, \mathcal{L}X_t\rangle = \langle X_t, \mathcal{S}X_t\rangle \leq -\lambda_S \|X_t\|^2. \tag{106}$$

Integrating this differential inequality yields $\|X_t\| \leq e^{-\lambda_S t}\|X_0\|$ for all such X_0. By density of $\mathrm{dom}(\mathcal{L}) \cap \mathcal{H}_0$ in $\mathcal{H}_0$ and continuity of $\mathcal{P}_t$, the same estimate extends to every $X_0 \in \mathcal{H}_0$. This is an exponential decay with prefactor $C = 1$ and rate λ_S. Thus, the system is coercive, not strictly hypocoercive.

(⇒): Assume the system is coercive ($C = 1$). By definition, there exists $\nu > 0$ such that $\|\mathcal{P}_t X\| \leq e^{-\nu t}\|X\|$ for all $X \in \mathcal{H}_0$. For $X \in \mathrm{dom}(\mathcal{L}) \cap \mathcal{H}_0$, divide the squared estimate by $2t$ and let $t \downarrow 0$:

$$\Re\langle X, \mathcal{L}X\rangle \leq -\nu\|X\|^2, \quad \forall X \in \mathrm{dom}(\mathcal{L}) \cap \mathcal{H}_0. \tag{107}$$

Substituting $\Re\langle X, \mathcal{L}X\rangle = \langle X, \mathcal{S}X\rangle$, we obtain:

$$\langle X, \mathcal{S}X\rangle \le -\nu\|X\|^2. \tag{108}$$

Now, suppose there exists a non-trivial vector $Y \in \ker(\mathcal{S}) \cap \mathcal{H}_0$. Then $\langle Y, \mathcal{S}Y\rangle = 0$. However, the inequality above requires $\langle Y, \mathcal{S}Y\rangle \le -\nu\|Y\|^2 < 0$, a contradiction. Therefore, $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$, which implies $\ker(\mathcal{S}) = \mathbb{C}\mathbf{1} = \ker(\mathcal{L})$.

Finally, if $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$ then the above argument shows that coercivity ($C = 1$ with some positive rate) is impossible. In that case, any exponential decay estimate (when it holds) must necessarily come with some prefactor $C > 1$. □

Appendix B. Proof of Lemma 4.7

This appendix establishes the structural bounds of Lemma 4.7.

Proof. In the unbounded setting, all identities below are first established for f in the common core $\mathcal{C}$ from Assumption 4.3; the resulting operator bounds then extend to the stated energy spaces by density and continuity. Recall the definition of the reference effective dynamics $Q_O = -P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S$. By Assumption 4.1.1, the pseudoinverse is bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \le \lambda_R^{-1}$, hence

$$\langle g, -\mathcal{R}^+ g\rangle \le \frac{1}{\lambda_R}\|g\|^2, \quad \forall g \in \mathcal{H}_F. \tag{109}$$

The complementary (lower) control needed below is supplied by the flux conditioning assumption (24), which applies to fluxes of the form $g = g_\psi = P_F\mathcal{V}_{\text{skew}}P_S\psi$.

B.1. **Proof of Properties 1 and 2 (Boundedness and Regularity). Solver boundedness and forward regularity.** Let $f \in \mathcal{H}$ and define $u := \mathcal{A}_\varpi f \in \mathcal{H}_S$. The defining equation is $(\varpi - Q_O)u = P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F f$. Taking the inner product with u and using the skew-adjointness $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$:

$$\varpi\|u\|^2 + \langle u, -Q_O u\rangle = \langle u, P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F f\rangle = -\langle \mathcal{V}_{\text{skew}}u, (-\mathcal{R}^+)P_F f\rangle. \tag{110}$$

Let $v := \mathcal{V}_{\text{skew}}u$. Since $P_S\mathcal{V}_{\text{skew}}P_S = 0$ (Centered Coupling), we strictly have $v \in \mathcal{H}_F$. Using the definition of Q_O and the pseudoinverse bound (109):

$$\langle u, -Q_O u\rangle = \langle v, -\mathcal{R}^+ v\rangle = \|(-\mathcal{R}^+)^{1/2}v\|^2. \tag{111}$$

Substituting this into the energy identity and applying Cauchy–Schwarz yields:

$$\varpi\|u\|^2 + \|(-\mathcal{R}^+)^{1/2}v\|^2 \le \|(-\mathcal{R}^+)^{1/2}v\|\,\|(-\mathcal{R}^+)^{1/2}P_F f\|. \tag{112}$$

Writing $a := \|(-\mathcal{R}^+)^{1/2}v\|$ and $b := \|(-\mathcal{R}^+)^{1/2}P_F f\|$, we have $\varpi\|u\|^2 + a^2 \le ab$. Completing the square,

$$a^2 - ab = \left(a - \tfrac{1}{2}b\right)^2 - \tfrac{1}{4}b^2 \ge -\tfrac{1}{4}b^2, \tag{113}$$

so $\varpi\|u\|^2 \le \frac{1}{4}b^2$, hence

$$\|u\| \le \frac{1}{2}\,\varpi^{-1/2}\,\|(-\mathcal{R}^+)^{1/2}P_F f\| \le \frac{1}{2\sqrt{\lambda_R}}\,\varpi^{-1/2}\,\|f\|. \tag{114}$$

Moreover, $a \le b$. In the bounded-operator setting one has $\mathrm{dom}(Q_O) = \mathcal{H}_S$, so $u \in \mathrm{dom}(Q_O)$. Then the vector $v = \mathcal{V}_{\text{skew}}u$ is a slow-induced flux of the form g_ψ with $\psi \in \mathrm{dom}(Q_O)$; we may therefore apply the flux conditioning bound (24) to v to obtain

$$\|v\|^2 \le M_R\,\langle v, -\mathcal{R}^+ v\rangle = M_R a^2 \le M_R b^2 \le \frac{M_R}{\lambda_R}\,\|f\|^2, \tag{115}$$

so $\|v\| \le \sqrt{M_R/\lambda_R}\,\|f\|$. This proves $\|\mathcal{A}_\varpi\| \le k_\mathcal{A}\varpi^{-1/2}$ and $\|\mathcal{V}_{\text{skew}}\mathcal{A}_\varpi\| \le k_{\text{reg}}$, with

$$k_\mathcal{A} := \frac{1}{2\sqrt{\lambda_R}}, \qquad k_{\text{reg}} := \sqrt{M_R/\lambda_R}. \tag{116}$$

Dual regularity. We consider the operator $\mathcal{T} := \mathcal{A}_\varpi\mathcal{V}_{\text{skew}}P_F$. Its norm equals that of its adjoint $\mathcal{T}^\dagger$. We compute the adjoint of the solver:

$$\mathcal{A}_\varpi^\dagger = \big((\varpi - Q_O)^{-1}P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F\big)^\dagger = -P_F(-\mathcal{R}^+)\mathcal{V}_{\text{skew}}P_S(\varpi - Q_O)^{-1}. \tag{117}$$

(Using $Q_O^\dagger = Q_O$ and $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$). Therefore:

$$\mathcal{T}^\dagger = P_F\mathcal{V}_{\text{skew}}^\dagger\mathcal{A}_\varpi^\dagger = P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)\mathcal{V}_{\text{skew}}P_S(\varpi - Q_O)^{-1}. \tag{118}$$

Let $g \in \mathcal{H}_S$ and $\psi := (\varpi - Q_O)^{-1}g$. Define the macroscopic flux

$$g_\psi := P_F\mathcal{V}_{\text{skew}}P_S\psi \in \mathcal{H}_F. \tag{119}$$

Then $\mathcal{T}^\dagger g = P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)g_\psi$. Invoking the **Flux Regularity** condition from Assumption 4.1.3:

$$\|P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)g_\psi\| \le \kappa_{\text{ell}}\|(-Q_O)\psi\|. \tag{120}$$

Substituting ψ back:

$$\|\mathcal{T}^\dagger g\| \le \kappa_{\text{ell}}\|(-Q_O)(\varpi - Q_O)^{-1}g\|. \tag{121}$$

Since Q_O is self-adjoint and non-positive, the operator $(-Q_O)(\varpi - Q_O)^{-1}$ is a spectral contraction ($\|\cdot\| \le 1$). Thus:

$$\|\mathcal{T}^\dagger g\| \le \kappa_{\text{ell}}\|g\|. \tag{122}$$

This proves $\|\mathcal{A}_\varpi\mathcal{V}_{\text{skew}}P_F\| \le \kappa_{\text{ell}}$.

B.2. **Proof of Property 3 (Perturbation Control).** We use the structural decomposition $\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$ from Assumption 4.1.4. We also recall the centered mapping property (40), which will be used later to sharpen the perturbation bookkeeping in the dissipation matrix. Recall the bound for the skew part (Assumption 4.1.4(a)):

$$\|\mathcal{K}_{\text{pert}}f\| \le a_1\,\|(-\mathcal{R})^{1/2}f\| + b_1\,\|f\|, \qquad f \in \text{dom}((-\mathcal{R})^{1/2}). \tag{123}$$

For the symmetric part, we use the energy bound (Assumption 4.1.4(b)):

$$\|\mathcal{S}_{\text{pert}}f\| \le a_2\,\|(-\mathcal{R})^{1/2}f\| + b_2\,\|f\|, \qquad f \in \text{dom}((-\mathcal{R})^{1/2}). \tag{124}$$

Reverse estimate ($\mathcal{V}_{\text{pert}}^\dagger\mathcal{A}_\varpi$). Let $f \in \mathcal{H}$ and $u := \mathcal{A}_\varpi f$. The adjoint is $\mathcal{V}_{\text{pert}}^\dagger = -\mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$. Applying the triangle inequality and the bounds above:

$$\|\mathcal{V}_{\text{pert}}^\dagger u\| \le \|\mathcal{K}_{\text{pert}}u\| + \|\mathcal{S}_{\text{pert}}u\| \tag{125}$$

$$\le b_1\,\|u\| + b_2\,\|u\| + (a_1 + a_2)\,\|(-\mathcal{R})^{1/2}u\|. \tag{126}$$

Since $u \in \mathcal{H}_S = \ker(\mathcal{R})$ (by definition of $\mathcal{A}_\varpi$), one has $(-\mathcal{R})^{1/2}u = 0$, hence

$$\|\mathcal{V}_{\text{pert}}^\dagger u\| \le (b_1 + b_2)\,\|u\|. \tag{127}$$

Substituting the Solver Bound ($\|u\| \le k_\mathcal{A}\varpi^{-1/2}\|f\|$):

$$\|\mathcal{V}_{\text{pert}}^\dagger\mathcal{A}_\varpi f\| \le (b_1 + b_2)k_\mathcal{A}\varpi^{-1/2}\,\|f\|. \tag{128}$$

This establishes (37a) with $k_P(\varpi) := (b_1 + b_2)k_\mathcal{A}\varpi^{-1/2}$.

Forward estimate ($\mathcal{A}_\varpi\mathcal{V}_{\rm pert}$). We aim to bound the operator norm of $\mathcal{W}_{\rm fwd} := \mathcal{A}_\varpi\mathcal{V}_{\rm pert}$. By adjoints on Hilbert spaces, $\|\mathcal{W}_{\rm fwd}\| = \|\mathcal{W}_{\rm fwd}^\dagger\| = \|\mathcal{V}_{\rm pert}^\dagger\mathcal{A}_\varpi^\dagger\|$. Let $g \in \mathcal{H}_S$ be arbitrary with $\|g\| = 1$ and set $w := \mathcal{A}_\varpi^\dagger g$. Using the bounds on $\mathcal{K}_{\rm pert}$ and the energy bound on $\mathcal{S}_{\rm pert}$,

$$\begin{aligned}\|\mathcal{V}_{\rm pert}^\dagger w\| &\le \|\mathcal{K}_{\rm pert} w\| + \|\mathcal{S}_{\rm pert} w\| \\ &\le (b_1+b_2)\,\|w\| + (a_1+a_2)\,\|(-\mathcal{R})^{1/2} w\|.\end{aligned} \tag{129}$$

We compute the norms on the right-hand side.

(1) **Solver term ($\|w\|$):**

$$\|w\| = \|\mathcal{A}_\varpi^\dagger g\| \le \|\mathcal{A}_\varpi\|\,\|g\| \le k_{\mathcal{A}}\,\varpi^{-1/2}. \tag{130}$$

(2) **Energy term ($\|(-\mathcal{R})^{1/2}w\|$):** Using $w = \mathcal{A}_\varpi^\dagger g$ and the functional calculus for the non-positive self-adjoint operator Q_O,

$$\|(-\mathcal{R})^{1/2}w\| = \|(-\mathcal{R})^{1/2}\mathcal{A}_\varpi^\dagger g\| = \|(-\mathcal{R})^{-1/2}P_F\mathcal{V}_{\rm skew}P_S(\varpi - Q_O)^{-1}g\|. \tag{131}$$

Setting $\psi := (\varpi - Q_O)^{-1}g \in \mathcal{H}_S$, we note the identity (for such slow-induced fluxes)

$$g_\psi := P_F\mathcal{V}_{\rm skew}P_S\psi, \tag{132}$$

$$\|(-\mathcal{R})^{-1/2}g_\psi\|^2 = \langle g_\psi, -\mathcal{R}^+ g_\psi\rangle = -\langle \psi, Q_O\psi\rangle = \|(-Q_O)^{1/2}\psi\|^2, \tag{133}$$

hence

$$\|(-\mathcal{R})^{1/2}w\| = \|(-Q_O)^{1/2}(\varpi - Q_O)^{-1}g\| \le \frac{1}{2}\,\varpi^{-1/2}\,\|g\|. \tag{134}$$

(The last step follows from the scalar bound $\sup_{x\ge 0}\frac{\sqrt{x}}{\varpi+x} = \frac{1}{2\sqrt{\varpi}}$.)

Substituting these estimates into (129) yields

$$\|\mathcal{V}_{\rm pert}^\dagger\mathcal{A}_\varpi^\dagger g\| \le \Big((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\,\varpi^{-1/2}\,\|g\|. \tag{135}$$

Since g is arbitrary, this establishes (37b) with $\tilde{k}_P(\varpi) := \big((b_1+b_2)k_{\mathcal{A}} + \frac{a_1+a_2}{2}\big)\varpi^{-1/2}$.

□

Appendix C. Proof of Theorem 5.1

This appendix derives the dissipation matrix governing the hypocoercive convergence. The computation is based on the modified Lyapunov functional with resolvent shift $\varpi > 0$ and mixing weight $\alpha > 0$:

$$\mathcal{E}_{\alpha,\varpi}(f) := \frac{1}{2}\|f\|^2 + \alpha\Re\langle \mathcal{A}_\varpi f, f\rangle, \tag{136}$$

where $\mathcal{A}_\varpi = (\varpi - Q_O)^{-1}P_S\mathcal{V}_{\rm skew}(-\mathcal{R}^+)P_F$ is the DMS corrector associated with the reference effective dynamics Q_O.

C.1. **Preliminaries.** This appendix uses the structural estimates of Lemma 4.7 (solver bound, regularity, and perturbation control) and the notation of equations (44)–(48). We set $\varpi := \varpi_* = \lambda(Q_O)$ throughout and use the effective perturbation coefficient

$$b_{2,\rm eff} := b_2 + \frac{a_2}{2}(1+\lambda_R), \tag{137}$$

as defined in (44). We also record the energy-admissibility threshold

$$\eta_{\rm sym}(\gamma) := \begin{cases} +\infty, & a_2 = 0, \\ 2\gamma/a_2, & a_2 > 0, \end{cases} \tag{138}$$

which is exactly the range for which the coefficient of $\|(-\mathcal{R})^{1/2}f\|^2$ in the basic energy estimate below remains nonnegative.

C.2. **Derivation of Differential Estimates. Standing regularity for the Lyapunov computation.** In the bounded-operator setting, for any $f_0 \in \mathcal{H}$ the solution $f_t = e^{t\mathcal{L}}f_0$ is continuously differentiable and satisfies $\partial_t f_t = \mathcal{L}f_t$. In unbounded settings, we first take $f_0 \in \mathcal{C} \cap \mathcal{H}_0$ on the common invariant core from Assumption 4.3; then $f_t = e^{t\mathcal{L}}f_0$ remains in $\mathcal{C}$, is classically differentiable, and all identities below are legitimate on $\mathcal{C}$. The resulting differential inequality is then extended to arbitrary $f_0 \in \mathcal{H}_0$ by density together with the norm equivalence from Proposition 4.9 and continuity of the semigroup.

Lemma C.1 (Energy identity for the fast dissipator)**.** *Assume $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$. Then for all $f \in \mathrm{dom}(\mathcal{R})$,*

$$\Re\langle \mathcal{R}f, f\rangle = \langle \mathcal{R}f, f\rangle = -\|(-\mathcal{R})^{1/2}f\|^2. \tag{139}$$

In particular, if $f = f_S + f_F$ with $f_S \in \ker(\mathcal{R})$ and $f_F \in \mathcal{H}_F$, then

$$\Re\langle \mathcal{R}f, f\rangle = \Re\langle \mathcal{R}f_F, f_F\rangle. \tag{140}$$

Proof. Since $\mathcal{R}$ is self-adjoint and non-positive, $-\mathcal{R} \geq 0$ admits a functional calculus square root $(-\mathcal{R})^{1/2}$ and $\langle \mathcal{R}f, f\rangle = -\langle(-\mathcal{R})f, f\rangle = -\|(-\mathcal{R})^{1/2}f\|^2$. Moreover $\mathcal{R}f_S = 0$ and $\mathcal{R}f = \mathcal{R}f_F$. □

Let f_t denote such a classical solution. By differentiating (42) along the flow and using that $\mathcal{A}_\varpi$ is time-independent, we obtain

$$\frac{d}{dt}\mathcal{E}_{\alpha,\varpi}(f_t) = \Re\langle \partial_t f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi \partial_t f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f_t, \partial_t f_t\rangle \tag{141}$$

$$= \Re\langle \mathcal{L}f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{L}f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f_t, \mathcal{L}f_t\rangle. \tag{142}$$

We now split the right-hand side into the "plain" dissipation coming from the generator itself and the hypocoercive correction generated by the cross term:

$$\frac{d}{dt}\mathcal{E}_{\alpha,\varpi}(f_t) = \mathcal{D}_I(f_t) + \mathcal{D}_{\mathrm{mod}}(f_t), \tag{143}$$

where we define

$$\mathcal{D}_I(f) := \Re\langle \mathcal{L}f, f\rangle, \tag{144}$$

$$\mathcal{D}_{\mathrm{mod}}(f) := \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{L}f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{L}f\rangle. \tag{145}$$

Expanding $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ in $\mathcal{D}_I$ then yields the decomposition displayed below.

Natural dissipation ($\mathcal{D}_I$).

$$\mathcal{D}_I = \gamma\Re\langle \mathcal{R}f, f\rangle + \Re\langle \mathcal{V}_{\mathrm{skew}}f, f\rangle + \eta\Re\langle \mathcal{V}_{\mathrm{pert}}f, f\rangle. \tag{146}$$

Using Assumption 4.1.1 ($\mathcal{R}$ coercivity) and Assumption 4.1.4 (Perturbation decomposition $\mathcal{V}_{\mathrm{pert}} = \mathcal{K}_{\mathrm{pert}} + \mathcal{S}_{\mathrm{pert}}$): the fast dissipator contributes $\gamma\Re\langle \mathcal{R}f, f\rangle \leq -\gamma\lambda_R\|f_F\|^2$, while $\Re\langle \mathcal{V}_{\mathrm{skew}}f, f\rangle = 0$ and $\Re\langle \mathcal{K}_{\mathrm{pert}}f, f\rangle = 0$ by skew-adjointness. For the symmetric part, using (30) and Cauchy–Schwarz,

$$\eta\Re\langle \mathcal{S}_{\mathrm{pert}}f, f\rangle \leq \eta\|\mathcal{S}_{\mathrm{pert}}f\|\,\|f\| \leq \eta\Big(b_2\|f\|^2 + a_2\,\|(-\mathcal{R})^{1/2}f\|\,\|f\|\Big) \tag{147}$$

$$\leq \eta\Big(\big(b_2 + \tfrac{a_2}{2}\big)\|f\|^2 + \tfrac{a_2}{2}\,\|(-\mathcal{R})^{1/2}f\|^2\Big). \tag{148}$$

Combining with the dissipator contribution, note that by Lemma C.1, $\Re\langle \mathcal{R}f, f\rangle = \Re\langle \mathcal{R}f_F, f_F\rangle = -\|(-\mathcal{R})^{1/2}f\|^2$. If $\eta \leq \eta_{\mathrm{sym}}(\gamma)$, i.e. if $\gamma - \eta a_2/2 \geq 0$, then we may use the lower bound

$\|(-\mathcal{R})^{1/2}f\|^2 \geq \lambda_R\|f_F\|^2$ without changing the inequality direction. Hence

$$\gamma\Re\langle\mathcal{R}f,f\rangle + \eta\Re\langle\mathcal{S}_{\text{pert}}f,f\rangle \leq -\big(\gamma - \tfrac{\eta a_2}{2}\big)\,\|(-\mathcal{R})^{1/2}f\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f\|^2 \tag{149}$$

$$\leq -\big(\gamma\lambda_R - \eta b_{2,\text{eff}}\big)\,\|f_F\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f_S\|^2, \tag{150}$$

where we used $\|(-\mathcal{R})^{1/2}f\|^2 \geq \lambda_R\|f_F\|^2$, the decomposition $\|f\|^2 = \|f_S\|^2 + \|f_F\|^2$, and the fact that $\frac{a_2\lambda_R}{2} + (b_2 + \frac{a_2}{2}) = b_{2,\text{eff}}$ by definition. The fast mode carries the full effective coefficient $b_{2,\text{eff}}$, while the slow mode retains only the smaller $b_2 + \frac{a_2}{2}$ (since $f_S \in \ker(\mathcal{R})$ does not benefit from the λ_R-amplification).

Thus:

$$\mathcal{D}_I \leq -\big(\gamma\lambda_R - \eta b_{2,\text{eff}}\big)\,\|f_F\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f_S\|^2. \tag{151}$$

Hypocoercive correction ($\mathcal{D}_{\text{mod}}$). Recall

$$\mathcal{D}_{\text{mod}}(f) = \alpha\,\Re\langle\mathcal{A}_\varpi\mathcal{L}f,f\rangle + \alpha\,\Re\langle\mathcal{A}_\varpi f,\mathcal{L}f\rangle. \tag{152}$$

Using $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ and linearity, we regroup $\mathcal{D}_{\text{mod}}$ according to which part of the generator appears:

$$\mathcal{D}_{\text{mod}}(f) = T_{\mathcal{R}}(f) + T_{\mathcal{V}}(f) + T_{\mathcal{P}}(f), \tag{153}$$

where

$$T_{\mathcal{R}}(f) := \alpha\gamma\,\Re\langle\mathcal{A}_\varpi\mathcal{R}f,f\rangle + \alpha\gamma\,\Re\langle\mathcal{A}_\varpi f,\mathcal{R}f\rangle, \tag{154}$$

$$T_{\mathcal{V}}(f) := \alpha\,\Re\langle\mathcal{A}_\varpi\mathcal{V}_{\text{skew}}f,f\rangle + \alpha\,\Re\langle\mathcal{A}_\varpi f,\mathcal{V}_{\text{skew}}f\rangle, \tag{155}$$

$$T_{\mathcal{P}}(f) := \alpha\eta\,\Re\langle\mathcal{A}_\varpi\mathcal{V}_{\text{pert}}f,f\rangle + \alpha\eta\,\Re\langle\mathcal{A}_\varpi f,\mathcal{V}_{\text{pert}}f\rangle. \tag{156}$$

Interaction with the fast dissipator. We start from the regrouped definition

$$T_{\mathcal{R}}(f) = \alpha\gamma\,\Re\langle\mathcal{A}_\varpi\mathcal{R}f,f\rangle + \alpha\gamma\,\Re\langle\mathcal{A}_\varpi f,\mathcal{R}f\rangle. \tag{157}$$

Since $\mathcal{R}f = \mathcal{R}f_F$ and $\mathcal{A}_\varpi f \in \mathcal{H}_S = \ker(\mathcal{R})$, the second term vanishes by self-adjointness of $\mathcal{R}$:

$$\Re\langle\mathcal{A}_\varpi f,\mathcal{R}f\rangle = \Re\langle\mathcal{R}\mathcal{A}_\varpi f,f\rangle = 0. \tag{158}$$

Moreover $\mathcal{A}_\varpi\mathcal{R}f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$, hence

$$T_{\mathcal{R}}(f) = \alpha\gamma\,\Re\langle\mathcal{A}_\varpi\mathcal{R}f_F,f_S\rangle. \tag{159}$$

Using $\mathcal{A}_\varpi = (\varpi - Q_O)^{-1}P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F$ and the pseudoinverse identity $\mathcal{R}^+\mathcal{R} = P_F$ on $\text{dom}(\mathcal{R})$, we obtain

$$\mathcal{A}_\varpi\mathcal{R}f_F = (\varpi - Q_O)^{-1}P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)\mathcal{R}f_F = -(\varpi - Q_O)^{-1}P_S\mathcal{V}_{\text{skew}}f_F. \tag{160}$$

Therefore, by skew-adjointness of $\mathcal{V}_{\text{skew}}$,

$$|T_{\mathcal{R}}| = \alpha\gamma\,\big|\Re\langle(\varpi - Q_O)^{-1}P_S\mathcal{V}_{\text{skew}}f_F,f_S\rangle\big| = \alpha\gamma\,\big|\Re\langle f_F,\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1}f_S\rangle\big| \tag{161}$$

$$\leq \alpha\gamma\,\|f_F\|\,\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1}f_S\|. \tag{162}$$

To bound $\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1}f_S\|$, set $\psi := (\varpi - Q_O)^{-1}f_S \in \text{dom}(Q_O) \subset \mathcal{H}_S$ and denote its induced microscopic flux

$$g_\psi := P_F\mathcal{V}_{\text{skew}}P_S\psi \in \mathcal{H}_F. \tag{163}$$

Since $\psi \in \mathcal{H}_S$, the centered-coupling assumption gives $P_S\mathcal{V}_{\text{skew}}\psi = 0$, hence $g_\psi = \mathcal{V}_{\text{skew}}\psi = \mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1}f_S$. Applying the flux conditioning bound (24) yields

$$\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1}f_S\|^2 = \|g_\psi\|^2 \leq M_R\,\langle g_\psi,-\mathcal{R}^+g_\psi\rangle. \tag{164}$$

The "extra Q_O" comes from rewriting the right-hand side using the definition $Q_O := -P_S\mathcal{V}_{\rm skew}\mathcal{R}^+\mathcal{V}_{\rm skew}P_S$: indeed, for $\psi \in \mathcal{H}_S$,

$$\langle g_\psi, -\mathcal{R}^+ g_\psi\rangle = \langle \mathcal{V}_{\rm skew}\psi, (-\mathcal{R}^+)\mathcal{V}_{\rm skew}\psi\rangle = \langle \psi, (-Q_O)\psi\rangle = \|(-Q_O)^{1/2}\psi\|^2. \tag{165}$$

Therefore

$$\|\mathcal{V}_{\rm skew}(\varpi - Q_O)^{-1} f_S\| \le \sqrt{M_R}\,\|(-Q_O)^{1/2}(\varpi - Q_O)^{-1} f_S\|. \tag{166}$$

Finally, by functional calculus for the self-adjoint operator $-Q_O \ge 0$,

$$\|(-Q_O)^{1/2}(\varpi - Q_O)^{-1}\| = \sup_{x\ge 0} \frac{\sqrt{x}}{\varpi + x} = \frac{1}{2\sqrt{\varpi}}, \tag{167}$$

which implies

$$\|\mathcal{V}_{\rm skew}(\varpi - Q_O)^{-1} f_S\| \le \frac{\sqrt{M_R}}{2}\,\varpi^{-1/2}\,\|f_S\|. \tag{168}$$

Hence

$$|T_{\mathcal{R}}| \le \alpha\gamma\,\frac{\sqrt{M_R}}{2}\,\varpi^{-1/2}\,\|f_F\|\,\|f_S\|. \tag{169}$$

Interaction with transport. Starting from the regrouped definition

$$T_{\mathcal{V}}(f) = \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\rm skew} f\rangle, \tag{170}$$

we decompose $f = f_S + f_F$ and use that $\mathcal{A}_\varpi$ maps into the slow space and annihilates the slow component: $\mathcal{A}_\varpi = \mathcal{A}_\varpi P_F$ (since P_F is on the right in (34)). Hence $\mathcal{A}_\varpi f = \mathcal{A}_\varpi f_F$. Expanding the *first* bracket gives

$$\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f, f\rangle = \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle \tag{171}$$

$$+ \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_F\rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_F\rangle. \tag{172}$$

The last term vanishes because $\mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$. Moreover, since $\mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S \in \mathcal{H}_S$, we have $\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_F\rangle = 0$. Therefore

$$\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f, f\rangle = \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle. \tag{173}$$

In the *second* bracket we use $\mathcal{A}_\varpi f = \mathcal{A}_\varpi f_F$ and expand $\mathcal{V}_{\rm skew} f$:

$$\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\rm skew} f\rangle = \Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_S\rangle + \Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_F\rangle. \tag{174}$$

Putting the pieces together, we obtain the explicit decomposition

$$T_{\mathcal{V}} = \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle \tag{175}$$

$$+ \alpha\,\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_S\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_F\rangle. \tag{176}$$

Main Coercivity. Since $f_S \in \mathcal{H}_S$ and $P_S\mathcal{V}_{\rm skew}P_S = 0$, we have $P_F\mathcal{V}_{\rm skew}f_S = \mathcal{V}_{\rm skew}f_S$. By the definition of $\mathcal{A}_\varpi$ and Q_O this yields

$$\alpha\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle = \alpha\langle (\varpi - Q_O)^{-1} Q_O f_S, f_S\rangle = -\alpha\,\langle (\varpi - Q_O)^{-1}(-Q_O) f_S, f_S\rangle \tag{177}$$

$$\le -\alpha\,\frac{\lambda(Q_O)}{\varpi + \lambda(Q_O)}\,\|f_S\|^2. \tag{178}$$

Here we used that $f \in \mathcal{H}_0$ implies $f_S = P_S f \in \mathcal{H}_S \cap \mathcal{H}_0 = \mathcal{H}_{S,0}$, so Assumption 4.2 applies to f_S.
Cross Terms. Since $\mathcal{A}_\varpi f_F \in \mathcal{H}_S$ (by the range of the corrector) and $\mathcal{V}_{\rm skew} f_S \in \mathcal{H}_F$ (by centering, $P_S\mathcal{V}_{\rm skew}P_S = 0$), the third inner product vanishes: $\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_S\rangle = 0$. The only surviving cross term is bounded by the dual regularity estimate:

$$\alpha\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle \le \alpha\|\mathcal{A}_\varpi \mathcal{V}_{\rm skew} P_F\|\|f_F\|\|f_S\| \le \alpha\kappa_{\rm ell}\|f_F\|\|f_S\|. \tag{179}$$

Fast–fast remainder. Using $\mathcal{V}_{\text{skew}}^{\dagger} = -\mathcal{V}_{\text{skew}}$ and Cauchy–Schwarz,

$$\left|\alpha \Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\text{skew}} f_F\rangle\right| = \alpha\left|\Re\langle -\mathcal{V}_{\text{skew}}\mathcal{A}_\varpi f_F, f_F\rangle\right| \le \alpha \left\|\mathcal{V}_{\text{skew}}\mathcal{A}_\varpi\right\| \left\|f_F\right\|^2 \le \alpha\, k_{\text{reg}} \left\|f_F\right\|^2. \tag{180}$$

Interaction with perturbation.

$$T_{\mathcal{P}} = \alpha\eta \Re\langle \mathcal{V}_{\text{pert}}^{\dagger}\mathcal{A}_\varpi f, f\rangle + \alpha\eta\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{pert}} f, f\rangle. \tag{181}$$

We estimate each term using Lemma 4.7 and the centered mapping property (40).
Reverse term. Since $\mathcal{A}_\varpi f \in \mathcal{H}_S$ and $\mathcal{V}_{\text{pert}}^{\dagger}\mathcal{H}_S \subset \mathcal{H}_F$, we have $\langle \mathcal{V}_{\text{pert}}^{\dagger}\mathcal{A}_\varpi f, f\rangle = \langle \mathcal{V}_{\text{pert}}^{\dagger}\mathcal{A}_\varpi f, f_F\rangle$. Therefore,

$$\left|\alpha\eta\Re\langle \mathcal{V}_{\text{pert}}^{\dagger}\mathcal{A}_\varpi f, f\rangle\right| \le \alpha\eta \left\|\mathcal{V}_{\text{pert}}^{\dagger}\mathcal{A}_\varpi f\right\| \left\|f_F\right\| \le \alpha\eta\, k_P(\varpi) \left\|f\right\| \left\|f_F\right\| \tag{182}$$

$$\le \alpha\eta\, k_P(\varpi)\left(\left\|f_F\right\|^2 + \left\|f_F\right\| \left\|f_S\right\|\right). \tag{183}$$

Forward term. Since $\mathcal{A}_\varpi \mathcal{H} \subset \mathcal{H}_S$, we have $\langle \mathcal{A}_\varpi \mathcal{V}_{\text{pert}} f, f\rangle = \langle \mathcal{A}_\varpi \mathcal{V}_{\text{pert}} f, f_S\rangle$. Therefore,

$$\left|\alpha\eta\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{pert}} f, f\rangle\right| \le \alpha\eta \left\|\mathcal{A}_\varpi \mathcal{V}_{\text{pert}} f\right\| \left\|f_S\right\| \le \alpha\eta\, \tilde{k}_P(\varpi) \left\|f\right\| \left\|f_S\right\| \tag{184}$$

$$\le \alpha\eta\, \tilde{k}_P(\varpi)\left(\left\|f_S\right\|^2 + \left\|f_F\right\| \left\|f_S\right\|\right). \tag{185}$$

C.3. **Dissipation Matrix Construction.** Summing all estimates and using Cauchy–Schwarz bounds of the form $|\langle X f_F, f_S\rangle| \le \|X\| \, \|f_F\| \, \|f_S\|$, we obtain a closed differential inequality of the form

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\varpi}(f_t) \le -\mathbf{v}(t)^T \mathbf{M}\, \mathbf{v}(t), \qquad \mathbf{v}(t) := \begin{pmatrix} \|P_F f_t\| \\ \|P_S f_t\| \end{pmatrix}. \tag{186}$$

Here $\mathbf{M}$ is a real symmetric 2×2 matrix whose entries depend only on the structural constants and the chosen parameters (α, ϖ). The entries of $\mathbf{M}$ are explicitly decomposed by powers of ϖ below.

Fast-Fast Block (M_{11}): From Natural Dissipation, the fast–fast transport remainder, and Reverse Perturbation.

$$\begin{aligned} M_{11} &= \big(\gamma\lambda_R - \eta b_{2,\text{eff}}\big) - \alpha k_{\text{reg}} - \alpha\eta k_P(\varpi) \\ &= \gamma\lambda_R - \eta b_{2,\text{eff}} - \alpha k_{\text{reg}} - \alpha\eta(b_1+b_2)k_{\mathcal{A}}\,\varpi^{-1/2}. \end{aligned} \tag{187}$$

Slow-Slow Block (M_{22}): From Main Coercivity, Natural Dissipation (gain), and both Perturbations.

$$\begin{aligned} M_{22} &= \alpha\frac{\lambda(Q_O)}{\varpi + \lambda(Q_O)} - \eta\big(b_2 + \tfrac{a_2}{2}\big) - \alpha\eta\,\tilde{k}_P(\varpi) \\ &= \alpha\frac{\lambda(Q_O)}{\varpi + \lambda(Q_O)} - \eta\big(b_2 + \tfrac{a_2}{2}\big) - \alpha\eta\left[\Big((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\varpi^{-1/2}\right]. \end{aligned} \tag{188}$$

Cross Block (M_{12}): From Mixing, Transport Cross terms, and Forward Perturbation.

$$\begin{aligned} M_{12} &= -\frac{1}{2}\alpha\left[\gamma\frac{\sqrt{M_R}}{2}\,\varpi^{-1/2} + \kappa_{\text{ell}} + \eta\big(k_P(\varpi) + \tilde{k}_P(\varpi)\big)\right] \\ &= -\frac{1}{2}\alpha\left[\kappa_{\text{ell}} + \left(\gamma\frac{\sqrt{M_R}}{2} + \eta\Big(2(b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\right)\varpi^{-1/2}\right]. \end{aligned} \tag{189}$$

This explicit form allows for direct asymptotic analysis by choosing $\varpi \sim \lambda(Q_O)$, confirming the stability for small η.

C.4. **Optimization of the Decay Rate.** We work in the unperturbed regime $\eta = 0$ to identify the dominant hypocoercive rate. Let $\mathbf{M}_0(\alpha)$ denote the dissipation matrix at $\eta = 0$. Since $\mathbf{M}_0$ is 2×2 symmetric, its smallest eigenvalue satisfies the determinant–trace bound

$$\lambda_{\min}(\mathbf{M}_0) \geq \frac{\det(\mathbf{M}_0)}{\mathrm{Tr}(\mathbf{M}_0)}. \tag{190}$$

Choice of resolvent shift. The slow-coercivity contribution in M^0_{22} is $\alpha\lambda(Q_O)/(\varpi+\lambda(Q_O))$, while the cross term $|M^0_{12}|$ contains $\gamma\Theta\varpi^{-1/2}$. A leading-order balance (maximising $\varpi(\varpi + \lambda(Q_O))^{-2}$ in the large-γ rate) motivates the gap-adapted choice

$$\varpi_* := \lambda(Q_O). \tag{191}$$

We do not claim here a separate global optimisation theorem over all admissible shifts; rather, ϖ_* is the canonical gap-scale choice that balances the slow coercive block against the cross term inside the determinant–trace estimate used in this paper. At this value, $\lambda(Q_O)/(\varpi_* + \lambda(Q_O)) = \frac{1}{2}$. Using the constants $\Theta, K_{\mathrm{geom}}, Z(\gamma)$ from (44)–(45), the unperturbed matrix entries at $\varpi = \varpi_*$ are:

$$M^0_{11} = \gamma\lambda_R - \alpha k_{\mathrm{reg}}, \tag{192}$$

$$M^0_{22} = \tfrac{1}{2}\,\alpha, \tag{193}$$

$$M^0_{12} = -\tfrac{1}{2}\,\alpha\, Z(\gamma). \tag{194}$$

Optimal mixing weight. The determinant $D(\alpha) := M^0_{11}M^0_{22} - (M^0_{12})^2$ is a concave quadratic in α:

$$D(\alpha) = \tfrac{1}{2}\gamma\lambda_R\,\alpha - \big(\tfrac{1}{2}k_{\mathrm{reg}} + \tfrac{1}{4}Z(\gamma)^2\big)\alpha^2. \tag{195}$$

It is maximised at

$$\alpha_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\mathrm{reg}}}, \tag{196}$$

in agreement with (46), giving

$$D(\alpha_*) = \frac{\frac{1}{4}(\gamma\lambda_R)^2}{Z(\gamma)^2 + 2k_{\mathrm{reg}}}. \tag{197}$$

Trace upper bound and rate estimate. The exact trace is $\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*} = (\gamma\lambda_R - \alpha_* k_{\mathrm{reg}}) + \frac{1}{2}\alpha_* = \gamma\lambda_R - \alpha_*(k_{\mathrm{reg}} - \frac{1}{2})$. Since $k_{\mathrm{reg}} = \sqrt{M_R/\lambda_R} \geq 1$ (the flux conditioning bound gives $M_R \geq \lambda_R$), we have $k_{\mathrm{reg}} - \frac{1}{2} \geq \frac{1}{2} > 0$, hence

$$\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*} \leq \gamma\lambda_R. \tag{198}$$

Applying (190) yields

$$\lambda_{\min}(\mathbf{M}_0) \geq \frac{D(\alpha_*)}{\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*}} \geq \frac{\frac{1}{4}\,\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\mathrm{reg}}} =: \Lambda_{\mathrm{hypo}}(\gamma), \tag{199}$$

consistent with definition (47). This provides a rigorous lower bound on the unperturbed decay rate for all $\gamma > 0$.

Asymptotic scaling analysis. Equation (199) reveals the precise scaling behavior across regimes without relying on ad-hoc approximations.
Overdamped Limit ($\gamma \to \infty$). The denominator is dominated by the $\mathcal{O}(\gamma^2)$ term, giving

$$\lim_{\gamma\to\infty} \gamma\Lambda_{\mathrm{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R\,\lambda(Q_O)}{\Theta^2} = \lambda(Q_O)\,\frac{\lambda_R}{M_R}. \tag{200}$$

Thus, $\Lambda_{\mathrm{hypo}}(\gamma) = \lambda(Q_O)\,\frac{\lambda_R}{M_R}\,\gamma^{-1} + \mathcal{O}(\gamma^{-2})$. In particular, $\Lambda_{\mathrm{hypo}}(\gamma) \asymp \lambda(Q_O)\,\gamma^{-1}$.

Underdamped/Intermediate Regime. For small or order-one γ, the denominator is controlled by the γ-independent structural constants K_{geom} and $2k_{\text{reg}}$. These arise from the slow–fast coupling through $\mathcal{V}_{\text{skew}}$ and prevent the estimate from degenerating even though $\mathcal{R}$ does not dissipate on $\mathcal{H}_S$ directly. The transport mixes slow directions into fast dissipative directions, yielding $\Lambda_{\text{hypo}}(\gamma) \asymp \gamma$ for small γ.

We now establish the stability condition by computing the operator norm of the perturbation matrix $\mathbf{E}(\gamma)$ as a function of γ, tracking the dependence on the macroscopic spectral gap $\varpi_* = \lambda(Q_O)$.

Explicit norm of the perturbation matrix. Recalling that $\mathbf{M} = \mathbf{M}_0 - \eta\mathbf{E}$, the perturbation matrix $\mathbf{E}$ has entries read off from the full $\mathbf{M}$ computation:

$$E_{11} = b_{2,\text{eff}} + \alpha_* \, k_P(\varpi_*), \tag{201}$$

$$E_{22} = \big(b_2 + \tfrac{a_2}{2}\big) + \alpha_* \, \tilde{k}_P(\varpi_*), \tag{202}$$

$$E_{12} = \frac{1}{2}\, \alpha_* \, \big(k_P(\varpi_*) + \tilde{k}_P(\varpi_*)\big). \tag{203}$$

Here E_{11} contains the full effective coefficient $b_{2,\text{eff}} = b_2 + \frac{a_2}{2}(1 + \lambda_R)$, while E_{22} contains only the slow-mode coefficient $b_2 + \frac{a_2}{2}$; the extra $\frac{a_2\lambda_R}{2}$ absorbed into $b_{2,\text{eff}}$ arises in the fast block from the bound $\|(-\mathcal{R})^{1/2} f\|^2 \geq \lambda_R \|f_F\|^2$ applied to the symmetric perturbation in $\mathcal{D}_I$ (see (151)). Since $\mathbf{E}$ is symmetric, we bound its spectral norm by the maximal absolute row sum (Gershgorin):

$$\|\mathbf{E}(\gamma)\| \leq \max\{E_{11} + |E_{12}|,\, E_{22} + |E_{12}|\}. \tag{204}$$

Using $b_{2,\text{eff}} \geq b_2 + \frac{a_2}{2}$ and $k_P(\varpi_*) \leq \tilde{k}_P(\varpi_*)$, both row sums admit the same uniform upper bound:

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma) \left[\tfrac{3}{2}\, \tilde{k}_P(\varpi_*) + \tfrac{1}{2}\, \tilde{k}_P(\varpi_*)\right] = b_{2,\text{eff}} + 2\alpha_*(\gamma)\, \tilde{k}_P(\varpi_*). \tag{205}$$

Recall from Lemma 4.7 that the bounds k_P and $\tilde{k}_P$ depend on $\varpi_*^{-1/2}$. We define the *strictly constant* structural coefficients (independent of ϖ and γ):

$$C_{\text{sing}} := 2(b_1 + b_2)k_{\mathcal{A}} + (a_1 + a_2). \tag{206}$$

Indeed,

$$2\tilde{k}_P(\varpi_*) = 2\Big((b_1 + b_2)k_{\mathcal{A}} + \frac{a_1 + a_2}{2}\Big)\varpi_*^{-1/2} = C_{\text{sing}}\, \varpi_*^{-1/2},$$

so the bound (205) becomes

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma)\, C_{\text{sing}}\, \varpi_*^{-1/2}. \tag{207}$$

Substituting $\alpha_*(\gamma)$ from (46), the bound becomes

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma)\, C_{\text{sing}}\, \lambda(Q_O)^{-1/2} = B_{\text{pert}}(\gamma), \tag{208}$$

consistent with definition (48).

Friction-dependent threshold. The stability condition $\lambda_{\min}(\mathbf{M}) > 0$ is ensured by $\eta\|\mathbf{E}(\gamma)\| \leq \frac{1}{2}\Lambda_{\text{hypo}}(\gamma)$, which yields the exact matrix threshold

$$\eta^\sharp(\gamma) := \frac{\Lambda_{\text{hypo}}(\gamma)}{2\|\mathbf{E}(\gamma)\|}. \tag{209}$$

Since $\|\mathbf{E}(\gamma)\| \leq B_{\text{pert}}(\gamma)$, the explicit certified threshold used in Theorem 5.1 is

$$\eta_{\text{mat}}(\gamma) := \frac{\Lambda_{\text{hypo}}(\gamma)}{2B_{\text{pert}}(\gamma)} \leq \eta^\sharp(\gamma). \tag{210}$$

Together with the energy-admissibility condition $\eta \le \eta_{\rm sym}(\gamma)$ coming from (151), this gives the theorem threshold

$$\eta^*(\gamma) := \min\{\eta_{\rm sym}(\gamma), \eta_{\rm mat}(\gamma)\}. \tag{211}$$

Substituting the explicit forms of $\Lambda_{\rm hypo}$ (47) and $B_{\rm pert}$ (48) yields

$$\eta_{\rm mat}(\gamma) = \frac{\frac{1}{4}\,\gamma\lambda_R}{\big(Z(\gamma)^2 + 2k_{\rm reg}\big)\big(2b_{2,\rm eff} + 2\alpha_*(\gamma)\, C_{\rm sing}\, \lambda(Q_O)^{-1/2}\big)}. \tag{212}$$

Asymptotic analysis of the certified threshold. We analyze first the scaling of $\eta_{\rm mat}(\gamma)$ in the limiting regimes. Note that $Z(\gamma) \asymp \gamma$ as $\gamma \to \infty$.
Overdamped Limit ($\gamma \to \infty$). The numerator scales as $\mathcal{O}(\gamma)$. In the denominator, the term $2b_{2,\rm eff} Z(\gamma)^2$ scales as $\mathcal{O}(\gamma^2)$, while the skew-term scales as $\mathcal{O}(\gamma)$. If $b_{2,\rm eff} > 0$ (symmetric perturbation present), the $\mathcal{O}(\gamma^2)$ term dominates and $\eta_{\rm mat}(\gamma) \asymp \gamma^{-1}$. If $b_{2,\rm eff} = 0$ (purely skew perturbation, i.e. $b_2 = a_2 = 0$), the denominator is dominated by the skew term $\mathcal{O}(\gamma)$ and $\eta_{\rm mat}(\gamma) \asymp \mathcal{O}(1)$.
Optimal Lift ($\gamma \sim \sqrt{\lambda(Q_O)}$). At the optimal friction $\gamma_{\rm opt}$, determined in the optimization step below, we have $Z(\gamma_{\rm opt}) = \mathcal{O}(1)$. Consider the singular term in the denominator:

$$\gamma_{\rm opt} \cdot C_{\rm sing}\lambda(Q_O)^{-1/2} \asymp \sqrt{\lambda(Q_O)} \cdot \lambda(Q_O)^{-1/2} \asymp \mathcal{O}(1). \tag{213}$$

The singularity thus cancels, and at the optimal friction the matrix-controlled threshold is well-behaved:

$$\eta_{\rm mat}(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}. \tag{214}$$

Since $\eta_{\rm sym}(\gamma_{\rm opt}) = +\infty$ when $a_2 = 0$ and $\eta_{\rm sym}(\gamma_{\rm opt}) = 2\gamma_{\rm opt}/a_2 \asymp \sqrt{\lambda(Q_O)}$ when $a_2 > 0$, the overall threshold $\eta^*(\gamma_{\rm opt}) = \min\{\eta_{\rm sym}(\gamma_{\rm opt}), \eta_{\rm mat}(\gamma_{\rm opt})\}$ has the same $\sqrt{\lambda(Q_O)}$ scaling.

From coercivity to global decay. Having established the positivity of the dissipation matrix, we convert the algebraic estimate into a global semigroup decay bound. This step involves explicit algebraic substitution to derive the final decay rate, ensuring that the singular scaling of the perturbation is correctly propagated.

Uniform norm equivalence. Recall the distortion parameter

$$\delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|. \tag{215}$$

Using $\|\mathcal{A}_{\varpi_*}\| \le k_{\mathcal{A}}\, \varpi_*^{-1/2}$ with $k_{\mathcal{A}} = (2\sqrt{\lambda_R})^{-1}$ and the explicit weight $\alpha_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2+2k_{\rm reg}}$, we obtain

$$\delta_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\rm reg}}\,\frac{1}{\sqrt{\lambda_R\,\varpi_*}} = \frac{\gamma\sqrt{\lambda_R}\,\varpi_*^{-1/2}}{\big(\gamma\Theta\varpi_*^{-1/2} + K_{\rm geom}\big)^2 + 2k_{\rm reg}}. \tag{216}$$

Setting $x := \gamma\Theta\varpi_*^{-1/2}$ gives

$$\delta_*(\gamma) = \frac{\sqrt{\lambda_R}}{\Theta} \cdot \frac{x}{(x + K_{\rm geom})^2 + 2k_{\rm reg}}. \tag{217}$$

The elementary optimisation

$$\sup_{x>0} \frac{x}{(x + K_{\rm geom})^2 + 2k_{\rm reg}} = \frac{1}{2\big(K_{\rm geom} + \sqrt{K_{\rm geom}^2 + 2k_{\rm reg}}\big)} \tag{218}$$

together with $M_R \geq \lambda_R$ and $\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}}} \geq \sqrt{2k_{\text{reg}}} \geq \sqrt{2}$ (the latter since $k_{\text{reg}} = \sqrt{M_R/\lambda_R} \geq 1$) therefore yields

$$\delta_*(\gamma) \leq \frac{\sqrt{\lambda_R}}{\sqrt{M_R}\big(K_{\text{geom}} + \sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}}}\big)} \leq 2^{-1/2}, \tag{219}$$

and therefore

$$\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\| = \frac{\delta_*(\gamma)}{2} \leq 2^{-3/2} < \frac{1}{2}. \tag{220}$$

Thus Proposition 4.9 applies with $(\alpha, \varpi) = (\alpha_*, \varpi_*)$. Hence the norm-equivalence prefactor satisfies

$$C_*(\gamma) := \sqrt{\frac{1 + \delta_*(\gamma)}{1 - \delta_*(\gamma)}} \leq 1 + \sqrt{2}. \tag{221}$$

Thus, for all f,

$$\frac{1}{2}(1 - \delta_*(\gamma))\|f\|^2 \leq \mathcal{E}_{\alpha_*,\varpi_*}(f) \leq \frac{1}{2}(1 + \delta_*(\gamma))\|f\|^2. \tag{222}$$

Explicit calculation of the decay rate. Set

$$c(\gamma, \eta) := \Lambda_{\text{hypo}}(\gamma) - \eta\, \|\mathbf{E}(\gamma)\|. \tag{223}$$

The algebraic dissipation estimate reads

$$\frac{d}{dt}\, \mathcal{E}_{\alpha_*,\varpi_*}(f_t) \leq -c(\gamma, \eta)\, \|f_t\|^2. \tag{224}$$

Next we use the norm equivalence from Proposition 4.9 in the convenient form

$$\frac{1}{2}(1 - \delta_*(\gamma))\, \|f\|^2 \leq \mathcal{E}_{\alpha_*,\varpi_*}(f) \leq \frac{1}{2}(1 + \delta_*(\gamma))\, \|f\|^2, \tag{225}$$

where $\delta_*(\gamma) := 2\alpha_*(\gamma)\, \|\mathcal{A}_{\varpi_*}\| \in (0, 1)$. The upper bound implies

$$\|f\|^2 \geq \frac{2}{1 + \delta_*(\gamma)}\, \mathcal{E}_{\alpha_*,\varpi_*}(f). \tag{226}$$

Substituting this into the dissipation inequality gives a closed differential inequality for the Lyapunov functional:

$$\frac{d}{dt}\, \mathcal{E}_{\alpha_*,\varpi_*}(f_t) \leq -\frac{2c(\gamma, \eta)}{1 + \delta_*(\gamma)}\, \mathcal{E}_{\alpha_*,\varpi_*}(f_t). \tag{227}$$

By Grönwall's inequality,

$$\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \leq \exp\Big(-\frac{2c(\gamma, \eta)}{1 + \delta_*(\gamma)}\, t\Big)\, \mathcal{E}_{\alpha_*,\varpi_*}(f_0). \tag{228}$$

Finally, converting back to the L^2 norm using the two-sided equivalence yields

$$\|f_t\|^2 \leq \frac{2}{1 - \delta_*(\gamma)}\, \mathcal{E}_{\alpha_*,\varpi_*}(f_t) \leq \frac{1 + \delta_*(\gamma)}{1 - \delta_*(\gamma)}\, e^{-\frac{2c(\gamma,\eta)}{1+\delta_*(\gamma)}t}\, \|f_0\|^2. \tag{229}$$

Therefore

$$\|f_t\| \leq \sqrt{\frac{1 + \delta_*(\gamma)}{1 - \delta_*(\gamma)}}\, e^{-\lambda_\eta t}\, \|f_0\|, \qquad \lambda_\eta := \frac{c(\gamma, \eta)}{1 + \delta_*(\gamma)}. \tag{230}$$

We now substitute the explicit expressions derived in the previous steps. For the unperturbed rate:

$$\Lambda_{\text{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R\, \gamma}{Z(\gamma)^2 + 2k_{\text{reg}}}, \quad \text{with } Z(\gamma) = \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}. \tag{231}$$

For the perturbation norm, we use the sharp bound (207) at $\varpi_* = \lambda(Q_O)$ to handle the macroscopic singularity:

$$\|\mathbf{E}(\gamma)\| \le b_{2,\text{eff}} + \underbrace{\frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}}_{\alpha_*(\gamma)} C_{\text{sing}}\, \lambda(Q_O)^{-1/2}. \tag{232}$$

Substituting these into the expression for λ_η and using the explicit distortion bound on $\delta_*(\gamma)$:

$$\lambda_\eta \ge \frac{\frac{1}{4}\lambda_R\,\gamma}{(1+\delta_*(\gamma))(Z(\gamma)^2 + 2k_{\text{reg}})} - \frac{\eta}{1+\delta_*(\gamma)}\left(b_{2,\text{eff}} + \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}} C_{\text{sing}}\, \lambda(Q_O)^{-1/2}\right). \tag{233}$$

Equivalently,

$$\lambda_\eta \ge \frac{\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)}{1+\delta_*(\gamma)} \ge (2-\sqrt{2})\big(\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)\big), \tag{234}$$

using $1+\delta_*(\gamma) \le 1 + 2^{-1/2}$, i.e. $(1+\delta_*(\gamma))^{-1} \ge (1+2^{-1/2})^{-1} = 2-\sqrt{2}$. This is the rigorous lower bound. The potentially singular factor $C_{\text{sing}}\lambda(Q_O)^{-1/2}$ is multiplied by the mixing weight $\alpha_*(\gamma) = \gamma\lambda_R/(Z(\gamma)^2 + 2k_{\text{reg}})$, which scales as $\sqrt{\lambda(Q_O)}$ at the optimal lift, ensuring the perturbation contribution remains bounded.

Finally, converting back to the L^2 norm via Grönwall's lemma yields the global estimate:

$$\|e^{t\mathcal{L}} f\|_{\mathcal{H}} \le C_*(\gamma)\, e^{-\lambda_\eta t}\|f\|_{\mathcal{H}} \le (1+\sqrt{2})\, e^{-\lambda_\eta t}\|f\|_{\mathcal{H}}. \tag{235}$$

This completes the proof.

C.5. **Optimization over Friction γ: Kinetic Acceleration.** We determine γ_{opt} maximising $\Lambda_{\text{hypo}}(\gamma)$. From (199), setting $x := \Theta\lambda(Q_O)^{-1/2}\gamma$, the rate function reads

$$\Lambda_{\text{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R}{\Theta}\sqrt{\lambda(Q_O)} \cdot \frac{x}{(x+K_{\text{geom}})^2 + 2k_{\text{reg}}}. \tag{236}$$

Optimal friction. The function $x \mapsto x/[(x+K_{\text{geom}})^2 + C_0]$ with $C_0 := 2k_{\text{reg}}$ is maximised when $(x+K_{\text{geom}})^2 + C_0 = 2x(x+K_{\text{geom}})$, equivalently $x^2 = K_{\text{geom}}^2 + C_0$. Translating back:

$$\gamma_{\text{opt}} = \frac{\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}}}}{\Theta}\sqrt{\lambda(Q_O)} =: C_{\text{lift}}\sqrt{\lambda(Q_O)}, \tag{237}$$

where $C_{\text{lift}} := \Theta^{-1}\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}}}$.

Maximal rate. At $\gamma = \gamma_{\text{opt}}$ one has $x_* := \Theta\lambda(Q_O)^{-1/2}\gamma_{\text{opt}} = \sqrt{K_{\text{geom}}^2 + C_0}$. Substituting into the denominator: $(x_* + K_{\text{geom}})^2 + C_0 = 2(K_{\text{geom}}^2 + C_0 + K_{\text{geom}}x_*)$. Hence

$$\Lambda_{\max} := \Lambda_{\text{hypo}}(\gamma_{\text{opt}}) = \frac{\frac{1}{4}\lambda_R}{2\Theta(x_* + K_{\text{geom}})}\sqrt{\lambda(Q_O)} \asymp \sqrt{\lambda(Q_O)}. \tag{238}$$

This confirms the kinetic acceleration: $\Lambda_{\max} \asymp \sqrt{\lambda(Q_O)}$, compared to the overdamped ceiling $\Lambda_{\text{hypo}}(\gamma) \asymp \lambda(Q_O)/\gamma$ as $\gamma \to \infty$.

Certified threshold at optimal lift. The admissible perturbation range certified by Theorem 5.1 is determined as follows. From (209), the exact matrix threshold depends on $\|\mathbf{E}(\gamma)\|$, while the theorem uses the explicit upper bound $B_{\text{pert}}(\gamma)$. The dominant term involves the mixing weight $\alpha_*(\gamma)$. Evaluating at γ_{opt}:

$$\alpha_*(\gamma_{\text{opt}}) = \frac{\gamma_{\text{opt}}\lambda_R}{Z(\gamma_{\text{opt}})^2 + 2k_{\text{reg}}}. \tag{239}$$

Using $\gamma_{\rm opt} = C_{\rm lift}\sqrt{\lambda(Q_O)}$ and $Z(\gamma_{\rm opt}) = \sqrt{K_{\rm geom}^2 + 2k_{\rm reg}} + K_{\rm geom} = \mathcal{O}(1)$, the numerator of $\alpha_*(\gamma_{\rm opt})$ is $\gamma_{\rm opt}\lambda_R \asymp \sqrt{\lambda(Q_O)}$ and the denominator $Z(\gamma_{\rm opt})^2 + 2k_{\rm reg} = \mathcal{O}(1)$ (independent of $\lambda(Q_O)$). Hence $\alpha_*(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}$, so $\alpha_*(\gamma_{\rm opt}) \cdot \lambda(Q_O)^{-1/2} = \mathcal{O}(1)$. Applying (207):

$$\|\mathbf{E}(\gamma_{\rm opt})\| \le b_{2,\rm eff} + \alpha_*(\gamma_{\rm opt})\, C_{\rm sing}\, \lambda(Q_O)^{-1/2} = \mathcal{O}(1). \tag{240}$$

Since $\Lambda_{\max} \asymp \sqrt{\lambda(Q_O)}$ and $B_{\rm pert}(\gamma_{\rm opt}) = \mathcal{O}(1)$, the matrix-controlled threshold satisfies

$$\eta_{\rm mat}(\gamma_{\rm opt}) = \frac{\Lambda_{\max}}{2B_{\rm pert}(\gamma_{\rm opt})} \asymp \sqrt{\lambda(Q_O)}. \tag{241}$$

Together with $\eta_{\rm sym}(\gamma_{\rm opt}) = +\infty$ when $a_2 = 0$ and $\eta_{\rm sym}(\gamma_{\rm opt}) = 2\gamma_{\rm opt}/a_2 \asymp \sqrt{\lambda(Q_O)}$ when $a_2 > 0$, this yields

$$\eta^*(\gamma_{\rm opt}) = \min\{\eta_{\rm sym}(\gamma_{\rm opt}), \eta_{\rm mat}(\gamma_{\rm opt})\} \asymp \sqrt{\lambda(Q_O)}. \tag{242}$$

This confirms that the accelerated regime retains an $\mathcal{O}(\sqrt{\lambda(Q_O)})$ admissible perturbation range.

Appendix D. Upper bound via singular value gap

This appendix proves Theorem 5.7.

D.1. **Exponential decay implies a singular value-gap bound.** For a (possibly unbounded) closed operator T on $\mathcal{H}_0$, define its injectivity modulus

$$s(T) := \inf_{f \in \mathrm{dom}(T)\cap\mathcal{H}_0\setminus\{0\}} \frac{\|Tf\|}{\|f\|}. \tag{243}$$

Lemma D.1 (From exponential decay to a singular value-gap bound)**.** *Assume* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$. *Assume moreover that* $e^{t\mathcal{L}}$ *is a contraction semigroup on* $\mathcal{H}_0$*, i.e.* $\|e^{t\mathcal{L}}f\| \le \|f\|$ *for all* $t \ge 0$ *and* $f \in \mathcal{H}_0$ *(equivalently,* $\mathcal{L}$ *is dissipative on* $\mathcal{H}_0$*:* $\mathrm{Re}\langle \mathcal{L}f, f\rangle \le 0$*). If there exist constants* $C \ge 1$ *and* $\nu > 0$ *such that*

$$\|e^{t\mathcal{L}}f\| \le C\, e^{-\nu t}\, \|f\|, \qquad \forall t \ge 0,\ \forall f \in \mathcal{H}_0, \tag{244}$$

then 0 *belongs to the resolvent set of* $\mathcal{L}|_{\mathcal{H}_0}$ *and*

$$\nu \le (1 + \log C)\, s(\mathcal{L}). \tag{245}$$

Proof. For $f \in \mathcal{H}_0$, define

$$u := -\int_0^\infty e^{t\mathcal{L}} f\, dt. \tag{246}$$

Set $t_* := (\log C)/\nu \ge 0$ (the time at which $Ce^{-\nu t}$ crosses 1). On $[0, t_*]$ use the contraction bound $\|e^{t\mathcal{L}}f\| \le \|f\|$; on $[t_*, \infty)$ use the exponential decay bound $\|e^{t\mathcal{L}}f\| \le Ce^{-\nu t}\|f\|$. Combining:

$$\|u\| \le \int_0^{t_*} \|f\|\, dt + \int_{t_*}^\infty Ce^{-\nu t}\|f\|\, dt = t_*\, \|f\| + \frac{C}{\nu} e^{-\nu t_*}\, \|f\| = \frac{\log C}{\nu}\, \|f\| + \frac{1}{\nu}\, \|f\| = \frac{1 + \log C}{\nu}\, \|f\|, \tag{247}$$

where we used $Ce^{-\nu t_*} = Ce^{-\log C} = 1$. Moreover, $u \in \mathrm{dom}(\mathcal{L})$ and differentiating under the integral gives

$$\mathcal{L}u = -\int_0^\infty \mathcal{L}e^{t\mathcal{L}} f\, dt = -\int_0^\infty \partial_t\big(e^{t\mathcal{L}} f\big)\, dt = f. \tag{248}$$

Thus $(\mathcal{L}|_{\mathcal{H}_0})^{-1}$ exists and satisfies $\|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\| \le (1 + \log C)/\nu$, hence $s(\mathcal{L}) = \|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\|^{-1} \ge \nu/(1 + \log C)$, i.e. $\nu \le (1 + \log C)\, s(\mathcal{L})$. □

Lemma D.2 (Relaxation time is bounded below by the singular value gap)**.** *Assume* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$*. Assume moreover that* $e^{t\mathcal{L}}$ *is a contraction semigroup on* $\mathcal{H}_0$*, i.e.* $\|e^{t\mathcal{L}} f\| \le \|f\|$ *for all* $t \ge 0$ *and* $f \in \mathcal{H}_0$*. Assume there exist constants* $C \ge 1$ *and* $\nu > 0$ *such that*

$$\|e^{t\mathcal{L}} f\| \le C\, e^{-\nu t}\, \|f\|, \qquad \forall t \ge 0,\ \forall f \in \mathcal{H}_0. \tag{249}$$

Define the L^2 *relaxation time*

$$t_{\mathrm{rel}}(\mathcal{L}) := \inf\Big\{ t \ge 0\,;\ \|e^{t\mathcal{L}} f\| \le e^{-1}\, \|f\| \text{ for all } f \in \mathcal{H}_0 \Big\}. \tag{250}$$

Then

$$t_{\mathrm{rel}}(\mathcal{L}) \ge \frac{1}{2\, s(\mathcal{L})}. \tag{251}$$

This L^2 *bound is a continuous-time analogue of discrete-time relaxation estimates based on singular values for non-reversible Markov chains; compare* [9].

Proof. By the semigroup property and the definition of $t_{\mathrm{rel}}(\mathcal{L})$, for any $t \ge 0$ write $t = n\, t_{\mathrm{rel}}(\mathcal{L}) + r$ with $n \in \mathbb{N}_0$ and $r \in [0, t_{\mathrm{rel}}(\mathcal{L}))$. Since $e^{t\mathcal{L}}$ is a contraction on $\mathcal{H}_0$, $\|e^{r\mathcal{L}}\| \le 1$. Combined with $\|e^{t_{\mathrm{rel}}(\mathcal{L})\,\mathcal{L}}\| \le e^{-1}$, the semigroup property gives

$$\|e^{t\mathcal{L}}\| \le \|e^{r\mathcal{L}}\|\, \|e^{t_{\mathrm{rel}}(\mathcal{L})\,\mathcal{L}}\|^n \le 1 \cdot e^{-n} = e^{-n}.$$

As in the proof of Lemma D.1, the inverse on $\mathcal{H}_0$ is given by $(\mathcal{L}|_{\mathcal{H}_0})^{-1} = -\int_0^\infty e^{t\mathcal{L}}\, dt$, hence

$$\|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\| \le \int_0^\infty \|e^{t\mathcal{L}}\|\, dt \le \sum_{n=0}^{\infty} \int_0^{t_{\mathrm{rel}}(\mathcal{L})} e^{-n}\, dr = \frac{t_{\mathrm{rel}}(\mathcal{L})}{1 - e^{-1}} \le 2\, t_{\mathrm{rel}}(\mathcal{L}).$$

Therefore $s(\mathcal{L}) = \|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\|^{-1} \ge (2\, t_{\mathrm{rel}}(\mathcal{L}))^{-1}$. □

D.2. **Test function and completion of the proof.**

Proof of Theorem 5.7. Let $\phi \in \mathrm{dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0$ be a normalized eigenfunction of Q_O with

$$Q_O \phi = -\lambda(Q_O)\, \phi, \qquad \|\phi\| = 1. \tag{252}$$

Define the first-order corrector

$$\psi := \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} \phi \in \mathcal{H}_F, \tag{253}$$

and the test function $f_\gamma := \phi - \gamma^{-1}\psi$. Since $\phi \in \mathcal{H}_S$ and $\psi \in \mathcal{H}_F$ we have $\langle \phi, \psi \rangle = 0$ and therefore

$$\|f_\gamma\|^2 = \|\phi\|^2 + \gamma^{-2}\|\psi\|^2 \ge 1. \tag{254}$$

Using $\mathcal{R}\phi = 0$ and $\mathcal{R}\mathcal{R}^+ = P_F$ on $\mathcal{H}$, together with centered coupling $\mathcal{V}_{\mathrm{skew}}\phi \in \mathcal{H}_F$, we obtain

$$\mathcal{R}\psi = \mathcal{R}\mathcal{R}^+ \mathcal{V}_{\mathrm{skew}}\phi = P_F \mathcal{V}_{\mathrm{skew}}\phi = \mathcal{V}_{\mathrm{skew}}\phi. \tag{255}$$

Consequently,

$$\mathcal{L} f_\gamma = \big(\gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}\big)\big(\phi - \gamma^{-1}\psi\big) \tag{256}$$

$$= \underbrace{\mathcal{V}_{\mathrm{skew}}\phi - \mathcal{R}\psi}_{=0} - \gamma^{-1}\mathcal{V}_{\mathrm{skew}}\psi \ + \ \eta\mathcal{V}_{\mathrm{pert}} f_\gamma. \tag{257}$$

We decompose the remaining transport term into slow/fast components:

$$\mathcal{V}_{\mathrm{skew}}\psi = P_S \mathcal{V}_{\mathrm{skew}}\psi + P_F \mathcal{V}_{\mathrm{skew}}\psi. \tag{258}$$

For the slow part, using $Q_O := -P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} P_S$ and $P_S\phi = \phi$, we have

$$P_S \mathcal{V}_{\mathrm{skew}}\psi = P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}}\phi = -Q_O\phi = \lambda(Q_O)\, \phi. \tag{259}$$

For the fast part, apply the flux regularity estimate (26) with the slow input ϕ. Writing $g_\phi := P_F\mathcal{V}_{\rm skew}P_S\phi = \mathcal{V}_{\rm skew}\phi$ and noting that $(-\mathcal{R}^+)g_\phi = -\psi$ on $\mathcal{H}_F$, we obtain

$$\|P_F\mathcal{V}_{\rm skew}\psi\| = \|P_F\mathcal{V}_{\rm skew}P_F\psi\| = \|P_F\mathcal{V}_{\rm skew}P_F(-\mathcal{R}^+)g_\phi\| \le \kappa_{\rm ell}\,\|(-Q_O)\phi\| = \kappa_{\rm ell}\,\lambda(Q_O). \tag{260}$$

Combining the slow and fast contributions yields

$$\|\mathcal{V}_{\rm skew}\psi\| \le (1+\kappa_{\rm ell})\,\lambda(Q_O). \tag{261}$$

Hence

$$\|\mathcal{L}f_\gamma\| \le \frac{(1+\kappa_{\rm ell})\,\lambda(Q_O)}{\gamma} + \eta\,\|\mathcal{V}_{\rm pert}f_\gamma\|. \tag{262}$$

To bound the perturbation, use Assumption 4.1.4 to estimate

$$\|\mathcal{V}_{\rm pert}f\| \le \|\mathcal{K}_{\rm pert}f\| + \|\mathcal{S}_{\rm pert}f\| \tag{263}$$

$$\le (a_1+a_2)\,\|(-\mathcal{R})^{1/2}f\| + (b_1+b_2)\,\|f\|, \qquad f\in{\rm dom}((-\mathcal{R})^{1/2}). \tag{264}$$

Since $\phi\in\ker(\mathcal{R})$ and $f_\gamma = \phi - \gamma^{-1}\psi$, we have

$$\|(-\mathcal{R})^{1/2}f_\gamma\| = \gamma^{-1}\,\|(-\mathcal{R})^{1/2}\psi\|. \tag{265}$$

Finally, using $\psi = \mathcal{R}^+\mathcal{V}_{\rm skew}\phi$ and the definition of Q_O,

$$\|(-\mathcal{R})^{1/2}\psi\|^2 = \langle(-\mathcal{R})\psi,\psi\rangle = \langle\mathcal{V}_{\rm skew}\phi,(-\mathcal{R}^+)\mathcal{V}_{\rm skew}\phi\rangle = -\langle Q_O\phi,\phi\rangle = \lambda(Q_O). \tag{266}$$

Therefore $\|(-\mathcal{R})^{1/2}f_\gamma\| = \sqrt{\lambda(Q_O)}/\gamma$, and

$$\|\mathcal{V}_{\rm pert}f_\gamma\| \le (b_1+b_2)\,\|f_\gamma\| + (a_1+a_2)\,\frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{267}$$

Dividing by $\|f_\gamma\|\ge 1$ gives

$$\frac{\|\mathcal{L}f_\gamma\|}{\|f_\gamma\|} \le \frac{(1+\kappa_{\rm ell})\,\lambda(Q_O)}{\gamma} + \eta\,(b_1+b_2) + \eta\,(a_1+a_2)\,\frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{268}$$

By definition of $s(\mathcal{L})$, this implies $s(\mathcal{L}) \le \|\mathcal{L}f_\gamma\|/\|f_\gamma\|$. Combining with Lemma D.1 yields the claimed bound on ν. □

Appendix E. Proof of Proposition 6.5

We verify Assumptions 4.1.1–4 and 4.3 for a uniformly convex potential $U\in C^2(\mathbb{R}^d)$ with $\nabla^2U \ge m > 0$,

$$\mathcal{V}_{\rm pert}f = F(q)\cdot\nabla_p f, \tag{269}$$

with $d\ge 2$, $m\in(0,1]$, and $F\in C_b^\infty(\mathbb{R}^d;\mathbb{R}^d)$ bounded. We work on the Schwartz core

$$\mathcal{C}_{\rm Lan} := \mathcal{S}(\mathbb{R}^d_q\times\mathbb{R}^d_p) \subset L^2(\mu), \tag{270}$$

where $\mu(dq,dp) = Z^{-1}e^{-\beta(\frac12|p|^2+U(q))}\,dq\,dp$. This space is invariant under P_S, P_F, $\mathcal{R}^+$, $\mathcal{V}_{\rm skew}$, and $\mathcal{V}_{\rm pert}$. Moreover, for every slow test function $\psi=\psi(q)$ one has

$$\mathcal{R}^+\mathcal{V}_{\rm skew}\psi = -p\cdot\nabla_q\psi \in \mathcal{C}_{\rm Lan}, \tag{271}$$

so all compositions in the definition of Q_O and in the corrector stay on the core. Since

$$Q_O = \beta^{-1}\Delta_q - \nabla_q U\cdot\nabla_q \tag{272}$$

is the weighted Laplacian on $L^2(\mu_q)$ with $\mu_q(dq)\propto e^{-\beta U(q)}\,dq$, and U is uniformly convex with $\nabla^2U\ge m>0$, the operator $(-Q_O)$ has a compact resolvent and $(\varpi - Q_O)^{-1}$ preserves $\mathcal{S}(\mathbb{R}^d)$ for every $\varpi>0$. Hence Assumption 4.3 holds.

E.1. **Microscopic spectral gap, centering, and slow gap.** The fast dissipator

$$\mathcal{R} = -p \cdot \nabla_p + \beta^{-1}\Delta_p \tag{273}$$

acts only on the Gaussian momentum variable and has spectral gap

$$\lambda_R = 1 \tag{274}$$

on $\mathcal{H}_F = \ker(\Pi_p)$. The Hamiltonian part $\mathcal{V}_{\text{skew}} = p \cdot \nabla_q - \nabla_q U \cdot \nabla_p$ is skew-adjoint in $L^2(\mu)$ because the Hamiltonian flow preserves μ. Since $\int p \, d\mu_p = 0$, one has

$$P_S \mathcal{V}_{\text{skew}} P_S = 0. \tag{275}$$

Also $P_S \mathcal{V}_{\text{pert}} P_S = 0$ because $\nabla_p \phi(q) = 0$ for slow functions.

For $\psi = \psi(q)$,

$$g_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi = p \cdot \nabla_q \psi. \tag{276}$$

Since $\mathcal{R}(p_i) = -p_i$, one has $(-\mathcal{R}^+) g_\psi = g_\psi$, so the conditioning constant is

$$M_R = 1. \tag{277}$$

The effective slow operator is

$$Q_O \psi = \beta^{-1}\Delta_q \psi - \nabla_q U \cdot \nabla_q \psi. \tag{278}$$

By the assumed Poincaré inequality for $\mu_q(dq) \propto e^{-\beta U(q)} \, dq$ with constant m (which follows from convexity $\nabla^2 U \geq m > 0$ by the Bakry–Émery criterion),

$$-\langle \psi, Q_O \psi \rangle_{L^2(\mu_q)} = \beta^{-1} \|\nabla_q \psi\|^2_{L^2(\mu_q)} \geq m \, \|\psi - \textstyle\int \psi \, d\mu_q\|^2_{L^2(\mu_q)}, \tag{279}$$

so Assumption 4.2 holds with

$$\lambda(Q_O) = m. \tag{280}$$

E.2. **Flux regularity (Assumption 4.1.3).** Define

$$Z_{ij}(p) := p_i p_j - \beta^{-1}\delta_{ij}. \tag{281}$$

Using $(-\mathcal{R}^+) g_\psi = g_\psi$ and a direct computation,

$$P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) g_\psi = \sum_{i,j=1}^{d} Z_{ij}(p) \, \partial_{q_i q_j} \psi. \tag{282}$$

The Gaussian fourth-moment identity

$$\mathbb{E}_{\mu_p}[Z_{ij} Z_{kl}] = \beta^{-2}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) \tag{283}$$

therefore gives

$$\|P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) g_\psi\|^2_{L^2(\mu)} = 2\beta^{-2} \, \|\nabla_q^2 \psi\|^2_{L^2(\mu_q)}. \tag{284}$$

To control the Hessian, we use the Bochner identity. Set $L := -\beta Q_O = -\Delta_q + \beta \nabla_q U \cdot \nabla_q$, the weighted Laplacian on $L^2(\mu_q)$. For every $\psi \in \mathcal{S}(\mathbb{R}^d)$, one has the commutator

$$[\partial_{q_i}, L] = \beta \, (\partial^2_{q_i q_k} U) \, \partial_{q_k}, \tag{285}$$

obtained from $[\partial_{q_i}, \nabla_q U \cdot \nabla_q] = (\partial^2_{q_i q_k} U)\partial_{q_k}$ and $[\partial_{q_i}, -\Delta_q] = 0$. Since L is self-adjoint in $L^2(\mu_q)$ and $\langle g, Lg\rangle_{L^2(\mu_q)} = \|\nabla_q g\|^2_{L^2(\mu_q)}$,

$$\begin{aligned}\|L\psi\|^2_{L^2(\mu_q)} &= \langle \psi, L^2\psi\rangle_{L^2(\mu_q)} = \sum_{i=1}^d \langle \partial_{q_i}\psi, \partial_{q_i}(L\psi)\rangle_{L^2(\mu_q)} \\ &= \sum_{i=1}^d \langle \partial_{q_i}\psi, L(\partial_{q_i}\psi)\rangle_{L^2(\mu_q)} + \beta\,\langle \nabla_q^2 U\,\nabla_q\psi, \nabla_q\psi\rangle_{L^2(\mu_q)} \\ &= \|\nabla_q^2\psi\|^2_{L^2(\mu_q)} + \beta\,\langle \nabla_q^2 U\,\nabla_q\psi, \nabla_q\psi\rangle_{L^2(\mu_q)}. \end{aligned} \tag{286}$$

Since U is convex ($\nabla_q^2 U \geq 0$), the second term is non-negative, giving the *Bochner inequality*

$$\|\nabla_q^2\psi\|_{L^2(\mu_q)} \leq \|L\psi\|_{L^2(\mu_q)} = \beta\,\|(-Q_O)\psi\|_{L^2(\mu_q)}. \tag{287}$$

Combining with (284) yields

$$\|P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) g_\psi\|_{L^2(\mu)} \leq \sqrt{2}\,\|(-Q_O)\psi\|_{L^2(\mu_q)}. \tag{288}$$

Thus Assumption 4.1.3 holds with

$$\kappa_{\text{ell}} = \sqrt{2}. \tag{289}$$

E.3. **Perturbation bounds (Assumption 4.1.4).** The $L^2(\mu)$-adjoint of $\mathcal{V}_{\text{pert}} = F(q)\cdot\nabla_p$ is

$$\mathcal{V}^\dagger_{\text{pert}} = -F(q)\cdot\nabla_p + \beta F(q)\cdot p, \tag{290}$$

so the skew/symmetric decomposition is

$$\mathcal{K}_{\text{pert}} = F(q)\cdot\nabla_p - \tfrac{\beta}{2}F(q)\cdot p, \qquad \mathcal{S}_{\text{pert}} = \tfrac{\beta}{2}F(q)\cdot p. \tag{291}$$

Since F is bounded,

$$\|F\cdot\nabla_p f\|_{L^2(\mu)} \leq \|F\|_{L^\infty}\,\|\nabla_p f\|_{L^2(\mu)} \leq \sqrt{\beta}\,\|F\|_{L^\infty}\,\|(-\mathcal{R})^{1/2} f\|_{L^2(\mu)}. \tag{292}$$

For the multiplier part, the Gaussian integration-by-parts estimate in momentum gives

$$\||p| f\|_{L^2(\mu)} \leq \sqrt{2d/\beta}\,\|f\|_{L^2(\mu)} + (2/\beta)\,\|\nabla_p f\|_{L^2(\mu)}. \tag{293}$$

Therefore

$$\|F\cdot p\, f\|_{L^2(\mu)} \leq \|F\|_{L^\infty}\Big(\sqrt{2d/\beta}\,\|f\|_{L^2(\mu)} + \tfrac{2}{\beta}\,\|\nabla_p f\|_{L^2(\mu)}\Big). \tag{294}$$

Combining the last two bounds shows that Assumption 4.1.4 holds with constants (a_1, b_1, a_2, b_2) depending only on $(\beta, d, \|F\|_{L^\infty})$ and independent of m. □

Appendix F. Proof of Proposition 6.7

All operators are bounded on the finite-dimensional space $\mathcal{H}$.

(1) Assumption 4.1.1 holds for $\mathcal{R}$ by construction, with $\lambda_R = 4\Gamma$.
(2) Assumption 4.1.2 holds since both $\mathcal{V}_{\text{skew}}$ and $\mathcal{V}_{\text{pert}}$ map $\mathcal{H}_S$ into $\mathcal{H}_F$.
(3) Assumption 4.1.3 and Assumption 4.2 are both verified explicitly. The key improvement over a single-channel coupling (e.g., $H_{\text{int}} = g\sigma_z \otimes \sigma_z$) is that the double commutator $\text{ad}^2_{\sigma_a}$ has kernel $\text{span}\{I, \sigma_a\}$; with two channels,

$$\ker(Q_O) \cap \mathcal{H}_0 = \ker(\text{ad}^2_{\sigma_x}) \cap \ker(\text{ad}^2_{\sigma_z}) \cap \{\text{tr} = 0\} = \{0\}, \tag{295}$$

so Assumption 4.2 holds with $\ker(Q_O) = \mathbb{C}\mathbf{1}$ and $\lambda(Q_O) = 4g^2/\kappa = g^2/\Gamma$; the numerical values in Table 2 simply reproduce this exact formula.

For the flux regularity bound (26), define the bounded operator

$$\mathcal{K} := P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) P_F \mathcal{V}_{\text{skew}} P_S : \mathcal{H}_S \to \mathcal{H}_F. \tag{296}$$

Because $P_S\mathcal{V}_{\text{skew}}P_S = 0$, both $\mathcal{K}$ and $-Q_O$ annihilate the stationary direction $\mathbb{C}\mathbf{1}$. On the finite-dimensional mean-zero slow space $\mathcal{H}_{S,0}$, the kernel computation above shows that $-Q_O|_{\mathcal{H}_{S,0}}$ is positive definite and therefore invertible. We may thus set

$$\mathcal{T} := \mathcal{K}\,(-Q_O|_{\mathcal{H}_{S,0}})^{-1} : \mathcal{H}_{S,0} \to \mathcal{H}_F, \qquad \kappa_{\text{ell}} := \|\mathcal{T}\| < \infty. \tag{297}$$

Writing any slow vector as $\psi = c\mathbf{1} + \psi_0$ with $\psi_0 \in \mathcal{H}_{S,0}$ and using $g_{\mathbf{1}} = 0$, we obtain

$$P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)g_\psi = \mathcal{K}\psi_0 = \mathcal{T}(-Q_O)\psi_0, \tag{298}$$

and hence

$$\|P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)g_\psi\| \le \kappa_{\text{ell}}\,\|(-Q_O)\psi_0\| = \kappa_{\text{ell}}\,\|(-Q_O)\psi\|. \tag{299}$$

Thus (26) holds analytically in this finite-dimensional model, and one may take $M_R := \|\mathcal{R}\|$.

(4) For Assumption 4.1.4, decompose $\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$ by $\mathcal{K}_{\text{pert}} := \frac{1}{2}(\mathcal{V}_{\text{pert}} - \mathcal{V}_{\text{pert}}^\dagger)$ and $\mathcal{S}_{\text{pert}} := \frac{1}{2}(\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}^\dagger)$. Since $\mathcal{V}_{\text{pert}}$ is bounded, the energy bounds (29)–(30) hold with $a_1 = a_2 = 0$ and $b_1 = \|\mathcal{K}_{\text{pert}}\|$, $b_2 = \|\mathcal{S}_{\text{pert}}\|$.

□

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